Alignment

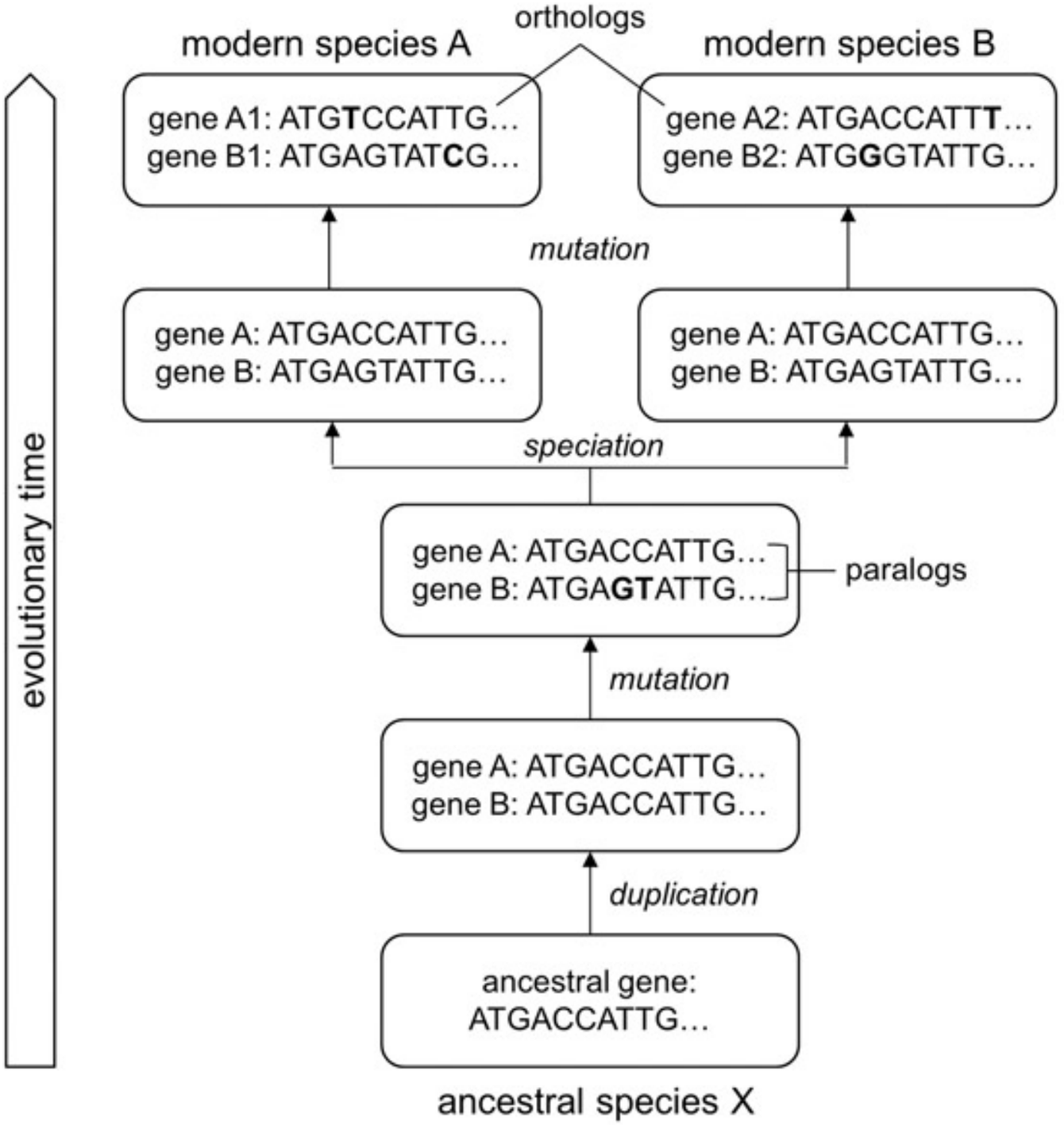
- Dynamic programming
- Global pairwise alignemnt (Needleman-Wunsch)
- Local pairwise alignemnt (Smith-Waterman)
- End gaps
- Linear and affine gap scoring

Alignments

```
TAAAGACCATAGGAAATAAAGATAA
TAACGACCAT-GGAAACAAAGATAA
```

Why do some sequences look alike?

Ortholog and paralog genes



What is a good global alignment?

```
Species 1: TAAAGACCATAGGAAATAAAGATAA
Species 2: TAACGACCAT-GGAAACAAAGATAA
```

Scoring

	A	C	G	T
A	**1**	**0**	**0**	**0**
C	**0**	**1**	**0**	**0**
G	**0**	**0**	**1**	**0**
T	**0**	**0**	**0**	**1**

Gap	**0**

```
ACGT-GTCAACG-T
ACGTCGTA--GCTA
```

```
ACGT-GTCAACGT-
ACGTCGT-AGC-TA
```

Hvad er scoren?

Scoring

	A	C	G	T
A	**1**	**0**	**0**	**0**
C	**0**	**1**	**0**	**0**
G	**0**	**0**	**1**	**0**
T	**0**	**0**	**0**	**1**

Gap	**0**

```
ACGT-GTCAACG-T
ACGTCGTA--GCTA
```

6

```
ACGT-GTCAACGT-
ACGTCGT-AGC-TA
```

9

Hvad er scoren?

Scoring

	A	C	G	T
A	**2**	**0**	**0**	**0**
C	**0**	**2**	**0**	**0**
G	**0**	**0**	**2**	**0**
T	**0**	**0**	**0**	**2**

Gap	**-1**

```
ACGT-GTCAACG-T
ACGTCGTA--GCTA
```

```
ACGT-GTCAACGT-
ACGTCGT-AGC-TA
```

Hvad er scoren?

Scoring

	A	C	G	T
A	**2**	**0**	**0**	**0**
C	**0**	**2**	**0**	**0**
G	**0**	**0**	**2**	**0**
T	**0**	**0**	**0**	**2**

Gap	**-1**

```
ACGT-GTCAACG-T
ACGTCGTA--GCTA
```

8

```
ACGT-GTCAACGT-
ACGTCGT-AGC-TA
```

14

Hvad er scoren?

What is an *optimal* alignment

	A	C	G	T
A	**2**	**0**	**0**	**0**
C	**0**	**2**	**0**	**0**
G	**0**	**0**	**2**	**0**
T	**0**	**0**	**0**	**2**

Gap	**-1**

```
Species 1: TAAAGACCATAGGAAATAAAGATAA
Species 2: TAACGACCAT-GGAAACAAAGATAA
```

Hvad mener vi med “optimalt”?
- og er et optimalt alignment “sandt”?

How many alignments are there?

På hvor mange måder kan man aligne flg. to sekvenser:

sekvens 1: **ag**

sekvens 2: **ct**

How many alignments are there?

```
ag      a-g     a-g     ag-
ct      ct-     -ct     -ct

        ag-     -ag     -ag
        c-t     c-t     ct-
```

```
-a-g    ag--    --ag
c-t-    --ct    ct--

a--g    a-g-    -ag-
-ct-    -c-t    c--t
```

How many alignments are there?

På hvor mange måder kan man aligne
to sekvenser med hver 20 baser?

$$\binom{2n}{n} = \frac{(2n)!}{(n!)^2}$$

How many alignments are there?

```
from math import factorial

for n in range(1, 21):
    nr = factorial(2*n) / factorial(n)**2
    print(n, nr)
```

1 2
2 6
3 20
4 70
5 252
6 924
7 3432
8 12870
9 48620
10 184756
11 705432
12 2704156
13 10400600
14 40116600
15 155117520
16 601080390
17 2333606220
18 9075135300
19 35345263800
20 137846528820

2 seqences of 40 bases

A:

GATCTACGGCGCAGCGGCCAGCTGTTCTAAGGTGACCGGAAATCCACGGGCCTAACGGCGAAGTCGGCGCGGT
GACCACCAGCCTCGTCTTCCAGCCACGGTTCAGAAATGATTTCAGCCCCCCCTGTGGCACGGTCCT

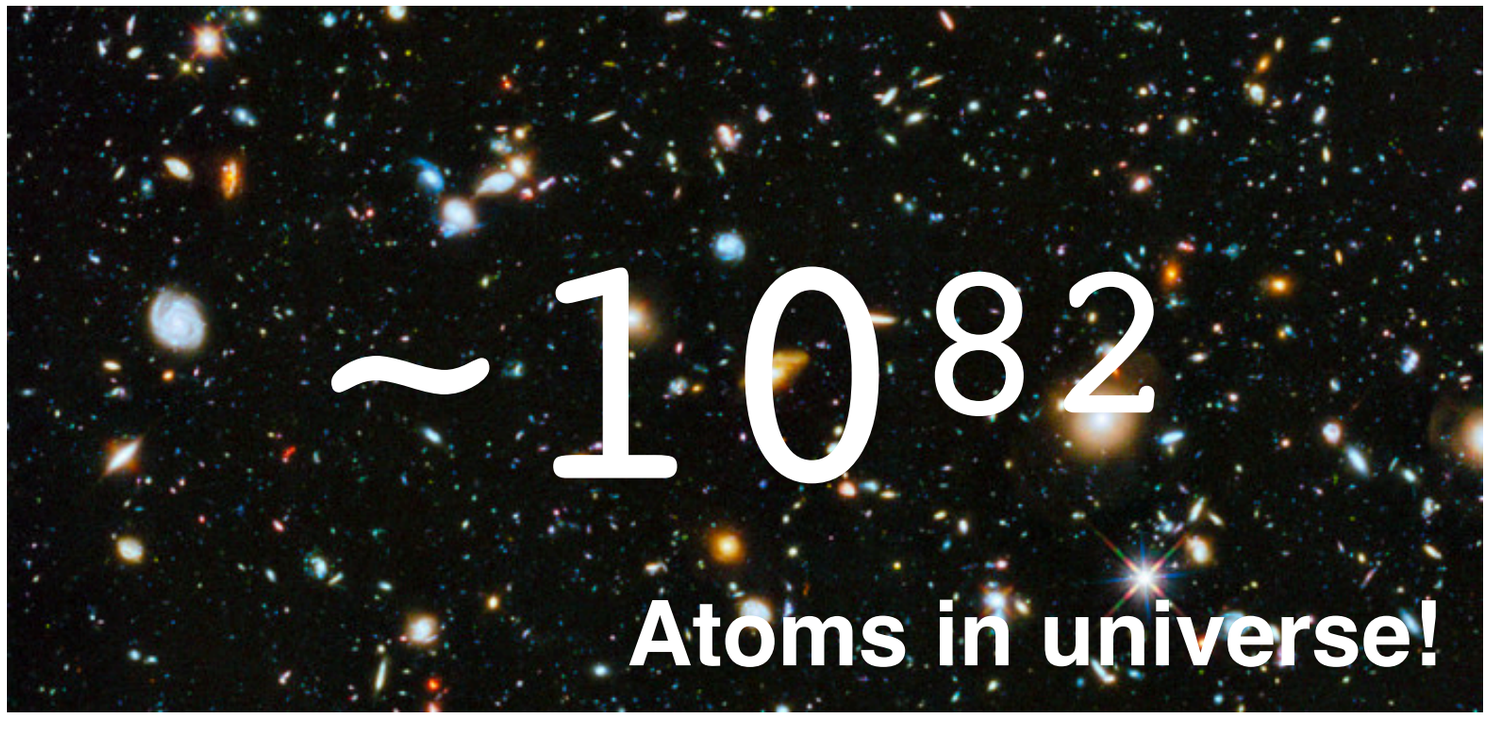
B:

GGGATCCACGGCGCTGCGGCCAGCCTGTTCCTCTAAGGTGACCGGACCACGGGCCACACGGCGAAGTCGGCGC
GGCGAAAACCACCAGCTCGTCTCCAGCCACGGTTCAGAAATGATTTCAGTGCTGTGGCACGGTCCT

2 seqences of 40 bases

A:
GATCTACGGCGCAGCGGCCAGCTGTTCTAAGGTGACCGGAAATCCACGGGCCTAACGGCGAAGTCGGCGCGGT
GACCACCAGCCTCGTCTTCCAGCCACGGTTCAGAAATGATTTCAGCCCCCCCTGTGGCACGGTCCT

B:
GGGATCCACGGCGCTGCGGCCAGCCTGTTCCTCTAAGGTGACCGGACCACGGGCCACACGGCGAAGTCGGCGC
GGCGAAAACCACCAGCTCGTCTCCAGCCACGGTTCAGAAATGATTTCAGTGCTGTGGCACGGTCCT



How do we find the score for the optimal alignment?

- The problem can be divided into smaller identical problems.
- Each part of an optimal alignment is itself an optimal alignemnt (If any part could be improved, the full alignment would not be optimal).

Optimal alignment



```
ga-ta
-acta
```

Optimal alignment of **ga** and **ac**



```
ga- ta
-ac ta
```

Optimal alignment of **gat** and **act**



```
ga-t a
-act a
```

g a t a **a c t a**

1 … … n 1 … … m

g a t a **a c t a**

1 … … n 1 … … m

score matrix:

	a	t	g	c
a	2	0	0	0
t	0	2	0	0
g	0	0	2	0
c	0	0	0	2

gap cost: -1

g a t a **a c t a**

1 … … n 1 … … m

score matrix:	a	t	g	c
a	2	0	0	0
t	0	2	0	0
g	0	0	2	0
c	0	0	0	2

gap cost: -1

S(n,m) = ?

g a t a **a c t a**

1 … … n 1 … … m

S(n,m) = ?

score matrix:

	a	t	g	c
a	2	0	0	0
t	0	2	0	0
g	0	0	2	0
c	0	0	0	2

gap cost: -1

2
a
a

-1
a
–

-1
–
a

g a t a **a c t a**

1 … … n 1 … … m

score matrix:				
	a	**t**	**g**	**c**
a	2	0	0	0
t	0	2	0	0
g	0	0	2	0
c	0	0	0	2

gap cost: -1

S(n,m) = ?

S(n-1, m-1) = 2

g a - t
- a c t

2

a
a

S(n-1, m) = 1

g a - t -
- a c t a

-1

a
-

S(n, m-1) = 1

g a - t a
- a c t -

-1

-
a

g a t a **a c t a**

1 … … n 1 … … m

score matrix:				
	a	t	g	c
a	2	0	0	0
t	0	2	0	0
g	0	0	2	0
c	0	0	0	2

gap cost: -1

S(n,m) = ?

S(n-1, m-1) = 2

```
g a - t
- a c t
```
+ 2
```
a
a
```
= 4

S(n-1, m) = 1

```
g a - t -
- a c t a
```
+ -1
```
a
-
```
= 0

S(n, m-1) = 1

```
g a - t a
- a c t -
```
+ -1
```
-
a
```
= 0

g a t a **a c t a**

1 … … n 1 … … m

score matrix:	a	t	g	c
a	2	0	0	0
t	0	2	0	0
g	0	0	2	0
c	0	0	0	2

gap cost: -1

$$S(n,m) = \max \begin{cases} S(n-1, m-1) = 2 & \begin{matrix} g\ a - t \\ - a\ c\ t \end{matrix} & + & \begin{matrix} 2 \\ a \\ a \end{matrix} & = 4 \\ S(n-1, m) = 1 & \begin{matrix} g\ a - t - \\ - a\ c\ t\ a \end{matrix} & + & \begin{matrix} -1 \\ a \\ - \end{matrix} & = 0 \\ S(n, m-1) = 1 & \begin{matrix} g\ a - t\ a \\ - a\ c\ t - \end{matrix} & + & \begin{matrix} -1 \\ - \\ a \end{matrix} & = 0 \end{cases}$$

g a t a **a c t a**

1 … … n 1 … … m

score matrix:	a	t	g	c
a	2	0	0	0
t	0	2	0	0
g	0	0	2	0
c	0	0	0	2

gap cost: -1

$$S(n,m) = \max \begin{cases} S(n-1, m-1) = ? \quad \mathbf{g\,a\,t} \;\; \mathbf{a\,c\,t} + \begin{matrix} \mathbf{a} \\ \mathbf{a} \end{matrix} \; (2) = ? \\ S(n-1, m) = ? \quad \mathbf{g\,a\,t} \;\; \mathbf{a\,c\,t\,a} + \begin{matrix} \mathbf{a} \\ \mathbf{-} \end{matrix} \; (-1) = ? \\ S(n, m-1) = ? \quad \mathbf{g\,a\,t\,a} \;\; \mathbf{a\,c\,t} + \begin{matrix} \mathbf{-} \\ \mathbf{a} \end{matrix} \; (-1) = ? \end{cases}$$

g a t a **a c t a**

1 … … n 1 … … m

score matrix:

	a	t	g	c
a	2	0	0	0
t	0	2	0	0
g	0	0	2	0
c	0	0	0	2

gap cost: -1

$$S(n,m) = \max \begin{cases} S(n-1,m-1) = \max \begin{cases} S(n-2,m-2) = ? \ \texttt{ga ac} & + \begin{matrix}2\\ t\\ t\end{matrix} \\ S(n-2,m-1) = ? \ \texttt{ga act} & + \begin{matrix}-1\\ t\\ -\end{matrix} \\ S(n-1,m-2) = ? \ \texttt{gat ac} & + \begin{matrix}-1\\ -\\ t\end{matrix} \end{cases} + \begin{matrix}2\\ a\\ a\end{matrix} = ? \\ S(n-1,m) = \max \begin{cases} S(n-2,m-1) = ? \ \texttt{ga act} & + \begin{matrix}2\\ t\\ a\end{matrix} \\ S(n-2,m) = ? \ \texttt{ga acta} & + \begin{matrix}-1\\ t\\ -\end{matrix} \\ S(n-1,m-1) = ? \ \texttt{gat act} & + \begin{matrix}-1\\ -\\ a\end{matrix} \end{cases} + \begin{matrix}-1\\ a\\ -\end{matrix} = ? \\ S(n,m-1) = \max \begin{cases} S(n-1,m-2) = ? \ \texttt{gat ac} & + \begin{matrix}2\\ a\\ t\end{matrix} \\ S(n-1,m-1) = ? \ \texttt{gat act} & + \begin{matrix}-1\\ a\\ -\end{matrix} \\ S(n,m-2) = ? \ \texttt{gata ac} & + \begin{matrix}-1\\ -\\ t\end{matrix} \end{cases} + \begin{matrix}-1\\ -\\ a\end{matrix} = ? \end{cases}$$

g a t a
1 … … n

a c t a
1 … … m

score matrix:

	a	t	g	c
a	2	0	0	0
t	0	2	0	0
g	0	0	2	0
c	0	0	0	2

gap cost: -1

$$S(n,m) = \max \begin{cases} S(n-1,m-1) = \max \{\ldots\} + \begin{matrix} 2 \\ \mathbf{a} \\ \mathbf{a} \end{matrix} = ? \\ S(n-1,\ m) = \max \{\ldots\} + \begin{matrix} -1 \\ \mathbf{a} \\ \mathbf{-} \end{matrix} = ? \\ S(n,\ m-1) = \max \{\ldots\} + \begin{matrix} -1 \\ \mathbf{-} \\ \mathbf{a} \end{matrix} = ? \end{cases}$$

A recursive solution

$$S(i,j) = \max \begin{cases} S(i-1,j-1) + s(a_i, b_j) & \text{if } i > 0 \text{ and } j > 0 \\ S(i-1,j) + g & \text{if } i > 0 \\ S(i,j-1) + g & \text{if } j > 0 \end{cases}$$

$$S(0,0) = 0$$

```
MAT = {'A': {'A': 2, 'T': 0, 'G': 0, 'C': 0},
       'T': {'A': 0, 'T': 2, 'G': 0, 'C': 0},
       'G': {'A': 0, 'T': 0, 'G': 2, 'C': 0},
       'C': {'A': 0, 'T': 0, 'G': 0, 'C': 2}}

G = -1

def score(a, b):
    if len(a) == 0 and len(b) == 0:
        return 0
    possible = []
    if len(a) > 0 and len(b) > 0:
        s = score(a[:-1], b[:-1]) + MAT[a[-1]][b[-1]]
        possible.append(s)
    if len(a) > 0:
        s = score(a[:-1], b) + G
        possible.append(s)
    if len(b) > 0:
        s = score(a, b[:-1]) + G
        possible.append(s)
    return max(possible)
```

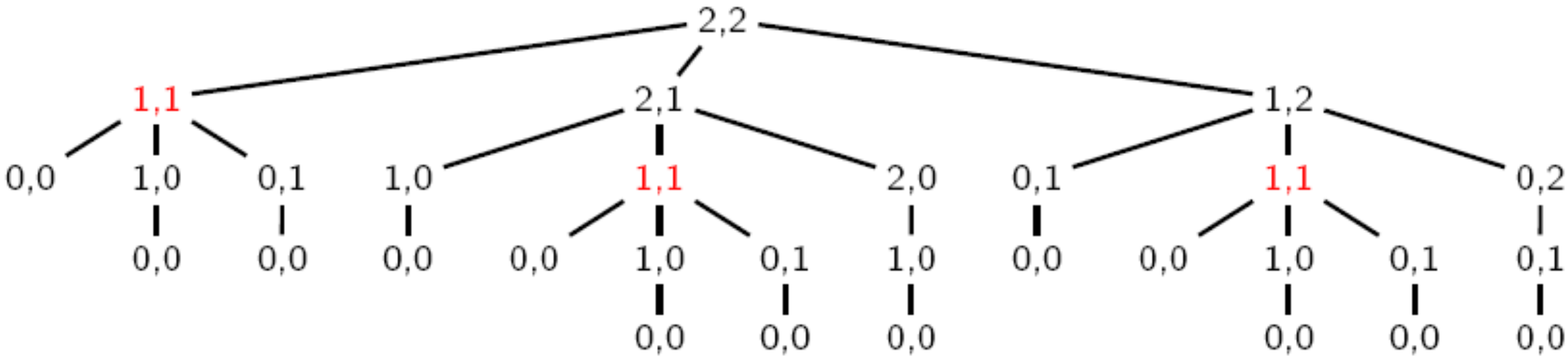
Can you see where we use recursion and where the base case is?

```
def score(a, b):
    if len(a) == 0 and len(b) == 0:
        return 0
    possible = []
    if len(a) > 0 and len(b) > 0:
        s = score(a[:-1], b[:-1]) + MAT[a[-1]][b[-1]]
        possible.append(s)
    if len(a) > 0:
        s = score(a[:-1], b) + G
        possible.append(s)
    if len(b) > 0:
        s = score(a, b[:-1]) + G
        possible.append(s)
    return max(possible)
```

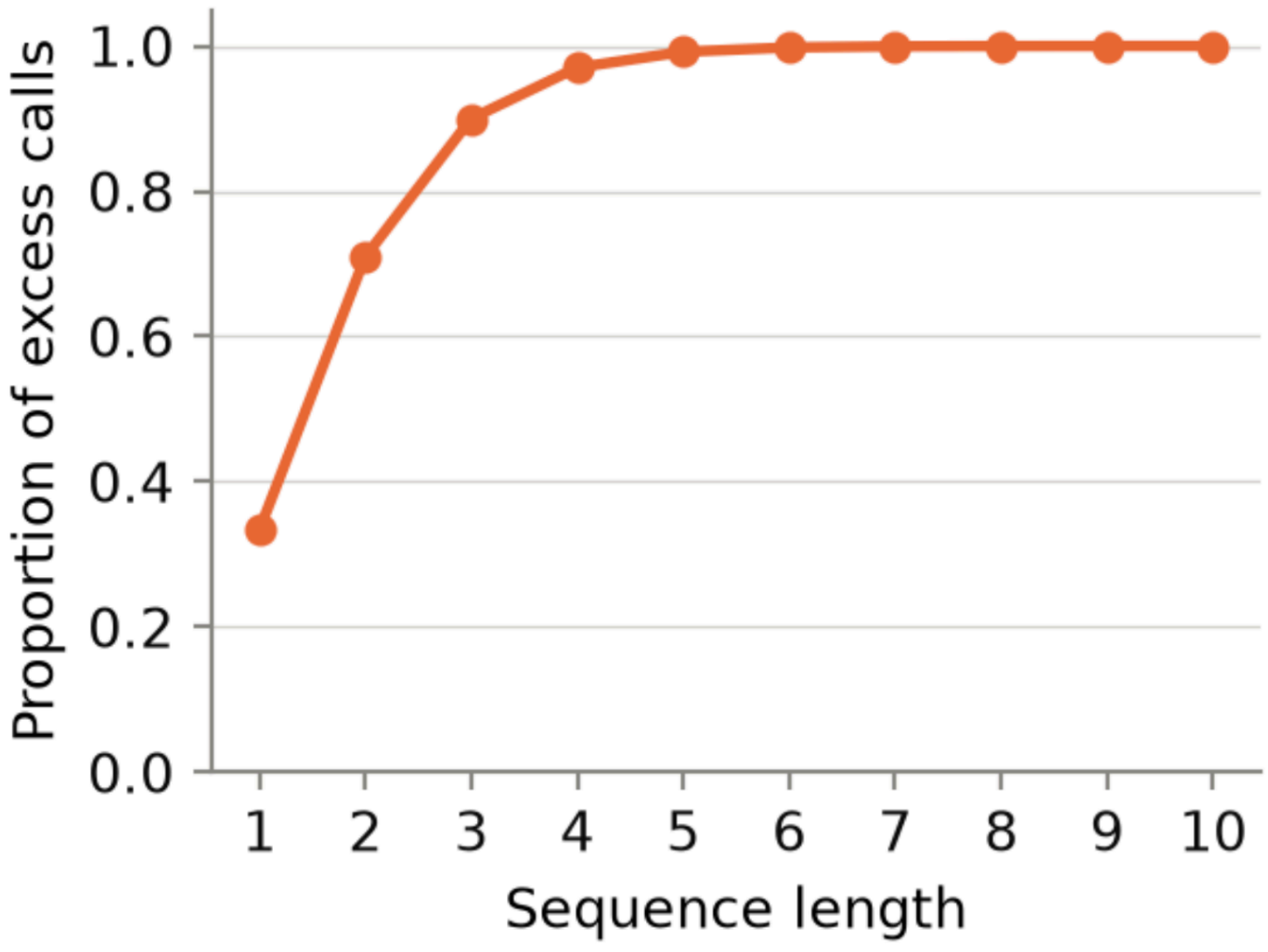
Why does it become so slow with longer sequences?

Why does it become so slow with longer sequences?

Alignment af to sekvenser med to baser:



How many function calls could we avoid?

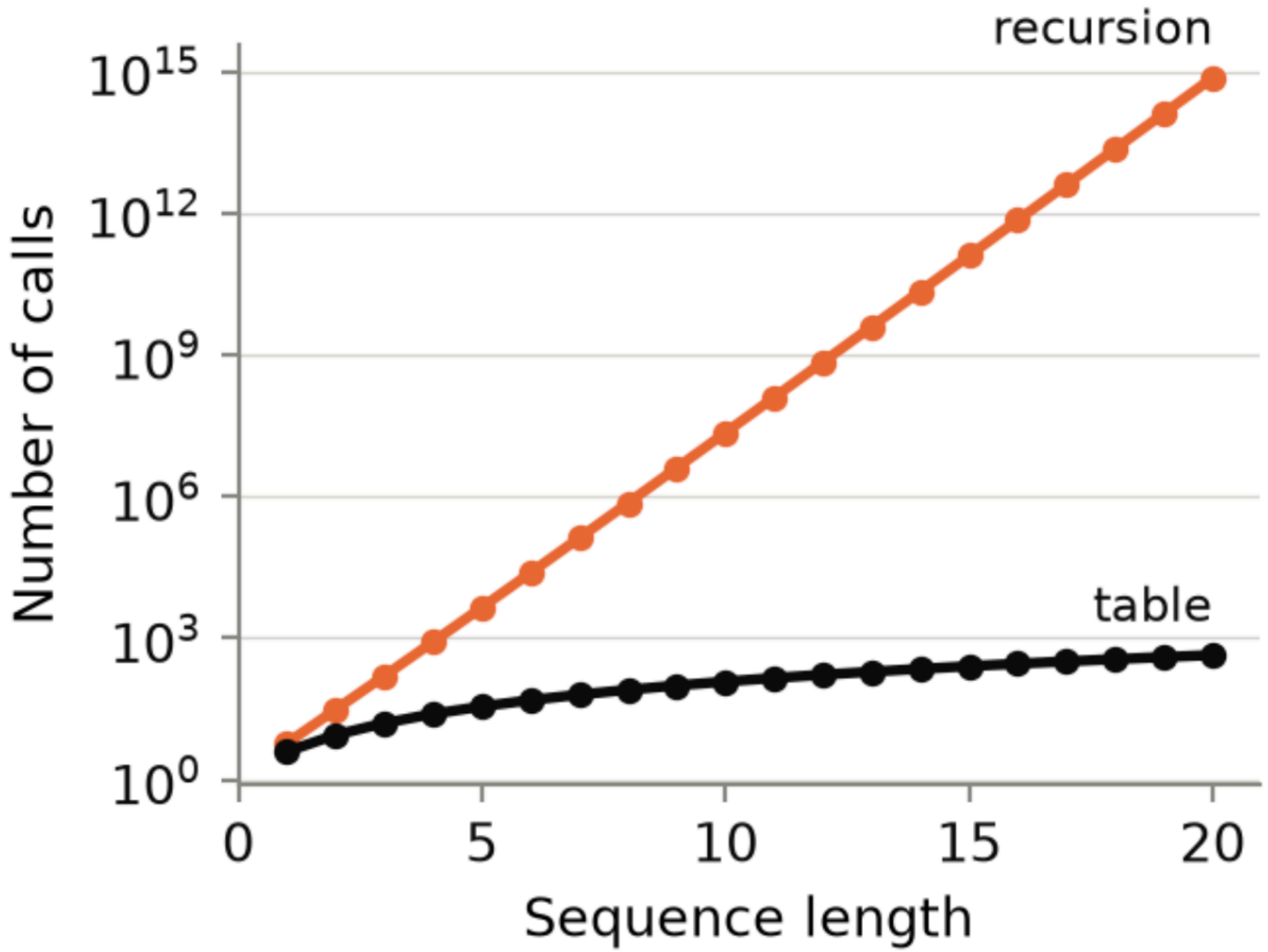


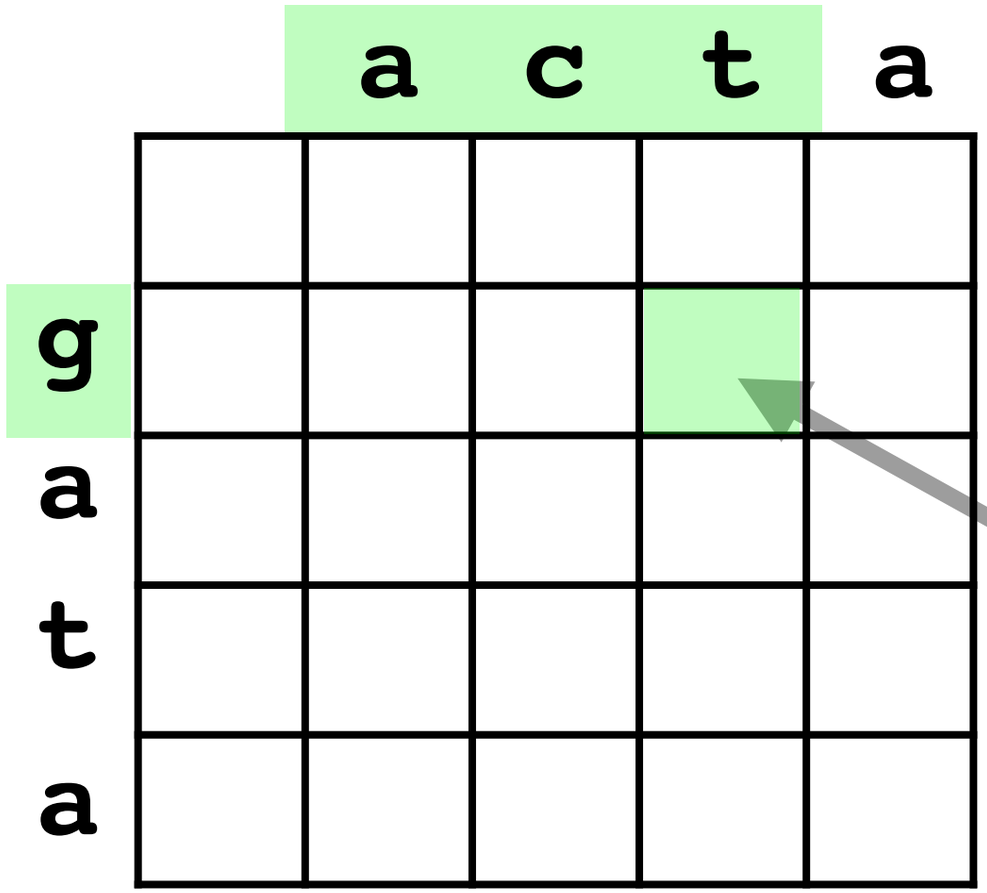
Needleman-Wunsch

- Dynamic programmering algoritm

- Finds a global alignment of to sequences with the highest score.

Needleman-Wunsch

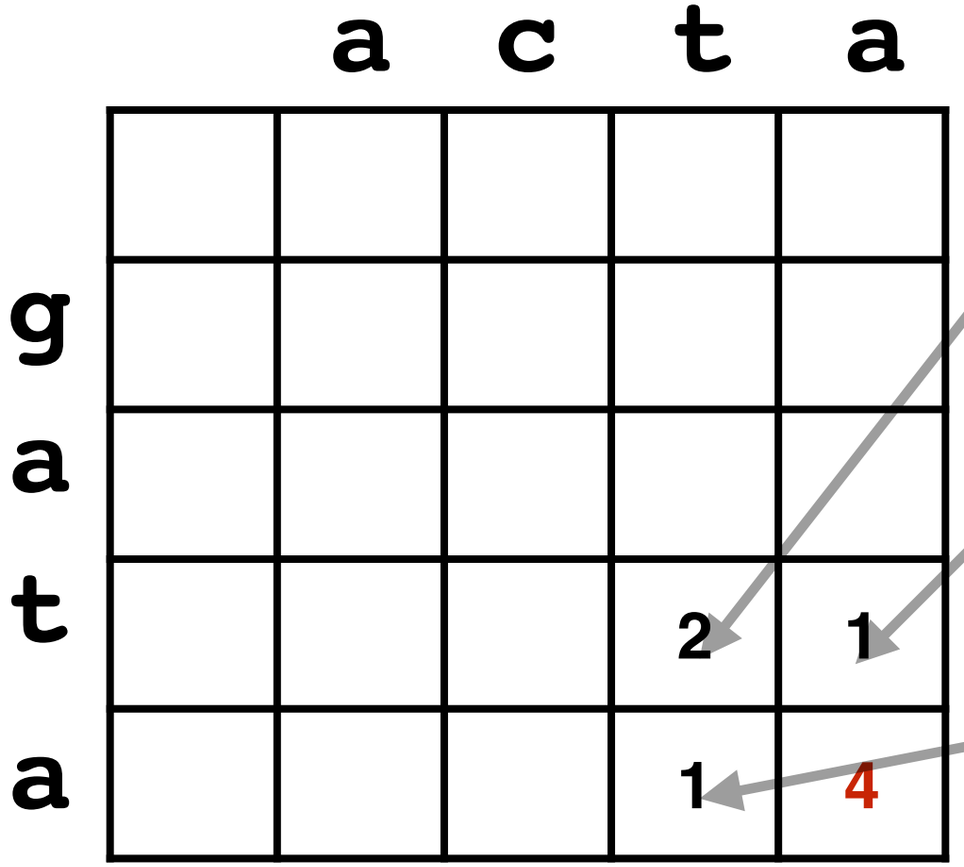




Tabel with scores for the optimal alignment of the first n bases of sequence 1 and m bases of sequence 2.

S(1,3)

Storing intermediate results in the table means we only need to compute them once.



S(n-1, m-1) = **2**

```
g a - t
- a c t
```

2 + a a = 4

S(n-1, m) = **1**

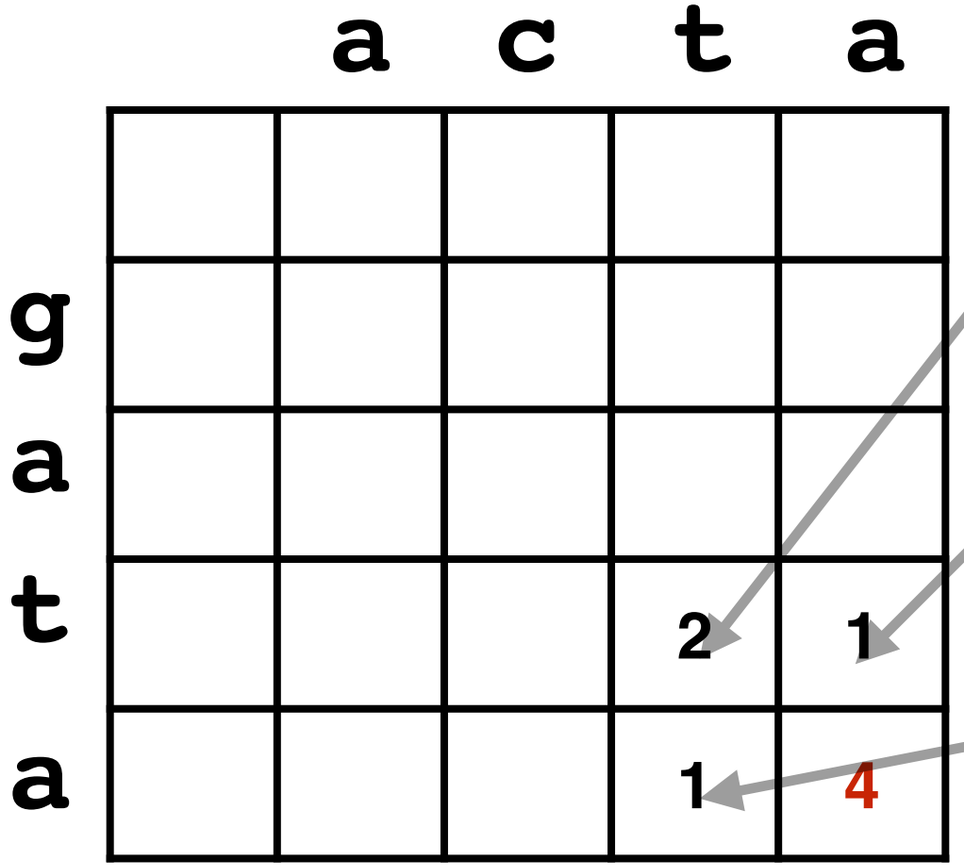
```
g a - t -
- a c t a
```

-1 + a - = 0

S(n, m-1) = **1**

```
g a - t a
- a c t -
```

-1 + - a = 0



S(n-1, m-1) = **2**

```
g a - t
- a c t
```

2

```
a
a
```

\+ = 4

S(n-1, m) = **1**

```
g a - t -
- a c t a
```

-1

```
a
-
```

\+ = 0

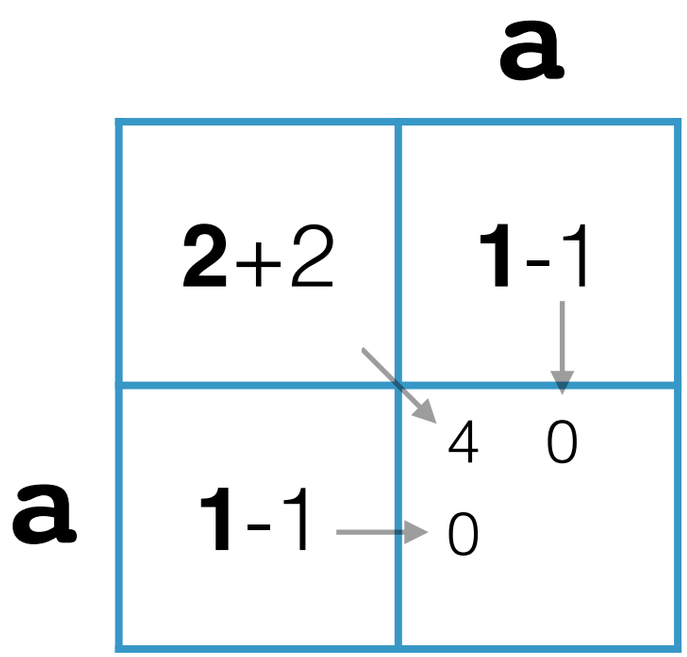
S(n, m-1) = **1**

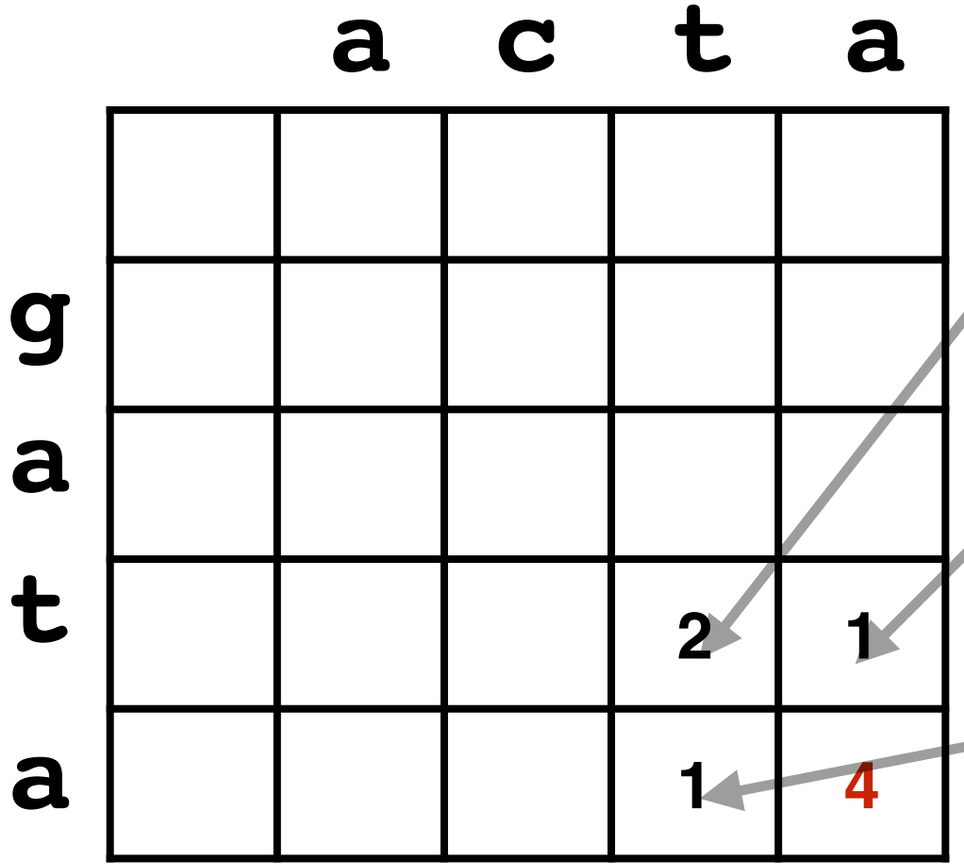
```
g a - t a
- a c t -
```

-1

```
-
a
```

\+ = 0





S(n-1, m-1) = **2**

```
g a - t
- a c t
```

\+ 2 (a / a) = 4

S(n-1, m) = **1**

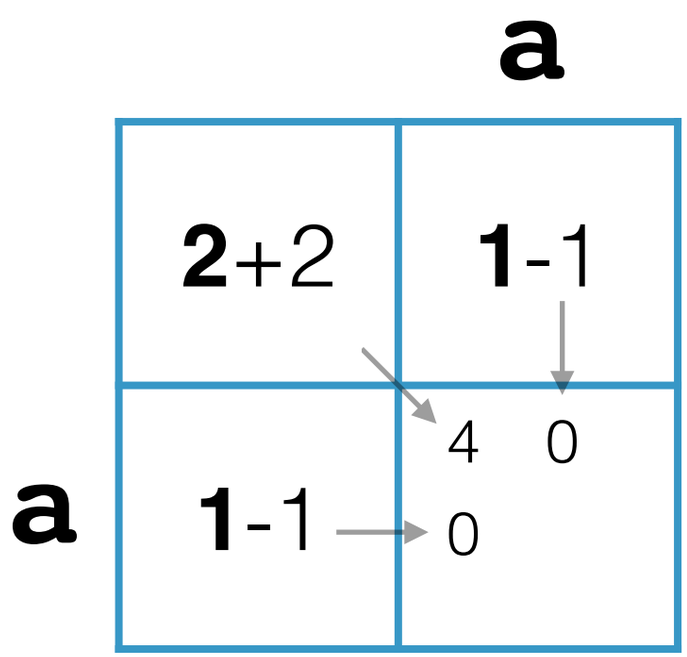
```
g a - t -
- a c t a
```

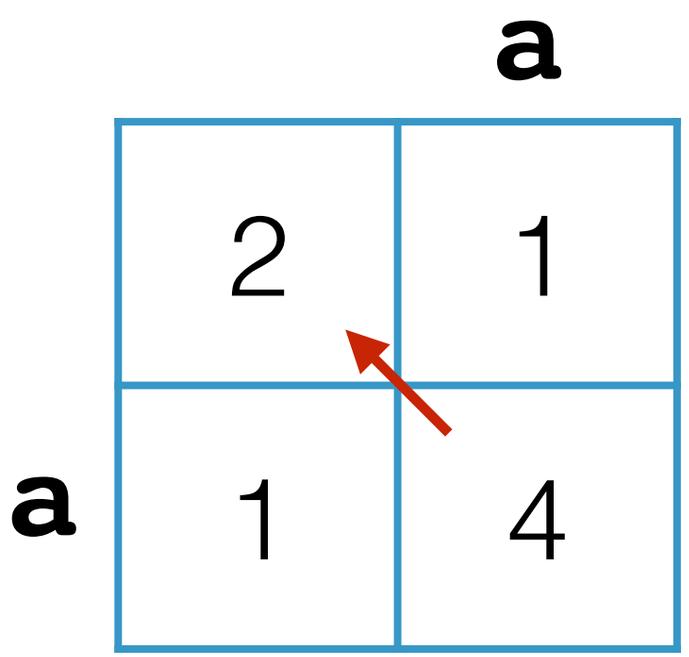
\+ -1 (a / -) = 0

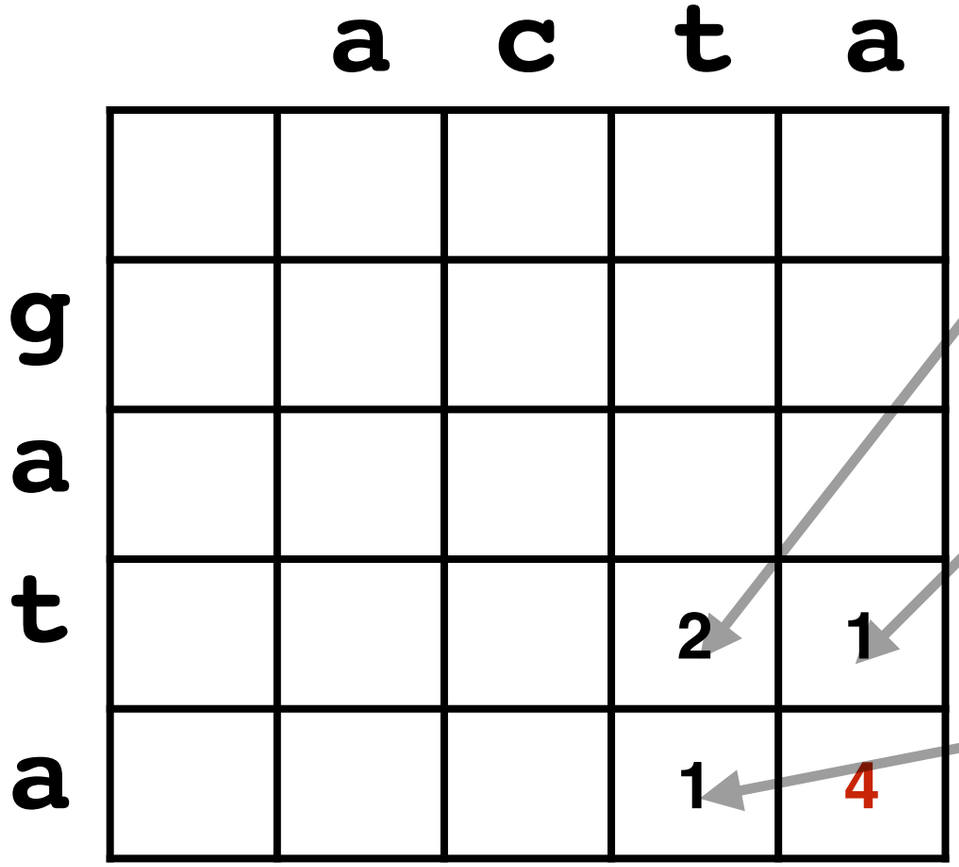
S(n, m-1) = **1**

```
g a - t a
- a c t -
```

\+ -1 (- / a) = 0







S(n-1, m-1) = **2**

```
g a - t
- a c t
```

2

```
a
a
```

= 4

S(n-1, m) = **1**

```
g a - t -
- a c t a
```

-1

```
a
-
```

= 0

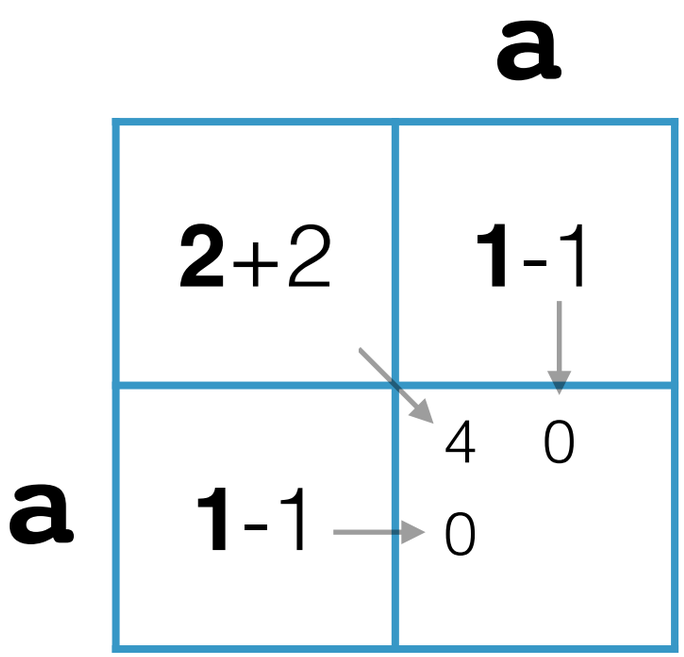
S(n, m-1) = **1**

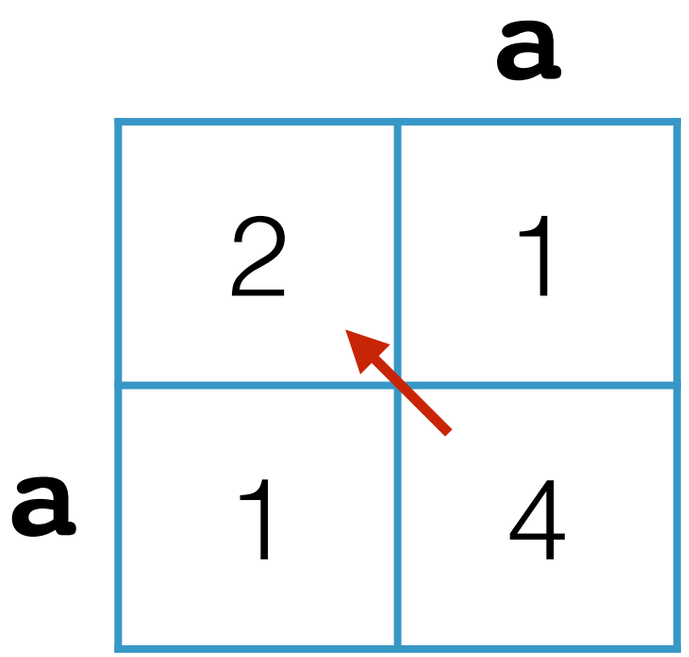
```
g a - t a
- a c t -
```

-1

```
-
a
```

= 0

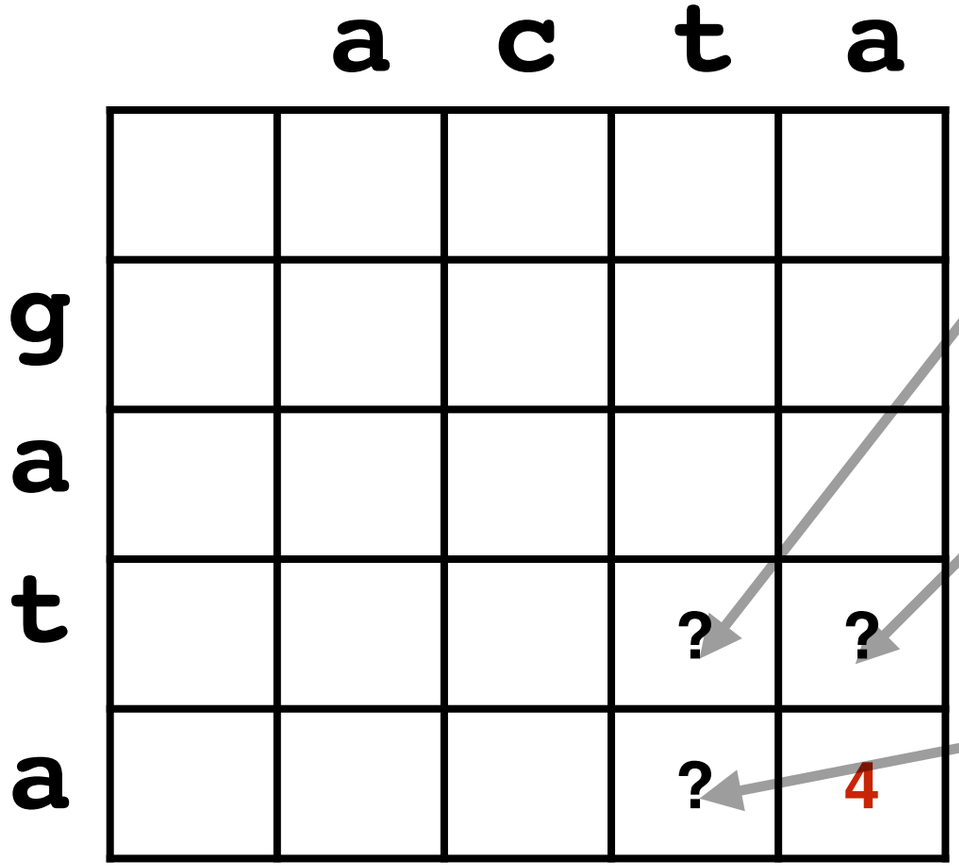




Last column
Would be:

```
a
a
```

…because that *column* along with an optimal alignment of the *rest* produces an optimal global alignment.



S(n-1, m-1) = ?

g a t a c t

2

a

\+ **a**

= ?

S(n-1, m) = ?

g a t a c t a

-1

a

\+ **–**

= ?

S(n, m-1) = ?

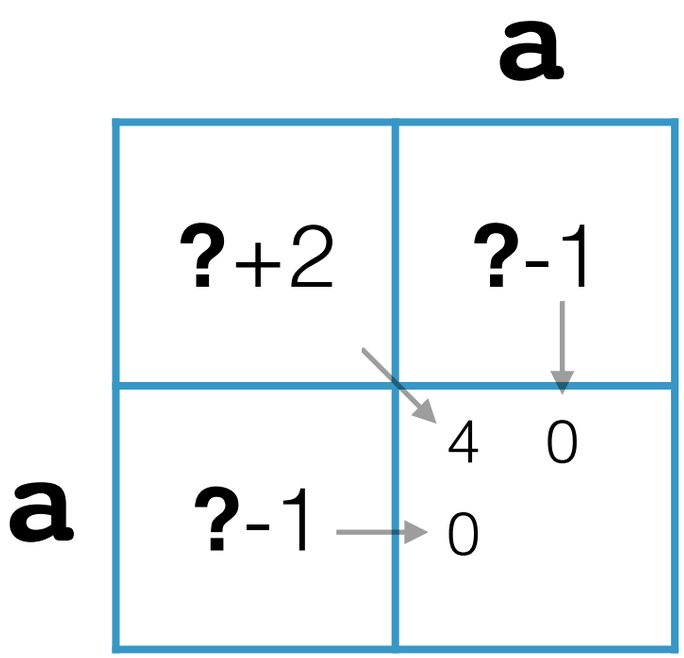
g a t a a c t

-1

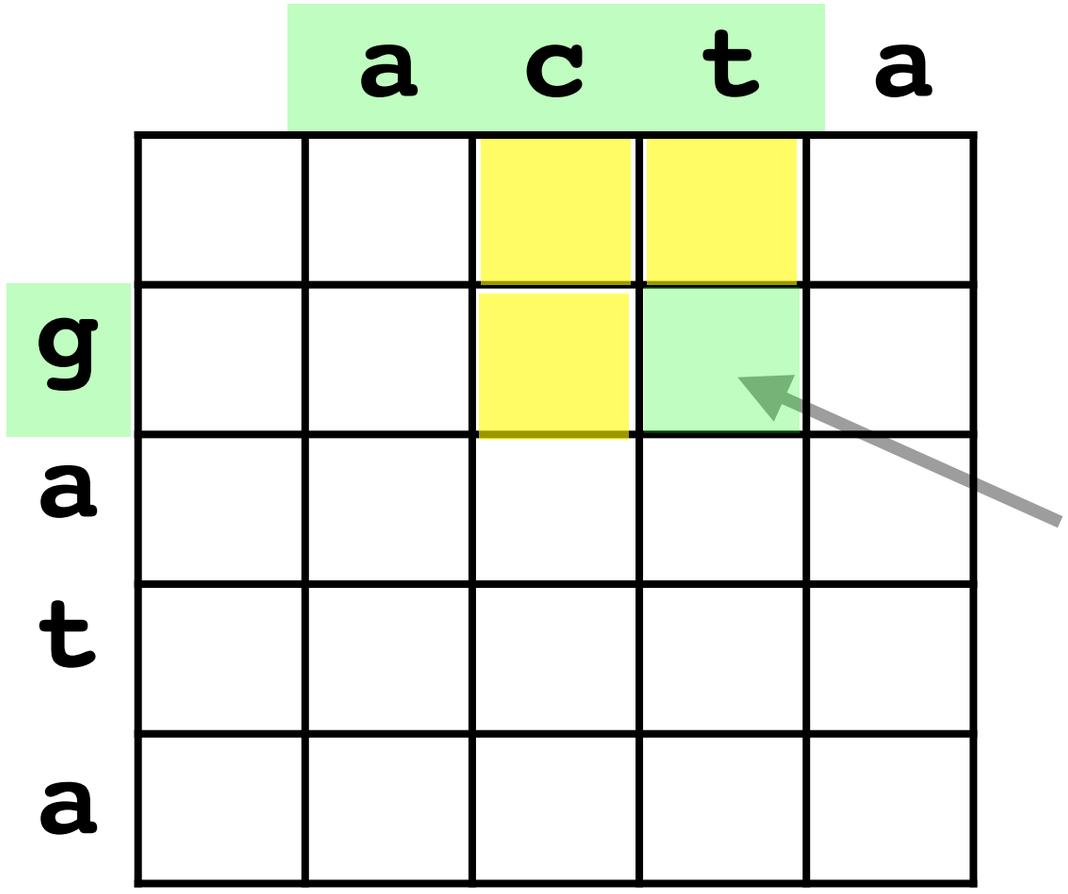
–

\+ **a**

= ?



But this is cheating because we don’t know the three intermediate results we need…



All the cells can be computed the same way as the last one as long we have the three yellow intermediate results.

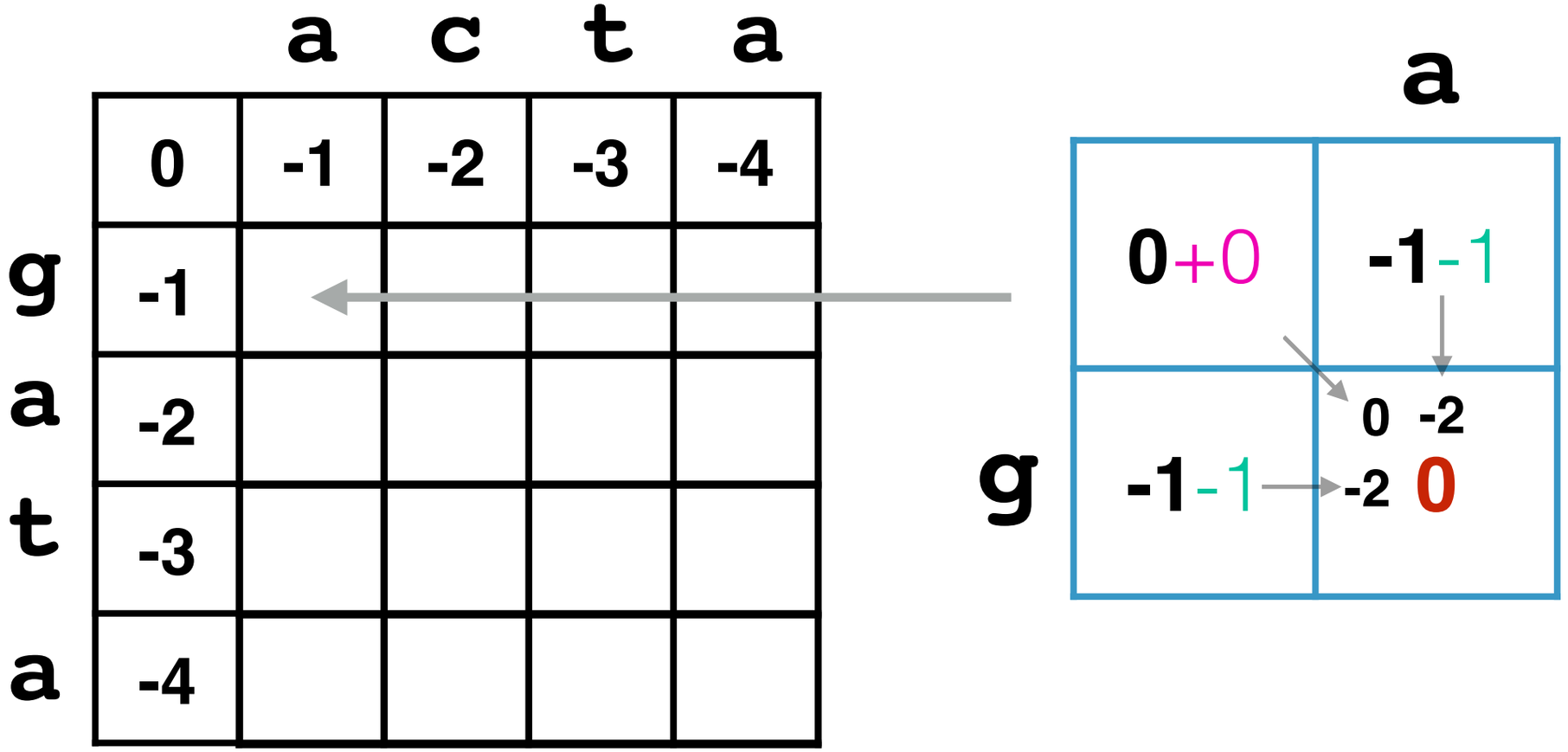
S(1,3)

Here the green cell has the score of the optimal alignment of the two seqeunces painted green.

		a	c	t	a
	0	-1	-2	-3	-4
g	-1				
a	-2				
t	-3				
a	-4				

We can exploit that the first column and row can be computed knowing only the gap score (here -1).

		a	c	t	a
	0	-1	-2	-3	-4
g	-1				
a	-2				
t	-3				
a	-4				



score matrix:

	a	t	g	c
a	2	0	0	0
t	0	2	0	0
g	0	0	2	0
c	0	0	0	2

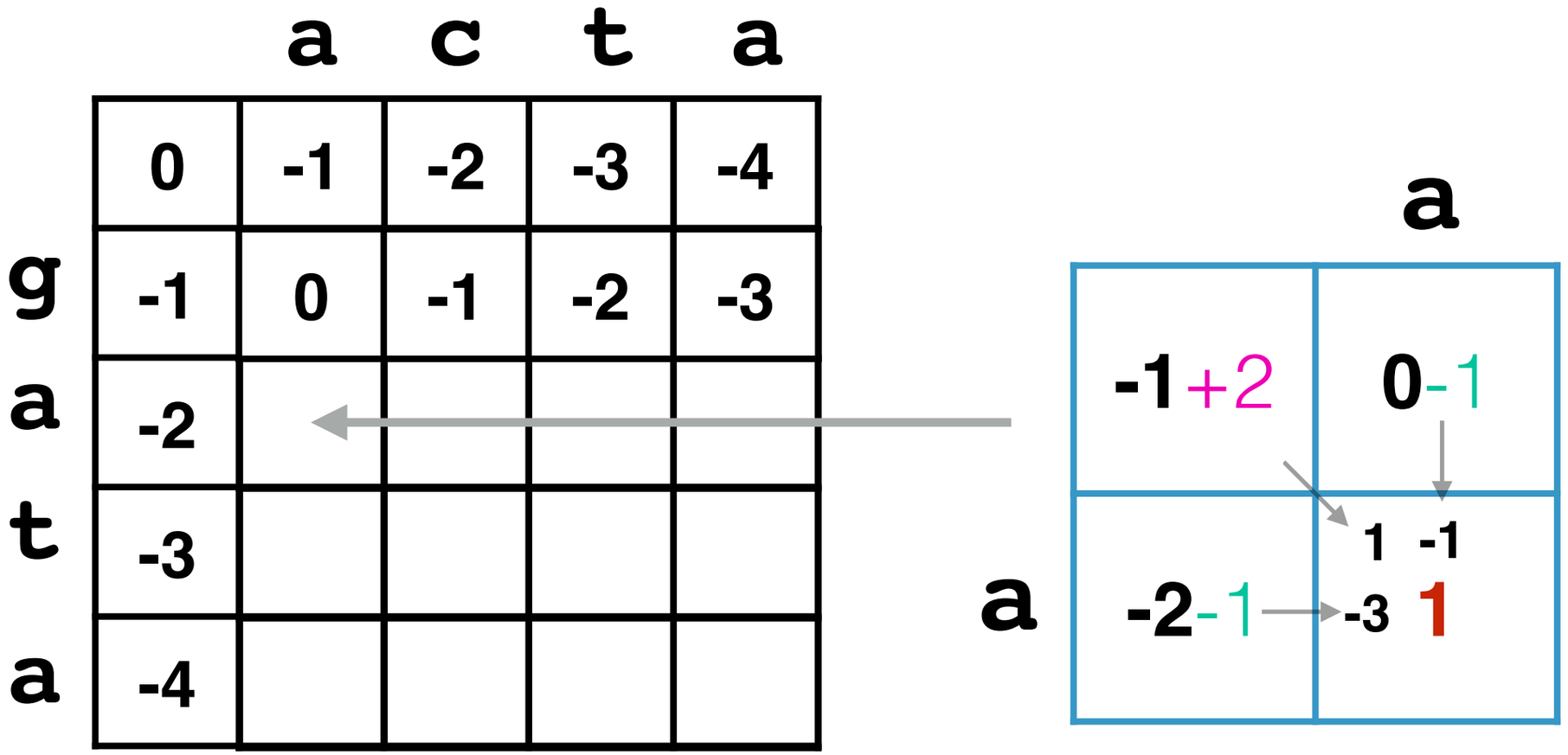
gap cost: -1

		a	c	t	a
	0	-1	-2	-3	-4
g	-1	0			
a	-2				
t	-3				
a	-4				

		a	c	t	a
	0	-1	-2	-3	-4
g	-1	0	-1		
a	-2				
t	-3				
a	-4				

		a	c	t	a
	0	-1	-2	-3	-4
g	-1	0	-1	-2	
a	-2				
t	-3				
a	-4				

		a	c	t	a
	0	-1	-2	-3	-4
g	-1	0	-1	-2	-3
a	-2				
t	-3				
a	-4				



score matrix:

	a	t	g	c
a	2	0	0	0
t	0	2	0	0
g	0	0	2	0
c	0	0	0	2

gap cost: -1

		a	c	t	a
	0	-1	-2	-3	-4
g	-1	0	-1	-2	-3
a	-2	1			
t	-3				
a	-4				

		a	c	t	a
	0	-1	-2	-3	-4
g	-1	0	-1	-2	-3
a	-2	1	0		
t	-3				
a	-4				

		a	c	t	a
	0	-1	-2	-3	-4
g	-1	0	-1	-2	-3
a	-2	1	0	-1	
t	-3				
a	-4				

		a	c	t	a
	0	-1	-2	-3	-4
g	-1	0	-1	-2	-3
a	-2	1	0	-1	0
t	-3				
a	-4				

		a	c	t	a
	0	-1	-2	-3	-4
g	-1	0	-1	-2	-3
a	-2	1	0	-1	0
t	-3	0			
a	-4				

		a	c	t	a
	0	-1	-2	-3	-4
g	-1	0	-1	-2	-3
a	-2	1	0	-1	0
t	-3	0	1		
a	-4				

		a	c	t	a
	0	-1	-2	-3	-4
g	-1	0	-1	-2	-3
a	-2	1	0	-1	0
t	-3	0	1	2	
a	-4				

		a	c	t	a
	0	-1	-2	-3	-4
g	-1	0	-1	-2	-3
a	-2	1	0	-1	0
t	-3	0	1	2	1
a	-4				

		a	c	t	a
	0	-1	-2	-3	-4
g	-1	0	-1	-2	-3
a	-2	1	0	-1	0
t	-3	0	1	2	1
a	-4	-1			

		a	c	t	a
	0	-1	-2	-3	-4
g	-1	0	-1	-2	-3
a	-2	1	0	-1	0
t	-3	0	1	2	1
a	-4	-1	0		

		a	c	t	a
	0	-1	-2	-3	-4
g	-1	0	-1	-2	-3
a	-2	1	0	-1	0
t	-3	0	1	2	1
a	-4	-1	0	1	

		a	c	t	a
	0	-1	-2	-3	-4
g	-1	0	-1	-2	-3
a	-2	1	0	-1	0
t	-3	0	1	2	1
a	-4	-1	0	1	4

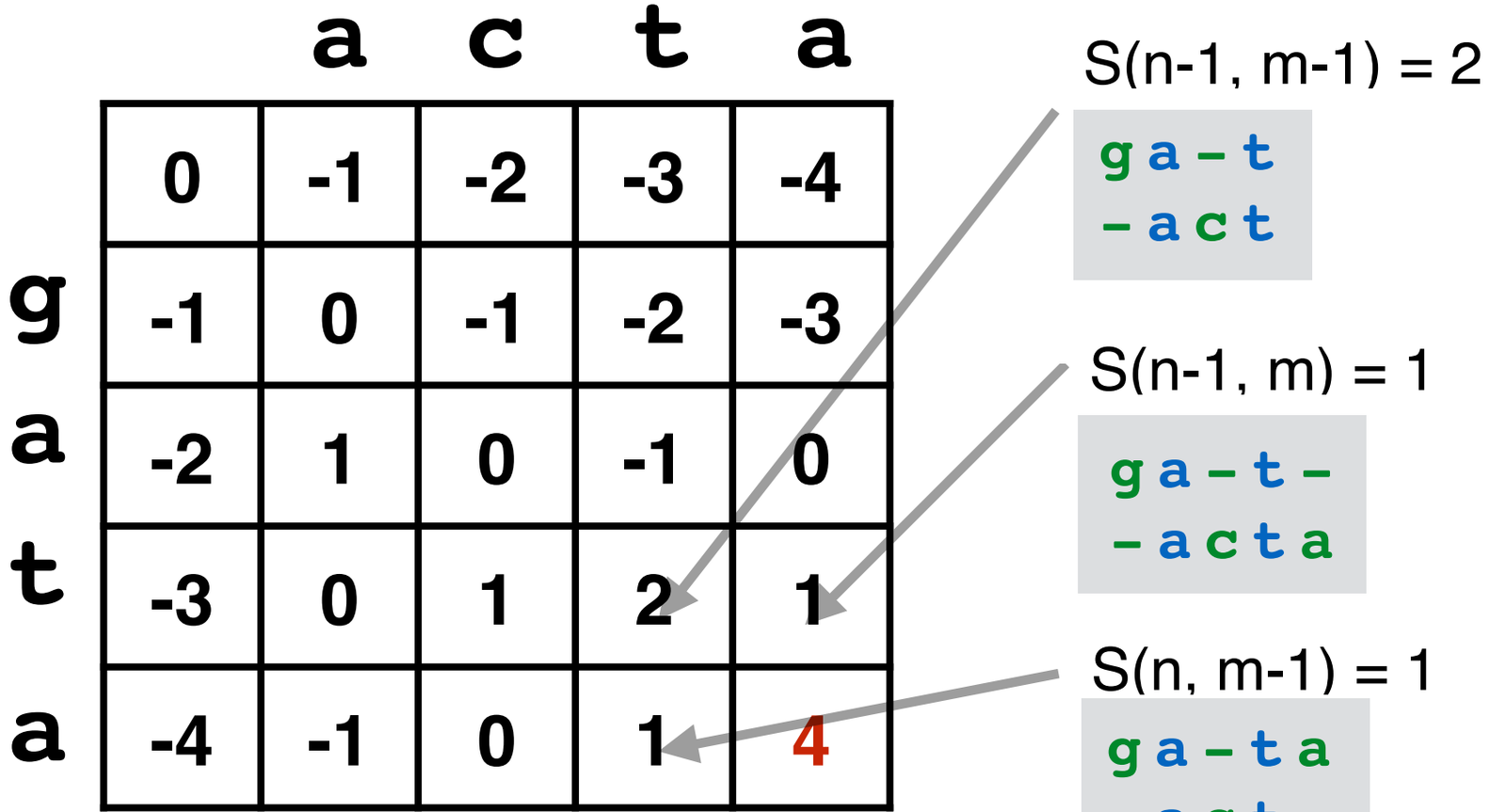
		a	c	t	a
	0	-1	-2	-3	-4
g	-1	0	-1	-2	-3
a	-2	1	0	-1	0
t	-3	0	1	2	1
a	-4	-1	0	1	4

Optimal alignment score is S(4, 4) = 4

Traceback

		a	c	t	a
	0	-1	-2	-3	-4
g	-1	0	-1	-2	-3
a	-2	1	0	-1	0
t	-3	0	1	2	1
a	-4	-1	0	1	4

Traceback


a c t a
0 -1 -2 -3 -4
g -1 0 -1 -2 -3
a -2 1 0 -1 0
t -3 0 1 2 1
a -4 -1 0 1 4
S(n-1, m-1) = 2
g a - t
- a c t
S(n-1, m) = 1
g a - t -
- a c t a
S(n, m-1) = 1
g a - t a
- a c t -

Traceback

		a	c	t	a
	0	-1	-2	-3	-4
g	-1	0	-1	-2	-3
a	-2	1	0	-1	0
t	-3	0	1	2	1
a	-4	-1	0	1	4

S(n-1, m-1) = 2

```
g a - t
- a c t
```

2 + a a = 4

S(n-1, m) = 1

```
g a - t -
- a c t a
```

-1 + a - = 0

S(n, m-1) = 1

```
g a - t a
- a c t -
```

-1 + - a = 0

Traceback

		a	c	t	a
	0	-1	-2	-3	-4
g	-1	0	-1	-2	-3
a	-2	1	0	-1	0
t	-3	0	1	2	1
a	-4	-1	0	1	4

S(n-1, m-1) = 2

```
g a - t
- a c t
```

2 + a a = 4

S(n-1, m) = 1

```
g a - t -
- a c t a
```

-1 + a - = 0

S(n, m-1) = 1

```
g a - t a
- a c t -
```

-1 + - a = 0

		a
	2+2	**1**-1
a	**1**-1	

Traceback

		a	c	t	a
	0	-1	-2	-3	-4
g	-1	0	-1	-2	-3
a	-2	1	0	-1	0
t	-3	0	1	2	1
a	-4	-1	0	1	4

S(n-1, m-1) = 2

```
g a - t
- a c t
```

2 + a a = 4

S(n-1, m) = 1

```
g a - t -
- a c t a
```

-1 + a - = 0

S(n, m-1) = 1

```
g a - t a
- a c t -
```

-1 + - a = 0

	a	
2+2	**1**-1	
a **1**-1	4 0 0	

Traceback

		a	c	t	a
	0	-1	-2	-3	-4
g	-1	0	-1	-2	-3
a	-2	1	0	-1	0
t	-3	0	1	2	1
a	-4	-1	0	1	4

S(n-1, m-1) = 2

```
g a - t
- a c t
```

2 + a a = 4

S(n-1, m) = 1

```
g a - t -
- a c t a
```

-1 + a - = 0

S(n, m-1) = 1

```
g a - t a
- a c t -
```

-1 + - a = 0

		a
a	2+2	1-1
	1-1	0

4 0

		a
a	2	1
	1	4

Traceback

		a	c	t	a
	0	-1	-2	-3	-4
g	-1	0	-1	-2	-3
a	-2	1	0	-1	0
t	-3	0	1	2	1
a	-4	-1	0	1	4

S(n-1, m-1) = 2

```
g a - t
- a c t
```

S(n-1, m) = 1

```
g a - t -
- a c t a
```

S(n, m-1) = 1

```
g a - t a
- a c t -
```

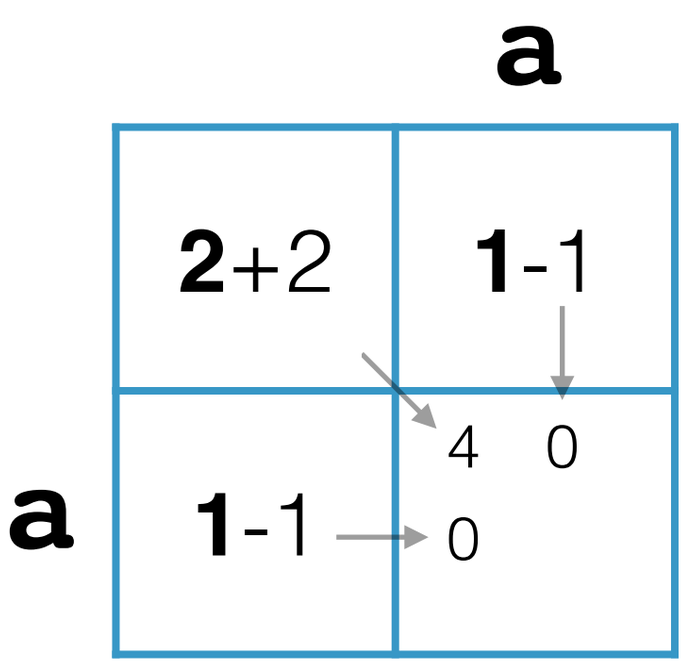
The last column is:

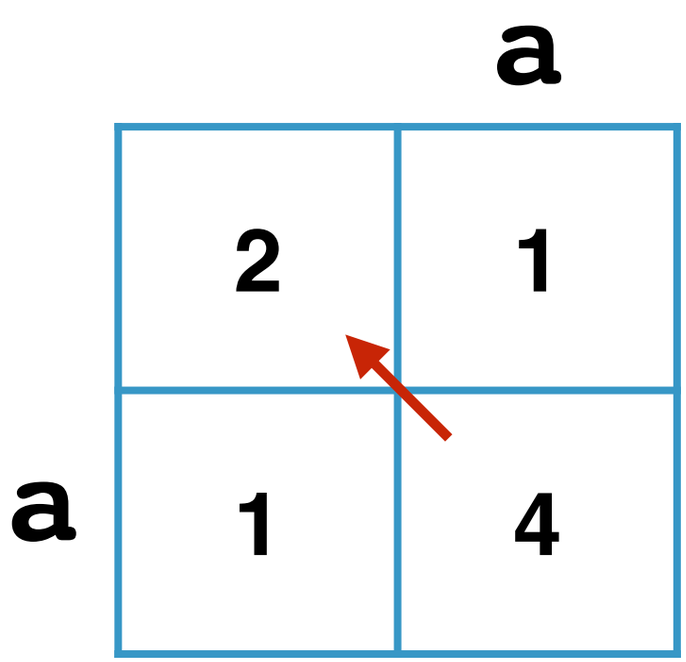
```
  2
+ a
  a
  = 4

 -1
+ a
  -
  = 0

 -1
+ -
  a
  = 0
```

```
a
a
```





Traceback

		a	c	t	a
	0	-1	-2	-3	-4
g	-1	0	-1	-2	-3
a	-2	1	0	-1	0
t	-3	0	1	2	1
a	-4	-1	0	1	4

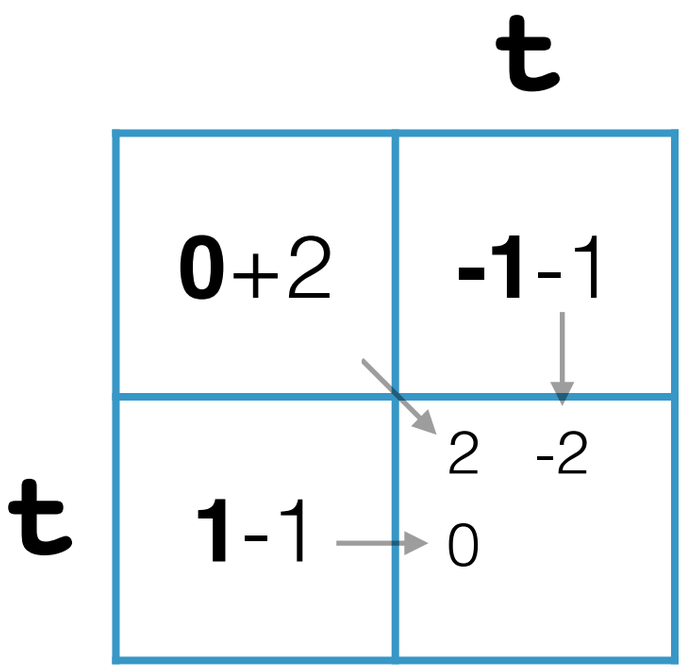


a

a

Traceback

		a	c	t	a
	0	-1	-2	-3	-4
g	-1	0	-1	-2	-3
a	-2	1	0	-1	0
t	-3	0	1	2	1
a	-4	-1	0	1	4



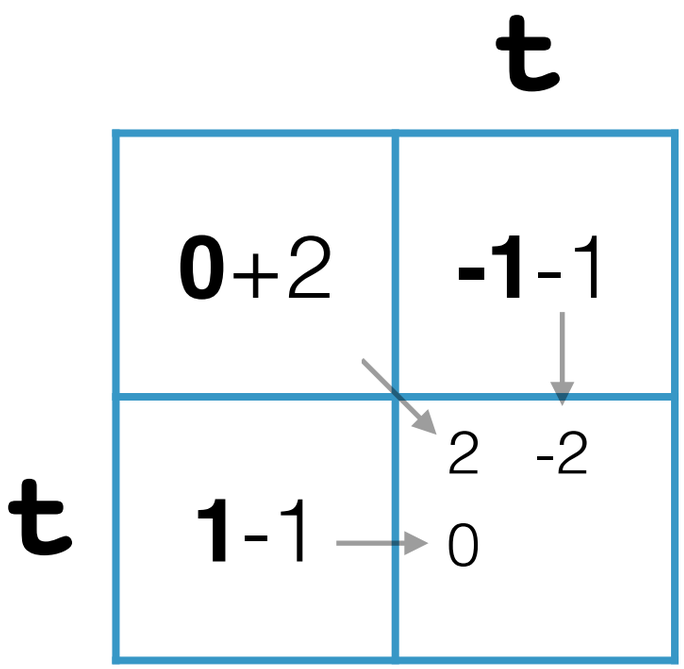
a

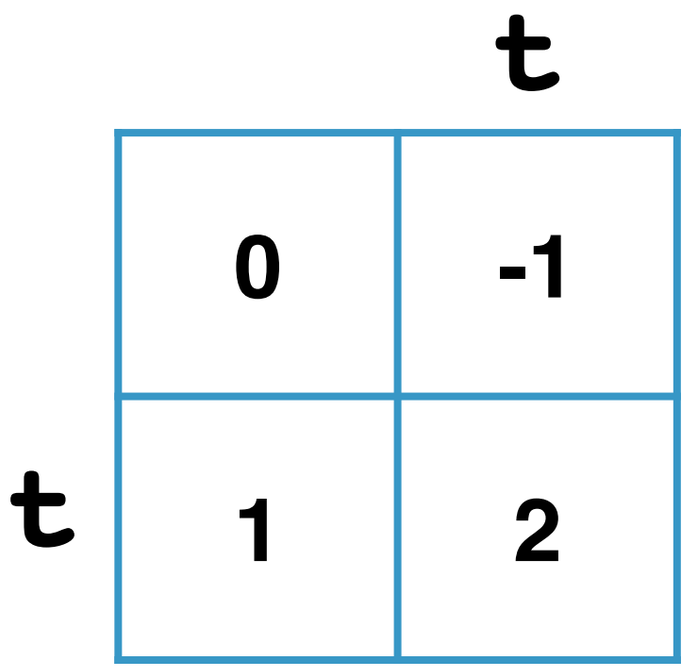
a

Traceback

		a	c	t	a
	0	-1	-2	-3	-4
g	-1	0	-1	-2	-3
a	-2	1	0	-1	0
t	-3	0	1	2	1
a	-4	-1	0	1	4

The optimal alignemnt score for **gat** and **act** is S(3, 3) = 2





a

a

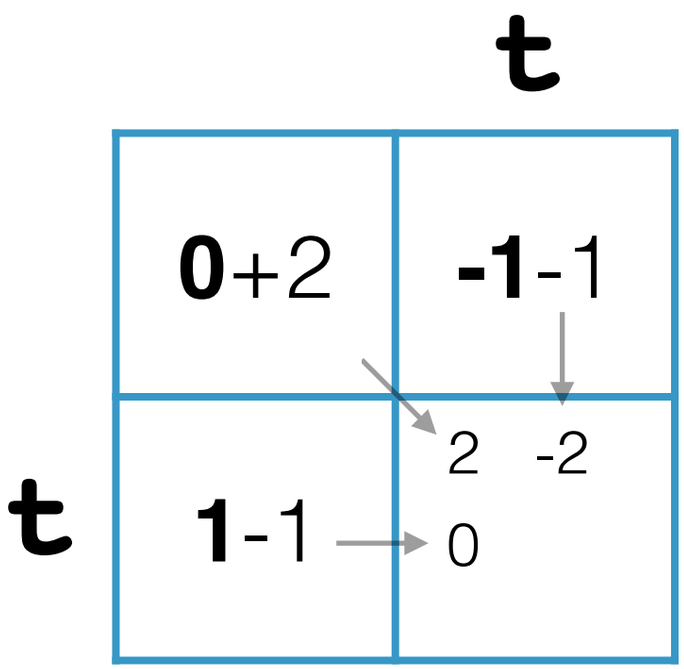
Traceback

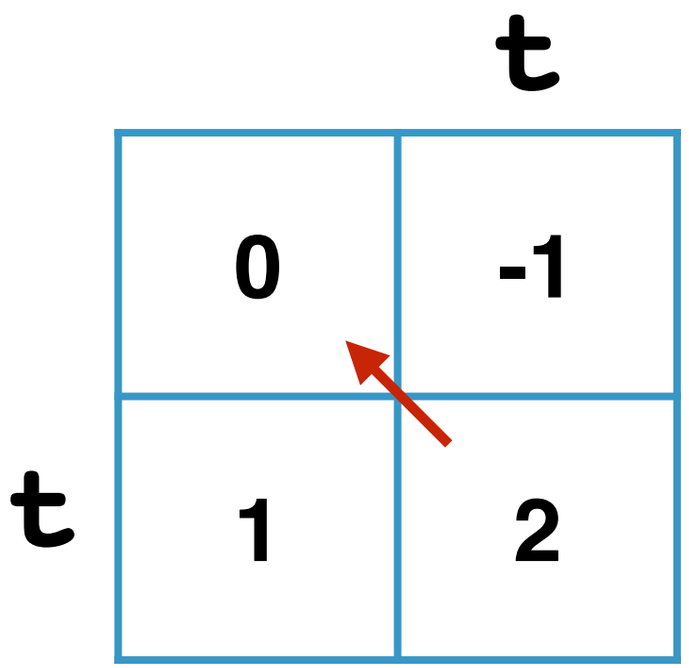
		a	c	t	a
	0	-1	-2	-3	-4
g	-1	0	-1	-2	-3
a	-2	1	0	-1	0
t	-3	0	1	2	1
a	-4	-1	0	1	4

The optimal alignemnt score for **gat** and **act** is S(3, 3) = 2

- and sidste kolonne i det alignment er:

```
t
t
```






```
a
a
```

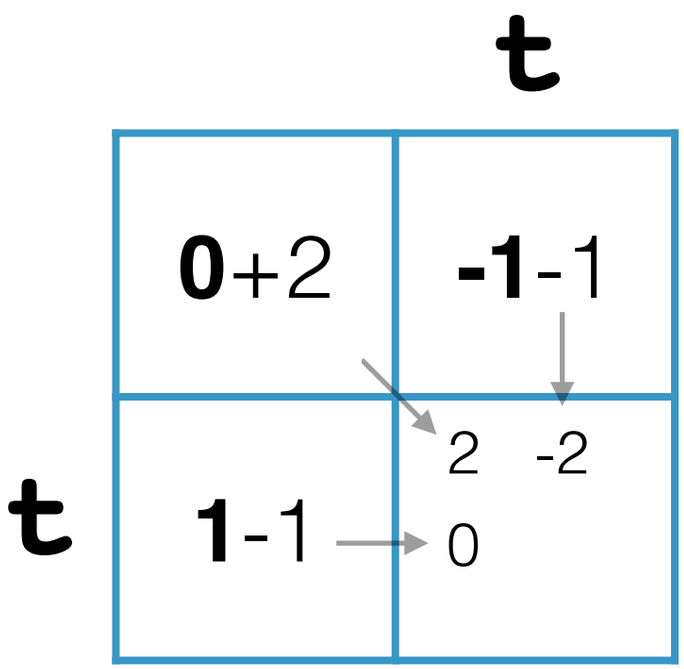
Traceback

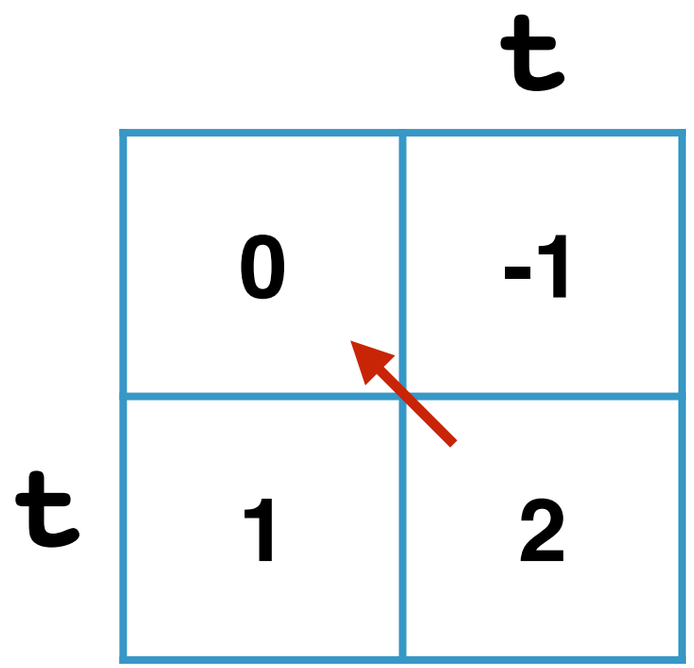
		a	c	t	a
	0	-1	-2	-3	-4
g	-1	0	-1	-2	-3
a	-2	1	0	-1	0
t	-3	0	1	2	1
a	-4	-1	0	1	4

The optimal alignemnt score for **gat** and **act** is S(3, 3) = 2

- and sidste kolonne i det alignment er:

```
t
t
```

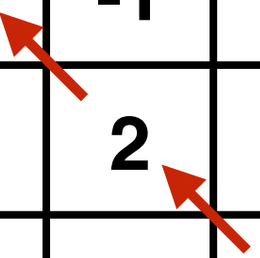





```
ta
ta
```

Traceback

		a	c	t	a
	0	-1	-2	-3	-4
g	-1	0	-1	-2	-3
a	-2	1	0	-1	0
t	-3	0	1	2	1
a	-4	-1	0	1	4



ta

ta

Traceback

		a	c	t	a
	0	-1	-2	-3	-4
g	-1	0	-1	-2	-3
a	-2	1	0	-1	0
t	-3	0	1	2	1
a	-4	-1	0	1	4

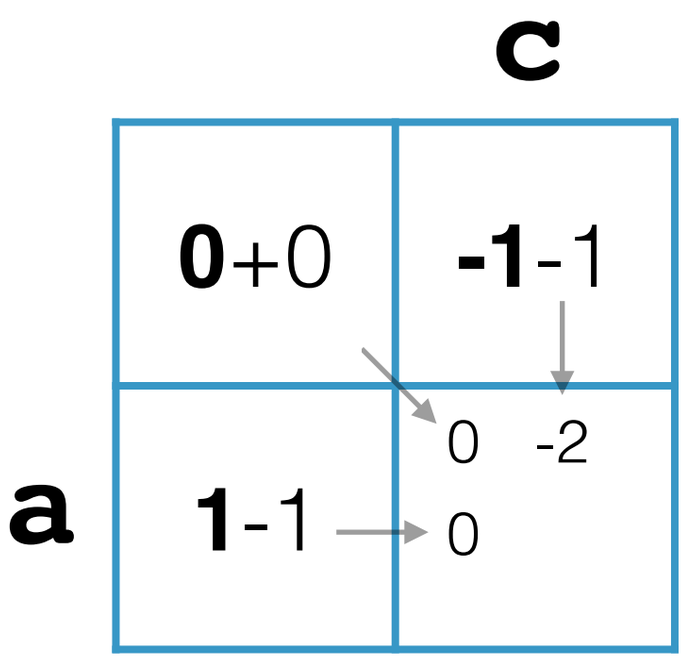
		c
	0+0	**-1**-1
a	**1**-1	

ta

ta

Traceback

		a	c	t	a
	0	-1	-2	-3	-4
g	-1	0	-1	-2	-3
a	-2	1	0	-1	0
t	-3	0	1	2	1
a	-4	-1	0	1	4



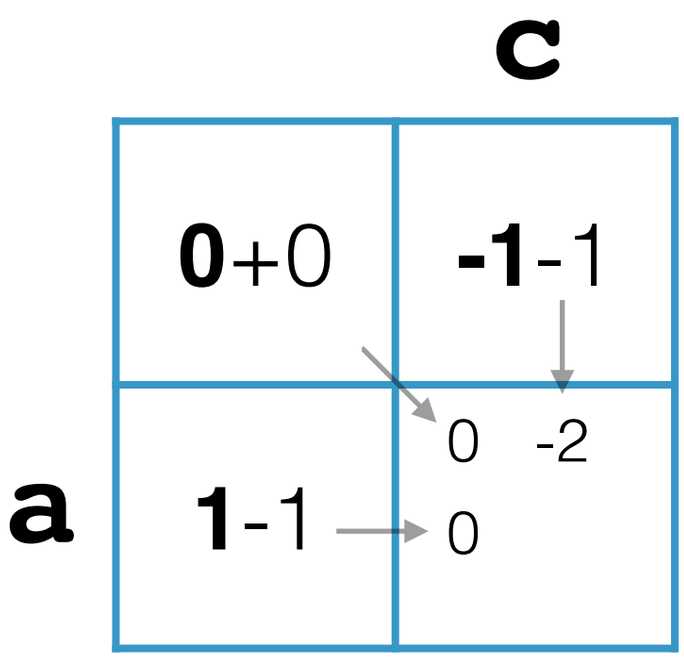
ta

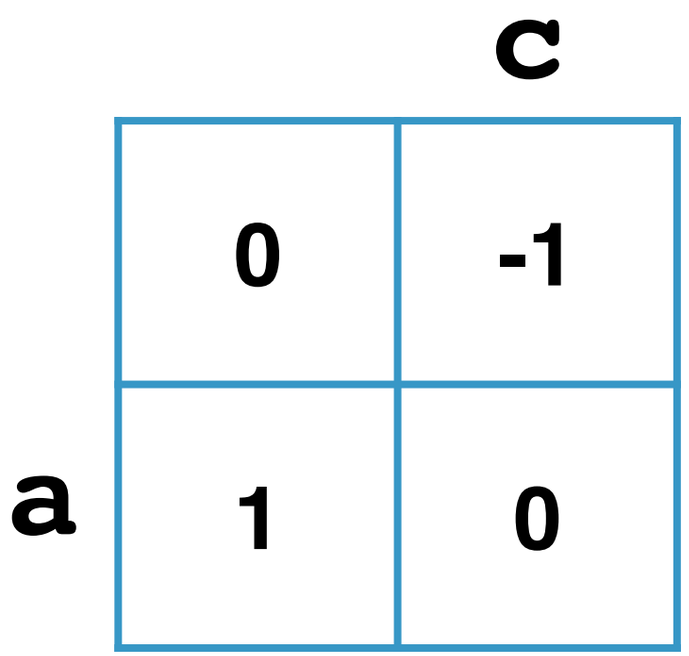
ta

Traceback

		a	c	t	a
	0	-1	-2	-3	-4
g	-1	0	-1	-2	-3
a	-2	1	0	-1	0
t	-3	0	1	2	1
a	-4	-1	0	1	4

The optimal alignemnt score for **ga** and **ac** is S(2, 2) = 0





ta

ta

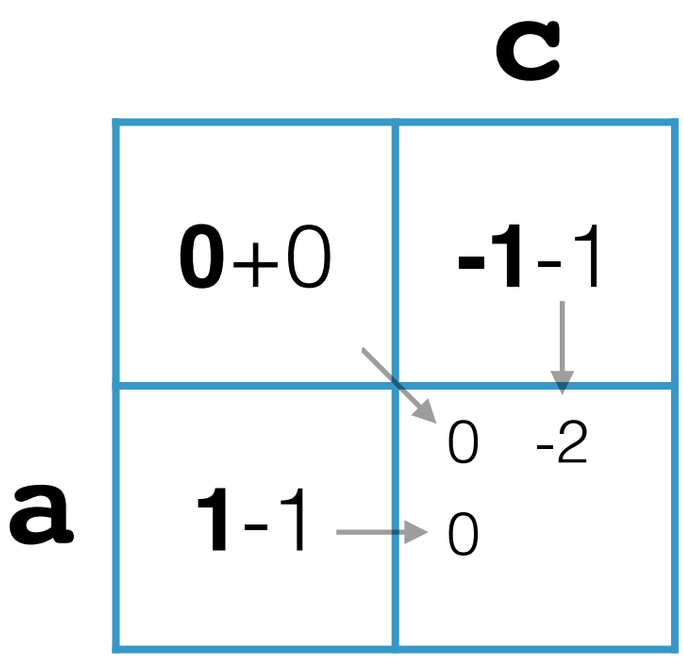
Traceback

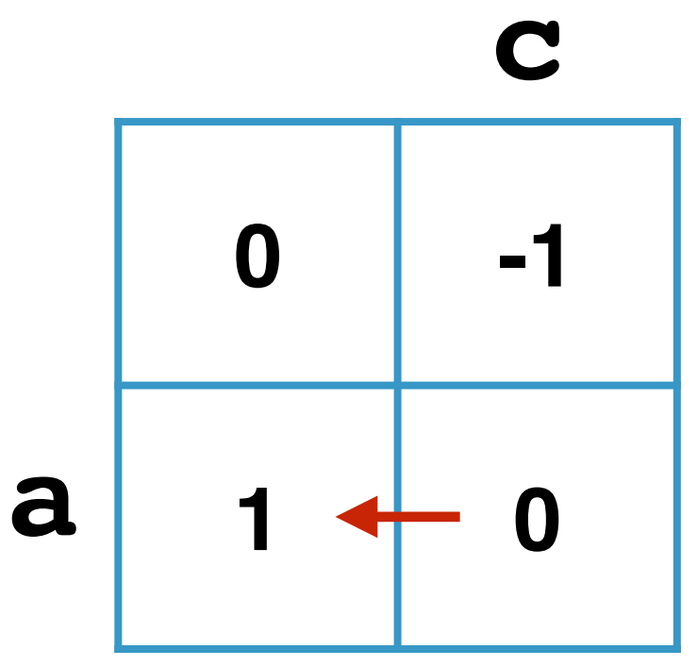
		a	c	t	a
	0	-1	-2	-3	-4
g	-1	0	-1	-2	-3
a	-2	1	0	-1	0
t	-3	0	1	2	1
a	-4	-1	0	1	4

The optimal alignemnt score for **ga** and **ac** is S(2, 2) = 0

- and the last column in that alignment is:

```
-
c
```






```
ta
ta
```

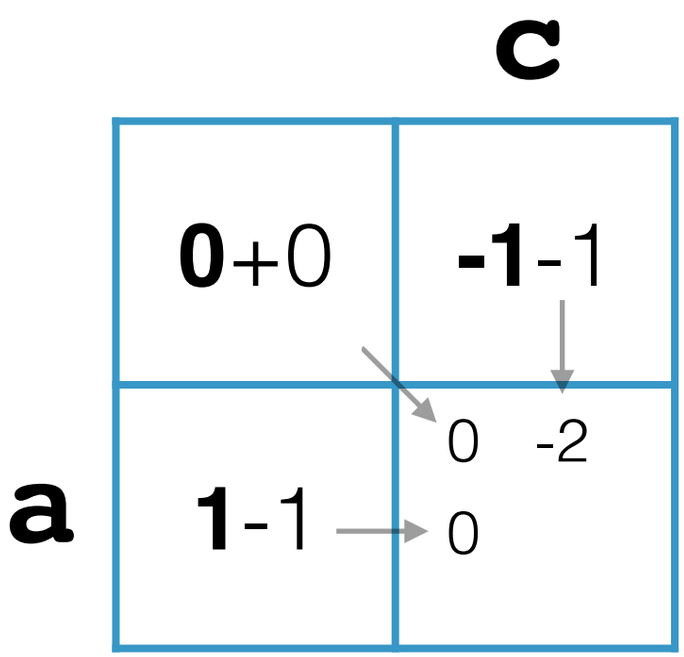
Traceback

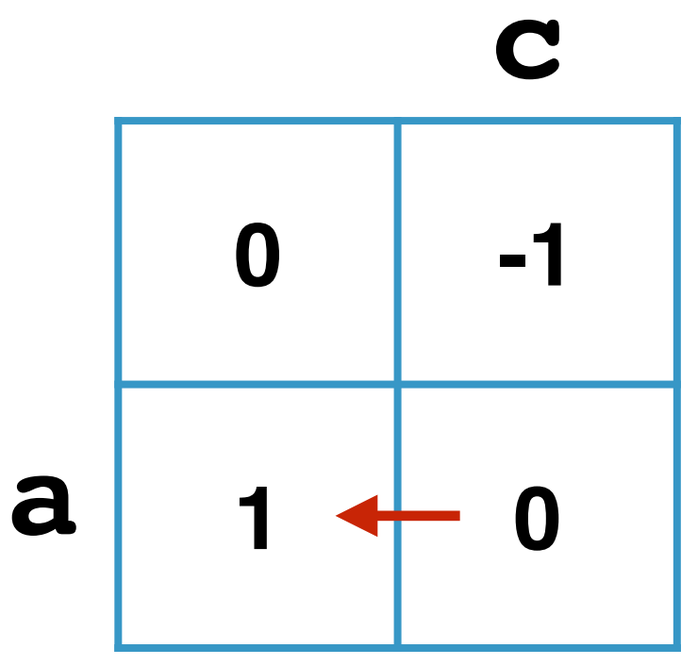
		a	c	t	a
	0	-1	-2	-3	-4
g	-1	0	-1	-2	-3
a	-2	1	0	-1	0
t	-3	0	1	2	1
a	-4	-1	0	1	4

The optimal alignemnt score for **ga** and **ac** is S(2, 2) = 0

- and the last column in that alignment is:

```
-
c
```






```
-ta
cta
```

Traceback

		a	c	t	a
	0	-1	-2	-3	-4
g	-1	0	-1	-2	-3
a	-2	1	0	-1	0
t	-3	0	1	2	1
a	-4	-1	0	1	4

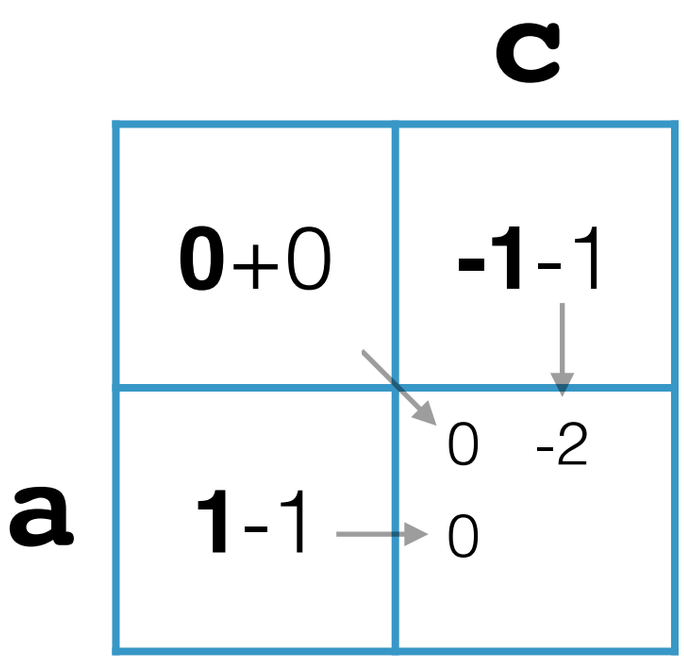
The optimal alignemnt score for **ga** and **ac** is S(2, 2) = 0

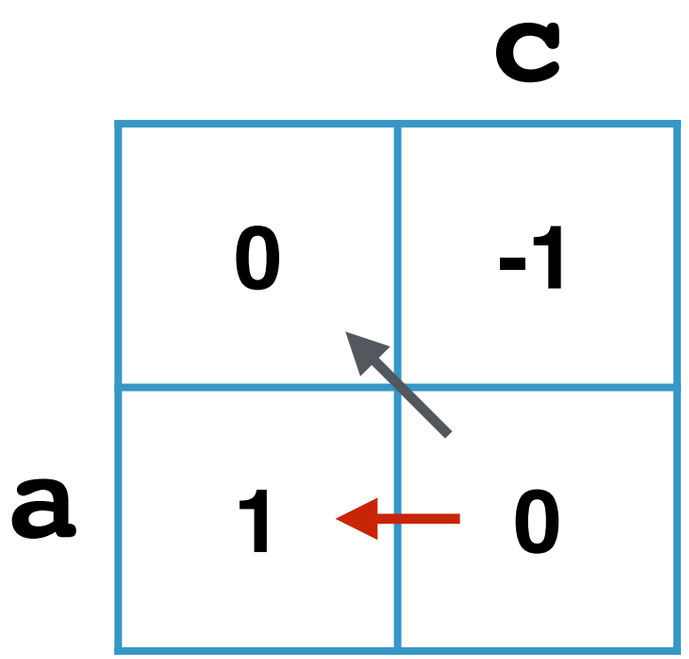
- and the last column in that alignment is:

```
-
c
```

- or alternatively:

```
a
c
```






```
-ta
cta
```

Traceback

		a	c	t	a
	0	-1	-2	-3	-4
g	-1	0	-1	-2	-3
a	-2	1	0	-1	0
t	-3	0	1	2	1
a	-4	-1	0	1	4

-ta

cta

Traceback

		a	c	t	a
	0	-1	-2	-3	-4
g	-1	0	-1	-2	-3
a	-2	1	0	-1	0
t	-3	0	1	2	1
a	-4	-1	0	1	4

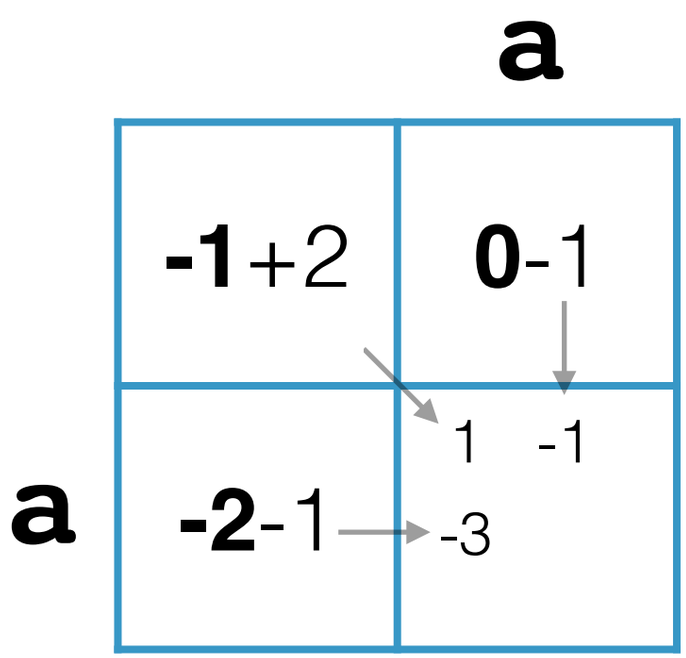
		a
	-1+2	**0**-1
a	**-2**-1	

-ta

cta

Traceback

		a	c	t	a
	0	-1	-2	-3	-4
g	-1	0	-1	-2	-3
a	-2	1	0	-1	0
t	-3	0	1	2	1
a	-4	-1	0	1	4



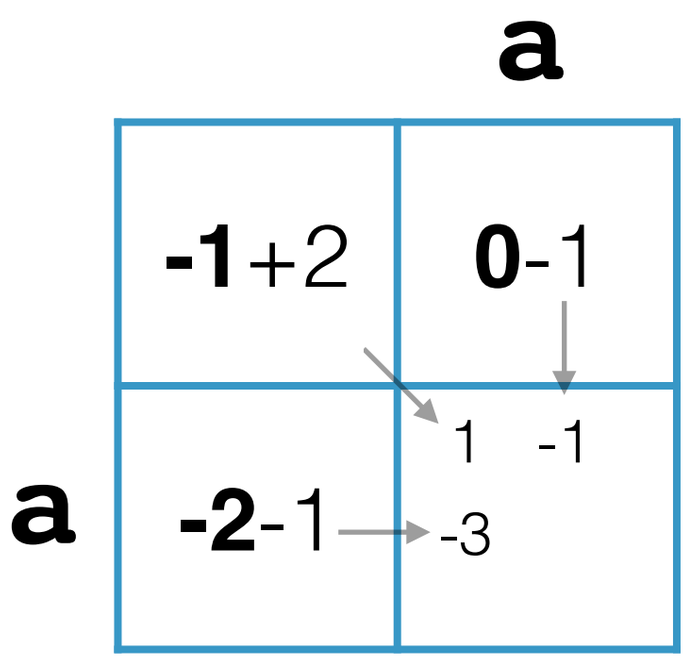
-ta

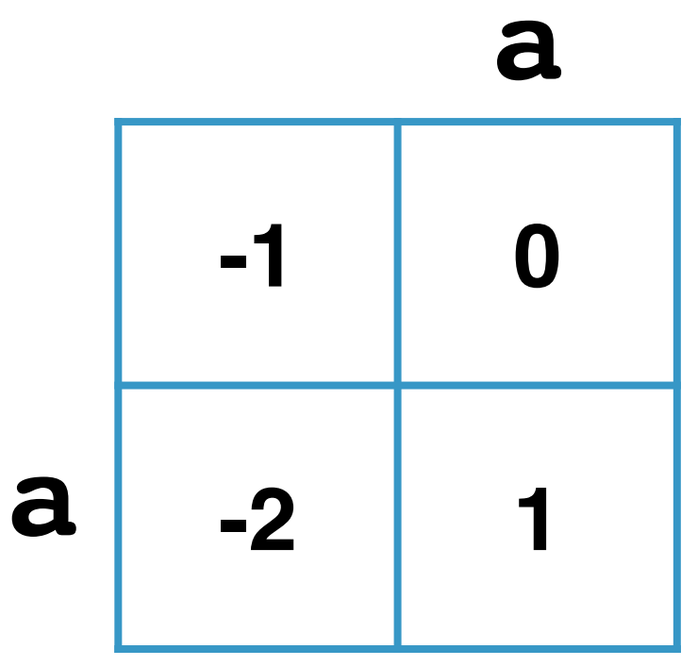
cta

Traceback

		a	c	t	a
	0	-1	-2	-3	-4
g	-1	0	-1	-2	-3
a	-2	1	0	-1	0
t	-3	0	1	2	1
a	-4	-1	0	1	4

The optimal alignemnt score for **ga** and **a** is S(2, 1) = 1





-ta

cta

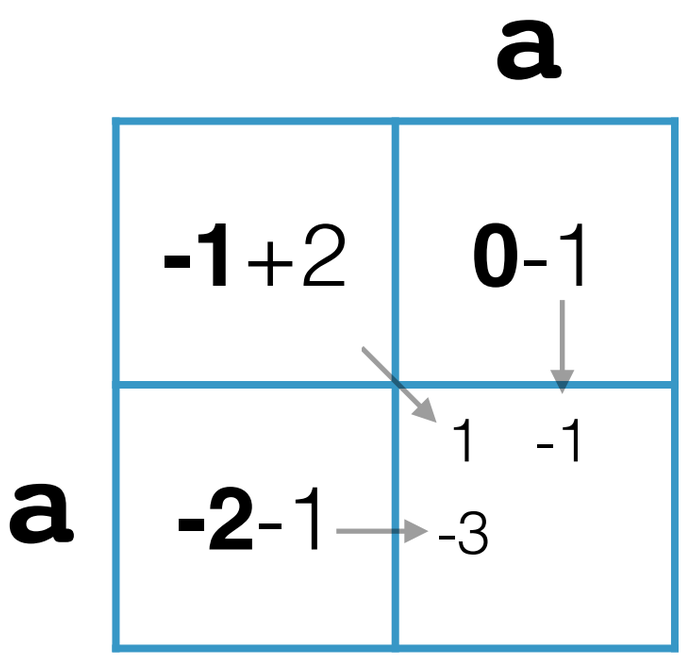
Traceback

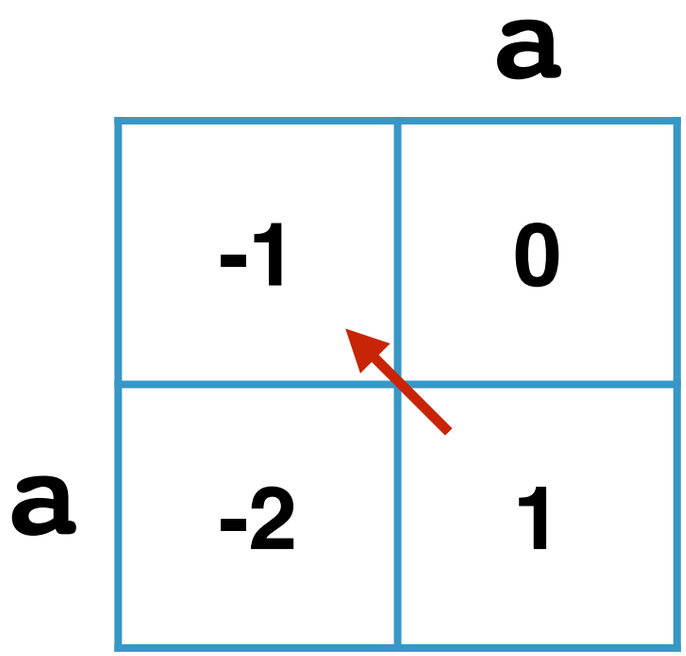
		a	c	t	a
	0	-1	-2	-3	-4
g	-1	0	-1	-2	-3
a	-2	1	0	-1	0
t	-3	0	1	2	1
a	-4	-1	0	1	4

The optimal alignemnt score for **ga** and **a** is S(2, 1) = 1

- and the last column in that alignment is:

a
a





-ta
cta

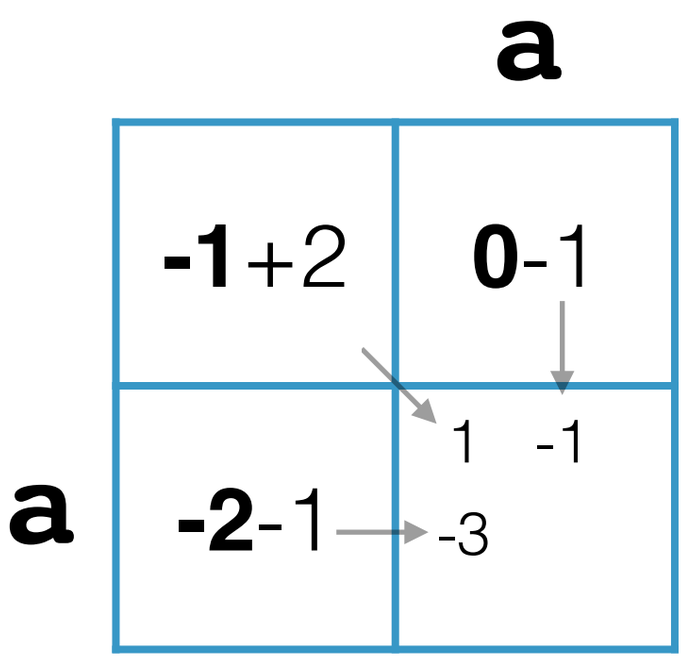
Traceback

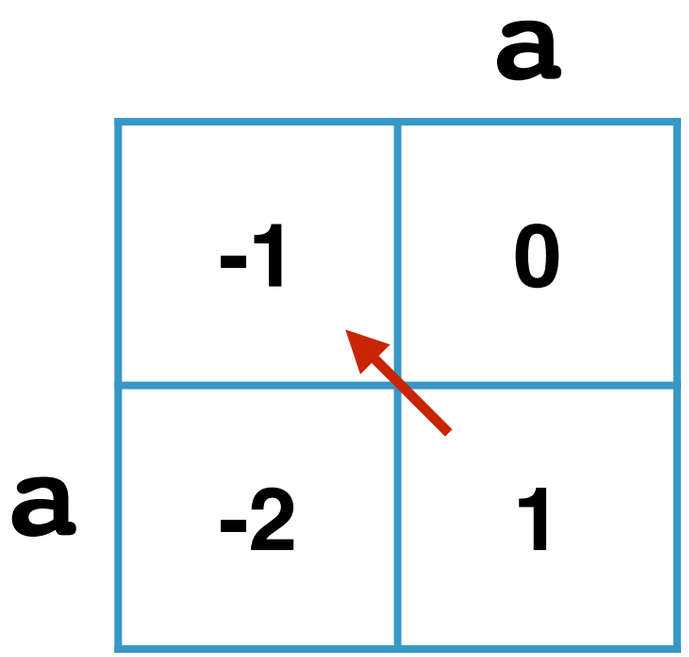
		a	c	t	a
	0	-1	-2	-3	-4
g	-1	0	-1	-2	-3
a	-2	1	0	-1	0
t	-3	0	1	2	1
a	-4	-1	0	1	4

The optimal alignemnt score for **ga** and **a** is S(2, 1) = 1

- and the last column in that alignment is:

a
a

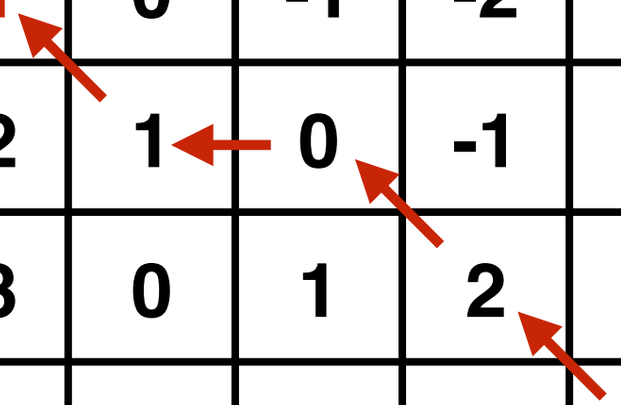





```
a-ta
acta
```

Traceback

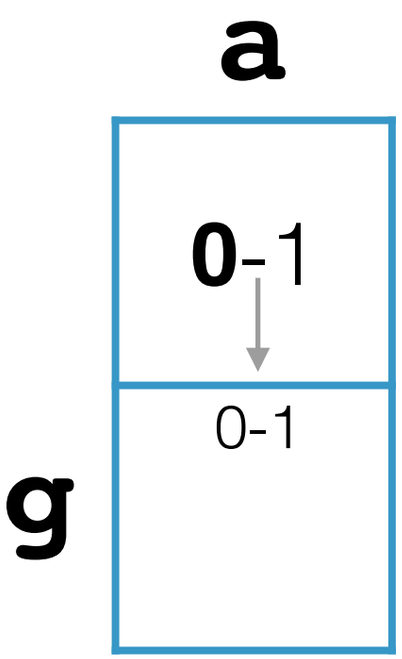
		a	c	t	a
	0	-1	-2	-3	-4
g	-1	0	-1	-2	-3
a	-2	1	0	-1	0
t	-3	0	1	2	1
a	-4	-1	0	1	4



a-ta
acta

Traceback

		a	c	t	a
	0	-1	-2	-3	-4
g	-1	0	-1	-2	-3
a	-2	1	0	-1	0
t	-3	0	1	2	1
a	-4	-1	0	1	4



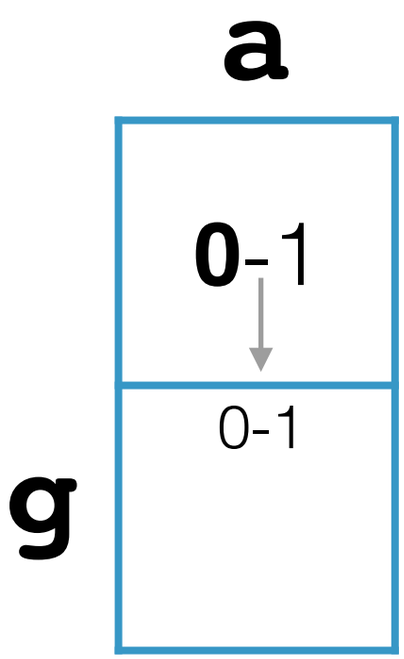
a-ta

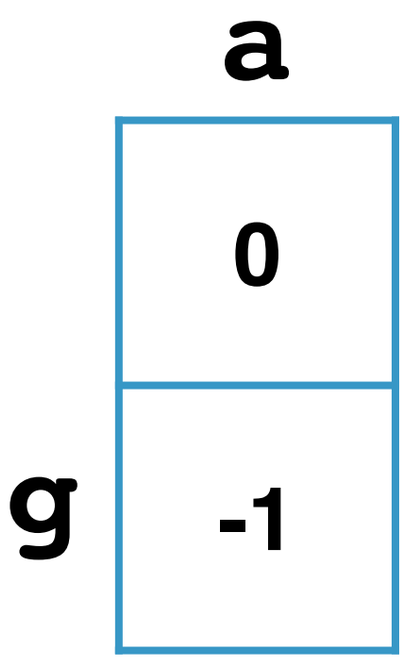
acta

Traceback

		a	c	t	a
	0	-1	-2	-3	-4
g	-1	0	-1	-2	-3
a	-2	1	0	-1	0
t	-3	0	1	2	1
a	-4	-1	0	1	4

The optimal alignemnt score for **g** and nothing er S(1, 0) = -1






```
a-ta
acta
```

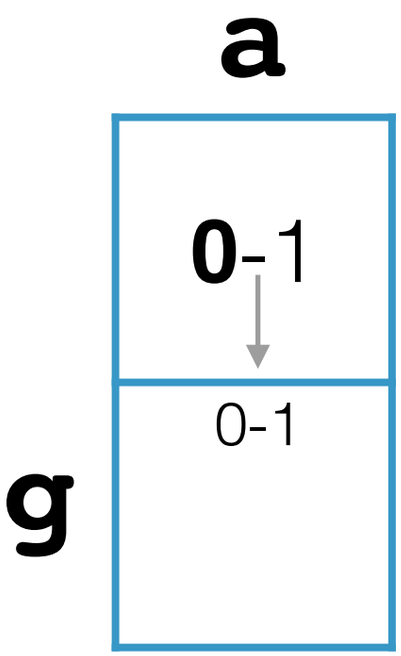
Traceback

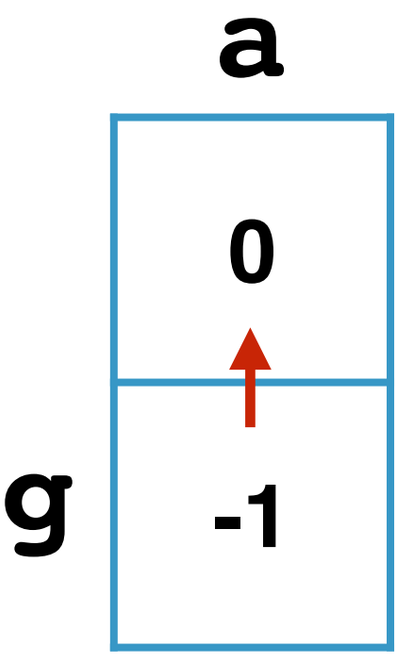
		a	c	t	a
	0	-1	-2	-3	-4
g	-1	0	-1	-2	-3
a	-2	1	0	-1	0
t	-3	0	1	2	1
a	-4	-1	0	1	4

The optimal alignemnt score for **g** and nothing er S(1, 0) = -1

- and the last column in that alignment is:

g
–





a-ta
acta

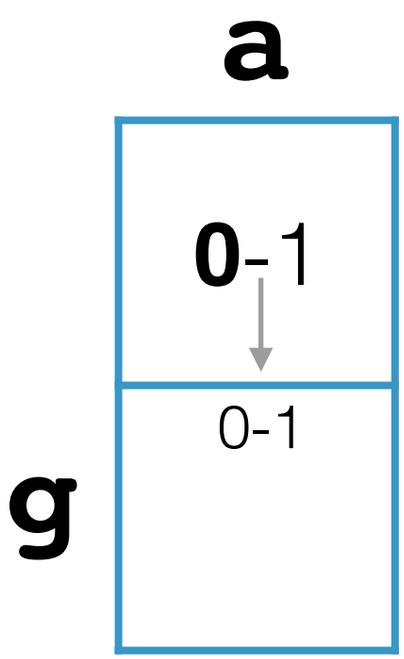
Traceback

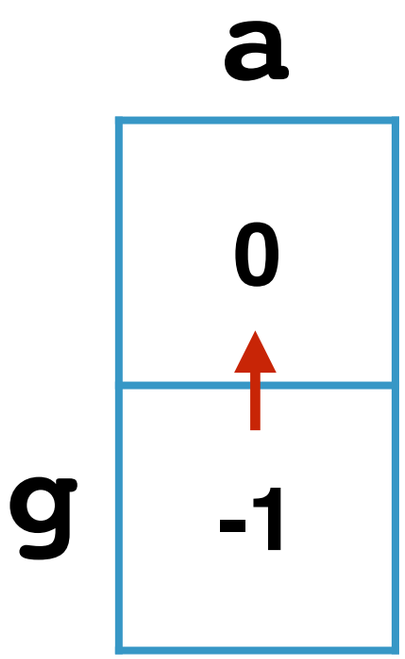
		a	c	t	a
	0	-1	-2	-3	-4
g	-1	0	-1	-2	-3
a	-2	1	0	-1	0
t	-3	0	1	2	1
a	-4	-1	0	1	4

The optimal alignemnt score for **g** and nothing er S(1, 0) = -1

- and the last column in that alignment is:

g
–





ga-ta
-acta

		a	c	t	a
	0	-1	-2	-3	-4
g	-1	0	-1	-2	-3
a	-2	1	0	-1	0
t	-3	0	1	2	1
a	-4	-1	0	1	4

Tadaaa!!

		a	c	t	a
	0	-1	-2	-3	-4
g	-1	0	-1	-2	-3
a	-2	1	0	-1	0
t	-3	0	1	2	1
a	-4	-1	0	1	4

Tadaaa!!

```
ga-ta
-acta
```

		a	c	t	a
	0	-1	-2	-3	-4
g	-1	0	-1	-2	-3
a	-2	1	0	-1	0
t	-3	0	1	2	1
a	-4	-1	0	1	4

Two optimal alignments:

```
gata
acta

ga-ta
-acta
```

Make a global alignemnt

`GGGCT` og `GAGGG`

```
score matrix:
   a  t  g  c
a  2 -1 -1 -1
t -1  2 -1 -1
g -1 -1  2 -1
c -1 -1 -1  2

 gap cost: -2
```

		G	A	G	G	G
	0	-2	-4	-6	-8	-10
G	-2					
G	-4					
G	-6					
C	-8					
T	-10					

```
score matrix:
   a  t  g  c
a  2 -1 -1 -1
t -1  2 -1 -1
g -1 -1  2 -1
c -1 -1 -1  2

 gap cost: -2
```

		G	A	G	G	G
	0	-2	-4	-6	-8	-10
G	-2					
G	-4					
G	-6					
C	-8					
T	-10					

```
score matrix:
   a  t  g  c
a  2 -1 -1 -1
t -1  2 -1 -1
g -1 -1  2 -1
c -1 -1 -1  2

 gap cost: -2
```

		G	A	G	G	G
	0	-2	-4	-6	-8	-10
G	-2					
G	-4					
G	-6					
C	-8					
T	-10					

```
score matrix:
   a  t  g  c
a  2 -1 -1 -1
t -1  2 -1 -1
g -1 -1  2 -1
c -1 -1 -1  2

 gap cost: -2
```

		G	A	G	G	G
	0	-2	-4	-6	-8	-10
G	-2	2				
G	-4					
G	-6					
C	-8					
T	-10					

```
score matrix:
   a  t  g  c
a  2 -1 -1 -1
t -1  2 -1 -1
g -1 -1  2 -1
c -1 -1 -1  2

 gap cost: -2
```

		G	A	G	G	G
	0	-2	-4	-6	-8	-10
G	-2	2	0			
G	-4					
G	-6					
C	-8					
T	-10					

```
score matrix:
   a  t  g  c
a  2 -1 -1 -1
t -1  2 -1 -1
g -1 -1  2 -1
c -1 -1 -1  2

 gap cost: -2
```

		G	A	G	G	G
	0	-2	-4	-6	-8	-10
G	-2	2	0	-2		
G	-4					
G	-6					
C	-8					
T	-10					

```
score matrix:
   a  t  g  c
a  2 -1 -1 -1
t -1  2 -1 -1
g -1 -1  2 -1
c -1 -1 -1  2

 gap cost: -2
```

		G	A	G	G	G
	0	-2	-4	-6	-8	-10
G	-2	2	0	-2	-4	
G	-4					
G	-6					
C	-8					
T	-10					

```
score matrix:
   a  t  g  c
a  2 -1 -1 -1
t -1  2 -1 -1
g -1 -1  2 -1
c -1 -1 -1  2

 gap cost: -2
```

		G	A	G	G	G
	0	-2	-4	-6	-8	-10
G	-2	2	0	-2	-4	-6
G	-4					
G	-6					
C	-8					
T	-10					

```
score matrix:
   a  t  g  c
a  2 -1 -1 -1
t -1  2 -1 -1
g -1 -1  2 -1
c -1 -1 -1  2

 gap cost: -2
```

		G	A	G	G	G
	0	-2	-4	-6	-8	-10
G	-2	2	0	-2	-4	-6
G	-4	0				
G	-6					
C	-8					
T	-10					

```
score matrix:
   a  t  g  c
a  2 -1 -1 -1
t -1  2 -1 -1
g -1 -1  2 -1
c -1 -1 -1  2

 gap cost: -2
```

		G	A	G	G	G
	0	-2	-4	-6	-8	-10
G	-2	2	0	-2	-4	-6
G	-4	0	1			
G	-6					
C	-8					
T	-10					

```
score matrix:
   a  t  g  c
a  2 -1 -1 -1
t -1  2 -1 -1
g -1 -1  2 -1
c -1 -1 -1  2

 gap cost: -2
```

		G	A	G	G	G
	0	-2	-4	-6	-8	-10
G	-2	2	0	-2	-4	-6
G	-4	0	1	2		
G	-6					
C	-8					
T	-10					

```
score matrix:
   a  t  g  c
a  2 -1 -1 -1
t -1  2 -1 -1
g -1 -1  2 -1
c -1 -1 -1  2

 gap cost: -2
```

		G	A	G	G	G
	0	-2	-4	-6	-8	-10
G	-2	2	0	-2	-4	-6
G	-4	0	1	2	0	
G	-6					
C	-8					
T	-10					

```
score matrix:
   a  t  g  c
a  2 -1 -1 -1
t -1  2 -1 -1
g -1 -1  2 -1
c -1 -1 -1  2

 gap cost: -2
```

		G	A	G	G	G
	0	-2	-4	-6	-8	-10
G	-2	2	0	-2	-4	-6
G	-4	0	1	2	0	-2
G	-6					
C	-8					
T	-10					

```
score matrix:
   a  t  g  c
a  2 -1 -1 -1
t -1  2 -1 -1
g -1 -1  2 -1
c -1 -1 -1  2

 gap cost: -2
```

		G	A	G	G	G
	0	-2	-4	-6	-8	-10
G	-2	2	0	-2	-4	-6
G	-4	0	1	2	0	-2
G	-6	-2				
C	-8					
T	-10					

```
score matrix:
   a  t  g  c
a  2 -1 -1 -1
t -1  2 -1 -1
g -1 -1  2 -1
c -1 -1 -1  2

 gap cost: -2
```

		G	A	G	G	G
	0	-2	-4	-6	-8	-10
G	-2	2	0	-2	-4	-6
G	-4	0	1	2	0	-2
G	-6	-2	-1			
C	-8					
T	-10					

```
score matrix:
   a  t  g  c
a  2 -1 -1 -1
t -1  2 -1 -1
g -1 -1  2 -1
c -1 -1 -1  2

 gap cost: -2
```

		G	A	G	G	G
	0	-2	-4	-6	-8	-10
G	-2	2	0	-2	-4	-6
G	-4	0	1	2	0	-2
G	-6	-2	-1	3		
C	-8					
T	-10					

```
score matrix:
   a  t  g  c
a  2 -1 -1 -1
t -1  2 -1 -1
g -1 -1  2 -1
c -1 -1 -1  2

 gap cost: -2
```

		G	A	G	G	G
	0	-2	-4	-6	-8	-10
G	-2	2	0	-2	-4	-6
G	-4	0	1	2	0	-2
G	-6	-2	-1	3	4	
C	-8					
T	-10					

```
score matrix:
   a  t  g  c
a  2 -1 -1 -1
t -1  2 -1 -1
g -1 -1  2 -1
c -1 -1 -1  2

 gap cost: -2
```

		G	A	G	G	G
	0	-2	-4	-6	-8	-10
G	-2	2	0	-2	-4	-6
G	-4	0	1	2	0	-2
G	-6	-2	-1	3	4	2
C	-8					
T	-10					

```
score matrix:
   a  t  g  c
a  2 -1 -1 -1
t -1  2 -1 -1
g -1 -1  2 -1
c -1 -1 -1  2

 gap cost: -2
```

		G	A	G	G	G
	0	-2	-4	-6	-8	-10
G	-2	2	0	-2	-4	-6
G	-4	0	1	2	0	-2
G	-6	-2	-1	3	4	2
C	-8	-4				
T	-10					

```
score matrix:
   a  t  g  c
a  2 -1 -1 -1
t -1  2 -1 -1
g -1 -1  2 -1
c -1 -1 -1  2

 gap cost: -2
```

		G	A	G	G	G
	0	-2	-4	-6	-8	-10
G	-2	2	0	-2	-4	-6
G	-4	0	1	2	0	-2
G	-6	-2	-1	3	4	2
C	-8	-4	-3			
T	-10					

```
score matrix:
   a  t  g  c
a  2 -1 -1 -1
t -1  2 -1 -1
g -1 -1  2 -1
c -1 -1 -1  2

 gap cost: -2
```

		G	A	G	G	G
	0	-2	-4	-6	-8	-10
G	-2	2	0	-2	-4	-6
G	-4	0	1	2	0	-2
G	-6	-2	-1	3	4	2
C	-8	-4	-3	1		
T	-10					

```
score matrix:
   a  t  g  c
a  2 -1 -1 -1
t -1  2 -1 -1
g -1 -1  2 -1
c -1 -1 -1  2

 gap cost: -2
```

		G	A	G	G	G
	0	-2	-4	-6	-8	-10
G	-2	2	0	-2	-4	-6
G	-4	0	1	2	0	-2
G	-6	-2	-1	3	4	2
C	-8	-4	-3	1	2	
T	-10					

```
score matrix:
   a  t  g  c
a  2 -1 -1 -1
t -1  2 -1 -1
g -1 -1  2 -1
c -1 -1 -1  2

 gap cost: -2
```

		G	A	G	G	G
	0	-2	-4	-6	-8	-10
G	-2	2	0	-2	-4	-6
G	-4	0	1	2	0	-2
G	-6	-2	-1	3	4	2
C	-8	-4	-3	1	2	3
T	-10					

```
score matrix:
   a  t  g  c
a  2 -1 -1 -1
t -1  2 -1 -1
g -1 -1  2 -1
c -1 -1 -1  2

 gap cost: -2
```

		G	A	G	G	G
	0	-2	-4	-6	-8	-10
G	-2	2	0	-2	-4	-6
G	-4	0	1	2	0	-2
G	-6	-2	-1	3	4	2
C	-8	-4	-3	1	2	3
T	-10	-6				

```
score matrix:
   a  t  g  c
a  2 -1 -1 -1
t -1  2 -1 -1
g -1 -1  2 -1
c -1 -1 -1  2

 gap cost: -2
```

		G	A	G	G	G
	0	-2	-4	-6	-8	-10
G	-2	2	0	-2	-4	-6
G	-4	0	1	2	0	-2
G	-6	-2	-1	3	4	2
C	-8	-4	-3	1	2	3
T	-10	-6	-5			

```
score matrix:
   a  t  g  c
a  2 -1 -1 -1
t -1  2 -1 -1
g -1 -1  2 -1
c -1 -1 -1  2

 gap cost: -2
```

		G	A	G	G	G
	0	-2	-4	-6	-8	-10
G	-2	2	0	-2	-4	-6
G	-4	0	1	2	0	-2
G	-6	-2	-1	3	4	2
C	-8	-4	-3	1	2	3
T	-10	-6	-5	-1		

```
score matrix:
   a  t  g  c
a  2 -1 -1 -1
t -1  2 -1 -1
g -1 -1  2 -1
c -1 -1 -1  2

 gap cost: -2
```

		G	A	G	G	G
	0	-2	-4	-6	-8	-10
G	-2	2	0	-2	-4	-6
G	-4	0	1	2	0	-2
G	-6	-2	-1	3	4	2
C	-8	-4	-3	1	2	3
T	-10	-6	-5	-1	0	

```
score matrix:
   a  t  g  c
a  2 -1 -1 -1
t -1  2 -1 -1
g -1 -1  2 -1
c -1 -1 -1  2

 gap cost: -2
```

		G	A	G	G	G
	0	-2	-4	-6	-8	-10
G	-2	2	0	-2	-4	-6
G	-4	0	1	2	0	-2
G	-6	-2	-1	3	4	2
C	-8	-4	-3	1	2	3
T	-10	-6	-5	-1	0	1

```
score matrix:
   a  t  g  c
a  2 -1 -1 -1
t -1  2 -1 -1
g -1 -1  2 -1
c -1 -1 -1  2

 gap cost: -2
```

		G	A	G	G	G
	0	-2	-4	-6	-8	-10
G	-2	2	0	-2	-4	-6
G	-4	0	1	2	0	-2
G	-6	-2	-1	3	4	2
C	-8	-4	-3	1	2	3
T	-10	-6	-5	-1	0	1

score matrix:

	a	t	g	c
a	2	-1	-1	-1
t	-1	2	-1	-1
g	-1	-1	2	-1
c	-1	-1	-1	2

gap cost: -2

		G	A	G	G	G
	0	-2	-4	-6	-8	-10
G	-2	2	0	-2	-4	-6
G	-4	0	1	2	0	-2
G	-6	-2	-1	3	4	2
C	-8	-4	-3	1	2	3
T	-10	-6	-5	-1	0	1

```
score matrix:
   a  t  g  c
a  2 -1 -1 -1
t -1  2 -1 -1
g -1 -1  2 -1
c -1 -1 -1  2

 gap cost: -2
```

GGGCT
GAGGG

G-GGCT
GAGGG-

G-GGCT
GAGG-G

EDNAFULL

	A	T	G	C	S	W	R	Y	K	M	B	V	H	D	N
A	5	-4	-4	-4	-4	1	1	-4	-4	1	-4	-1	-1	-1	-2
T	-4	5	-4	-4	-4	1	-4	1	1	-4	-1	-4	-1	-1	-2
G	-4	-4	5	-4	1	-4	1	-4	1	-4	-1	-1	-4	-1	-2
C	-4	-4	-4	5	1	-4	-4	1	-4	1	-1	-1	-1	-4	-2
S	-4	-4	1	1	-1	-4	-2	-2	-2	-2	-1	-1	-3	-3	-1
W	1	1	-4	-4	-4	-1	-2	-2	-2	-2	-3	-3	-1	-1	-1
R	1	-4	1	-4	-2	-2	-1	-4	-2	-2	-3	-1	-3	-1	-1
Y	-4	1	-4	1	-2	-2	-4	-1	-2	-2	-1	-3	-1	-3	-1
K	-4	1	1	-4	-2	-2	-2	-2	-1	-4	-1	-3	-3	-1	-1
M	1	-4	-4	1	-2	-2	-2	-2	-4	-1	-3	-1	-1	-3	-1
B	-4	-1	-1	-1	-1	-3	-3	-1	-1	-3	-1	-2	-2	-2	-1
V	-1	-4	-1	-1	-1	-3	-1	-3	-3	-1	-2	-1	-2	-2	-1
H	-1	-1	-4	-1	-3	-1	-3	-1	-3	-1	-2	-2	-1	-2	-1
D	-1	-1	-1	-4	-3	-1	-1	-3	-1	-3	-2	-2	-2	-1	-1
N	-2	-2	-2	-2	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1

Local alignment

Lokalt alignment

- What is the difference between a global and local alignment?
- When to use a local alignment?
- What are the strengths of local alignment?

(A) local

```
PI3-kinase  DRHNSNIMVKDDGQLFHIDFG
cAMP PK     DLKPENLLIDQQGYIQVTDFG
```

(B) global

```
                    10                  20                  30                  40                  50
PI3-kinase  HQLGNLR--LEECRI---MSSAKRPLWLNWENPDIMSELLFQNNEIIFKNGDDLRQDMLT
cAMP PK     GNAAAAKKGXEQESVKEFLAKAKEDFLKKWENPAQNTAHLDQFERIKTLGTGSFGRVML-
                  10                  20                  30                  40                  50

                    60                  70                  80                  90                  100                 110
PI3-kinase  LQIIRIME--NIWQNQGLDLRMLPYGCLSIGDCVGLIEVVRNSHTIMQ-IQCKGGLKGAL
cAMP PK     ---VKHMETGNHYAMKILDKQKVVK--------LKQIEHTLNEKRILQAVNFPFLVKLEF
                    60                  70                  80                  90                  100

                  120                 130                 140                 150                 160
PI3-kinase  QFNSHT-LHQWLKDKNKGEIYDAA--IDLFTRSCAGYCVATFILGIGDRHNSNIMVKD-D
cAMP PK     SFKDNSNLYMVMEYVPGGEMFSHLRRIGRFSEPHARFYAAQIVLTFEYLHSLDLIYRDLK
                  110                 120                 130                 140                 150                 160

                  170                 180                 190                 200                 210                 220
PI3-kinase  GQLFHIDFGHFLDHKKKKFGYKRERVP-----FVLTQDFL---IVISKGAQECTKTREFE
cAMP PK     PENLLIDQQGYI--QVTDFGFAK-RVKGRTWXLCGTPEYLAPEIILSKGYNKAVDWWALG
                  170                 180                 190                 200                 210                 220

                    230                 240                 250                 260                 270
PI3-kinase  RF-QEMC--YKAYLAIRQHANLFINLFSMMLGSGMPELQSFDDIAYIRKTLALDKTEQEA
cAMP PK     VLIYEMAAGYPPFFA-DQPIQIYEKIVSGKVR--FPSHFSSDLKDLLRNLLQVDLTKR--
                    230                 240                 250                 260                 270                 280

                  280                 290                 300                                                 310
PI3-kinase  LEYFMKQMNDAHHGGWTTKMDWI----------------------------FHTIKQHALN----
cAMP PK     FGNLKNGVNDIKNHKWFATTDWIAIYQRKVEAPFIPKFKGPGDTSNFDDYEEEEIRVXIN
                  290                 300                 310                 320                 330                 340
```

Smith-Waterman

- Almost the same as Needleman-Wunch
- You just reset the score to zero when a sub-alignment becomes too small (lower than zero).
- In the traceback, you stop extending the alignment when when the Vi forlænger ikke alignments med en score på 0 eller mindre.

Smith-Waterman

- The score matrix used must result in negative scores for random matches.

- I.e., the score of a random alignment must be negative:

$$\sum_{a,b} q_a q_b s(a,b) < 0$$

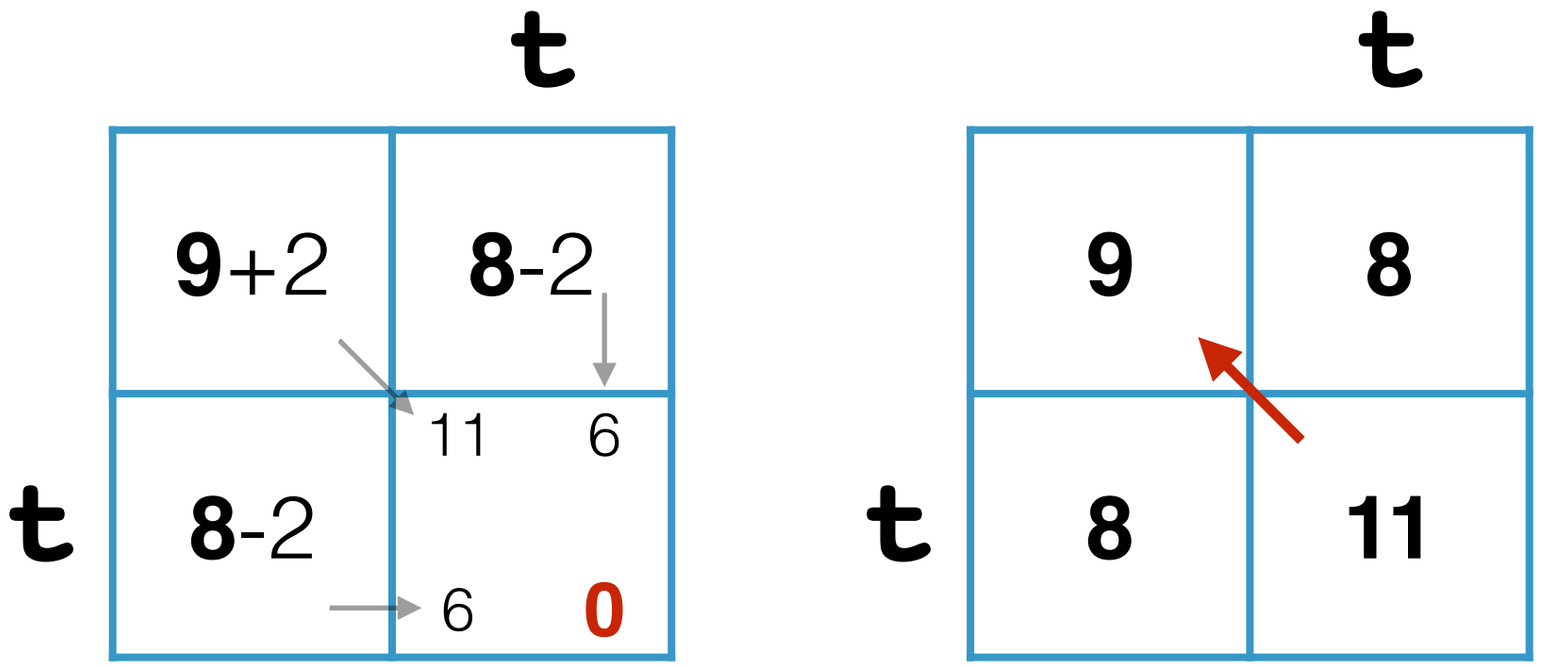
q_a = frequency of base a

s = score matrix

```
score matrix:
   a  t  g  c
a  2 -1 -1 -1
t -1  2 -1 -1
g -1 -1  2 -1
c -1 -1 -1  2

 gap cost: -2
```

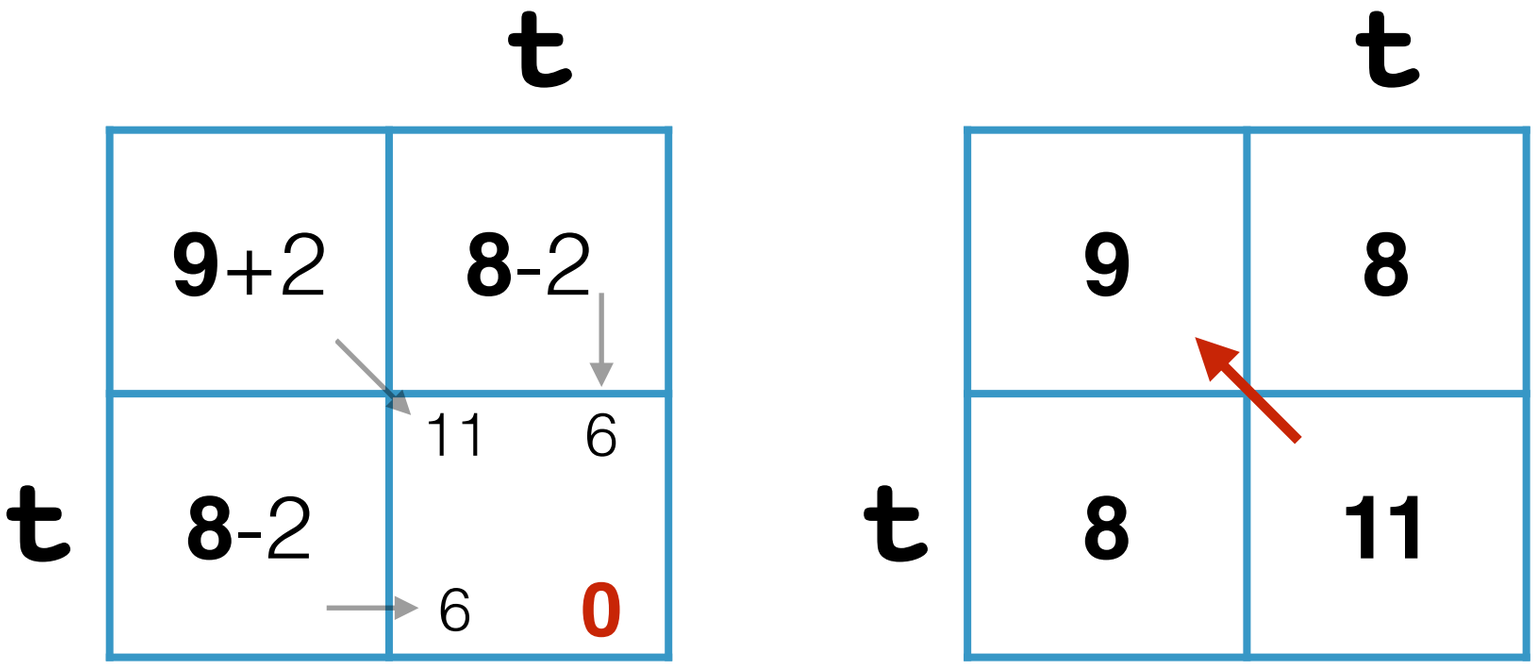
Maximum of three scores **and 0**

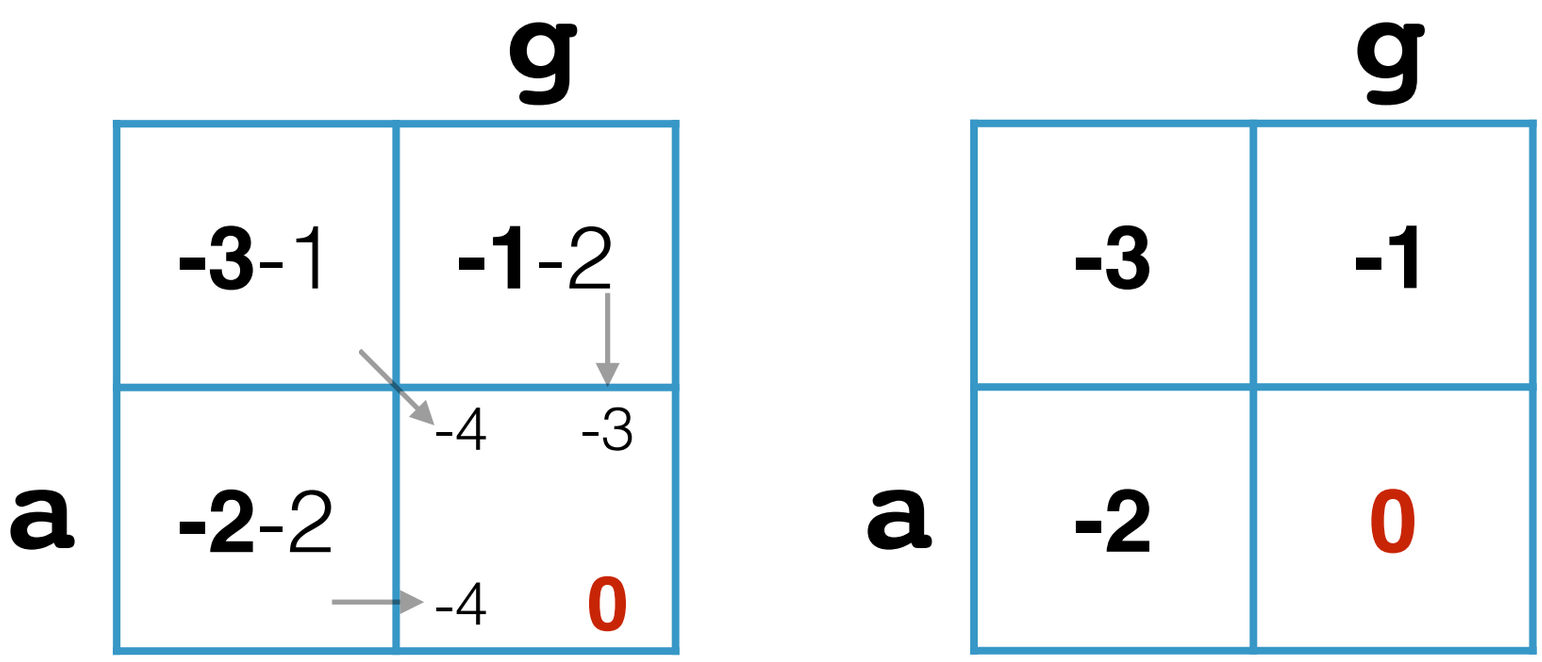



```
score matrix:
   a  t  g  c
a  2 -1 -1 -1
t -1  2 -1 -1
g -1 -1  2 -1
c -1 -1 -1  2

 gap cost: -2
```

Maximum of three scores **and 0**

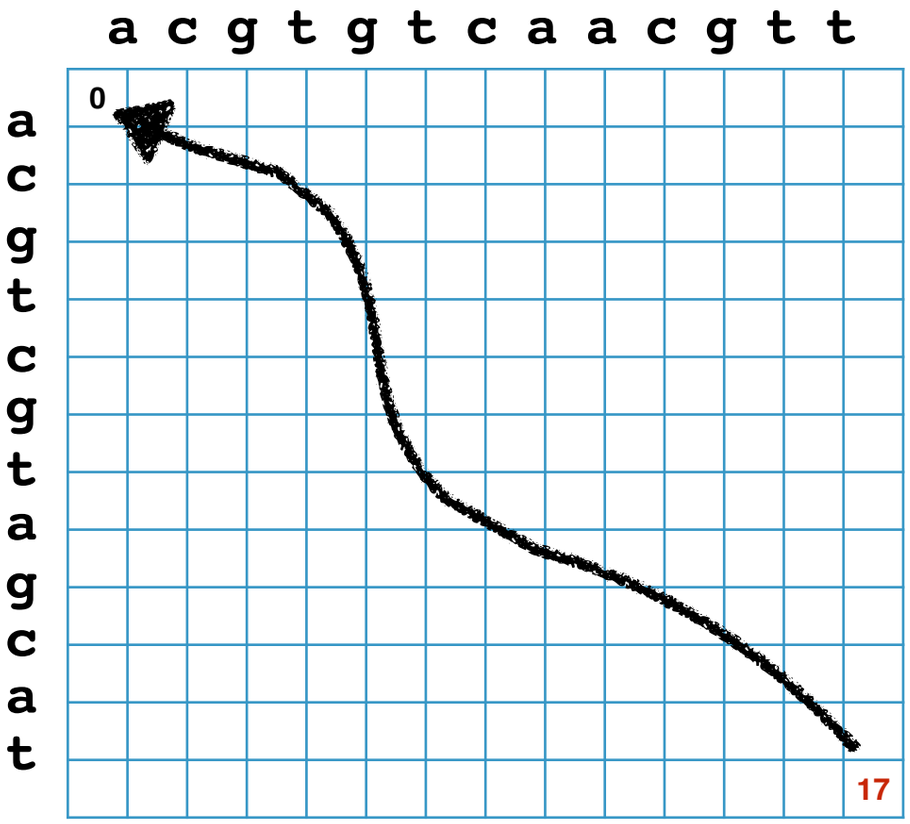





```
score matrix:
   a  t  g  c
a  2 -1 -1 -1
t -1  2 -1 -1
g -1 -1  2 -1
c -1 -1 -1  2

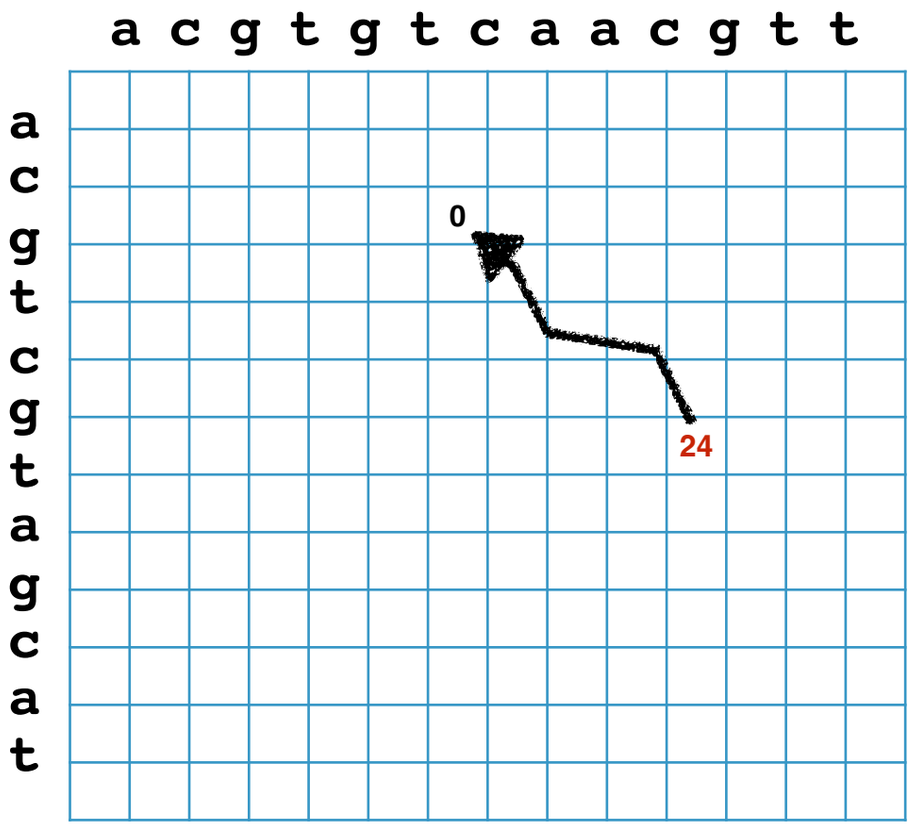
 gap cost: -2
```

Global alignment



“Traceback” from last to first cell.

Lokal alignment



“Traceback” from cell with largest score to the first cell with zero score.

Make a local alignment

`GGGCT` og `GAGGG`

```
score matrix:
   a  t  g  c
a  2 -1 -1 -1
t -1  2 -1 -1
g -1 -1  2 -1
c -1 -1 -1  2

 gap cost: -2
```

Lokal alignment

		G	A	G	G	G
	0	0	0	0	0	0
G	0					
G	0					
C	0					
G	0					
C	0					
T	0					

Top row and left column are all zeros (because they cannot be negative)

Lokal alignment

		G	A	G	G	G
	0	0	0	0	0	0
G	0					
G	0					
C	0					
G	0					
T	0					

Top row and left column are all zeros (because they cannot be negative)

Lokal alignment

		G	A	G	G	G
	0	0	0	0	0	0
G	0	2				
G	0					
C	0					
G	0					
T	0					

Top row and left column are all zeros (because they cannot be negative)

Lokal alignment

		G	A	G	G	G
	0	0	0	0	0	0
G	0	2	0			
G	0					
G	0					
C	0					
T	0					

Top row and left column are all zeros (because they cannot be negative)

Lokal alignment

		G	A	G	G	G
	0	0	0	0	0	0
G	0	2	0	2		
G	0					
G	0					
C	0					
T	0					

Top row and left column are all zeros (because they cannot be negative)

Lokal alignment

		G	A	G	G	G
	0	0	0	0	0	0
G	0	2	0	2	2	
G	0					
G	0					
C	0					
T	0					

Top row and left column are all zeros (because they cannot be negative)

Lokal alignment

		G	A	G	G	G
	0	0	0	0	0	0
G	0	2	0	2	2	2
G	0					
G	0					
C	0					
T	0					

Top row and left column are all zeros (because they cannot be negative)

Lokal alignment

		G	A	G	G	G
	0	0	0	0	0	0
G	0	2	0	2	2	2
G	0	2				
G	0					
C	0					
T	0					

Top row and left column are all zeros (because they cannot be negative)

Lokal alignment

		G	A	G	G	G
	0	0	0	0	0	0
G	0	2	0	2	2	2
G	0	2	1			
G	0					
C	0					
T	0					

Top row and left column are all zeros (because they cannot be negative)

Lokal alignment

		G	A	G	G	G
	0	0	0	0	0	0
G	0	2	0	2	2	2
G	0	2	1	2		
G	0					
C	0					
T	0					

Top row and left column are all zeros (because they cannot be negative)

Lokal alignment

		G	A	G	G	G
	0	0	0	0	0	0
G	0	2	0	2	2	2
G	0	2	1	2	4	
G	0					
C	0					
T	0					

Top row and left column are all zeros (because they cannot be negative)

Lokal alignment

		G	A	G	G	G
	0	0	0	0	0	0
G	0	2	0	2	2	2
G	0	2	1	2	4	4
G	0					
C	0					
T	0					

Top row and left column are all zeros (because they cannot be negative)

Lokal alignment

		G	A	G	G	G
	0	0	0	0	0	0
G	0	2	0	2	2	2
G	0	2	1	2	4	4
G	0	2				
C	0					
T	0					

Top row and left column are all zeros (because they cannot be negative)

Lokal alignment

		G	A	G	G	G
	0	0	0	0	0	0
G	0	2	0	2	2	2
G	0	2	1	2	4	4
G	0	2	1			
C	0					
T	0					

Top row and left column are all zeros (because they cannot be negative)

Lokal alignment

		G	A	G	G	G
	0	0	0	0	0	0
G	0	2	0	2	2	2
G	0	2	1	2	4	4
G	0	2	1	3		
C	0					
T	0					

Top row and left column are all zeros (because they cannot be negative)

Lokal alignment

		G	A	G	G	G
	0	0	0	0	0	0
G	0	2	0	2	2	2
G	0	2	1	2	4	4
G	0	2	1	3	4	
C	0					
T	0					

Top row and left column are all zeros (because they cannot be negative)

Lokal alignment

		G	A	G	G	G
	0	0	0	0	0	0
G	0	2	0	2	2	2
G	0	2	1	2	4	4
G	0	2	1	3	4	6
C	0					
T	0					

Top row and left column are all zeros (because they cannot be negative)

Lokal alignment

		G	A	G	G	G
	0	0	0	0	0	0
G	0	2	0	2	2	2
G	0	2	1	2	4	4
G	0	2	1	3	4	6
C	0	0				
T	0					

Top row and left column are all zeros (because they cannot be negative)

Lokal alignment

		G	A	G	G	G
	0	0	0	0	0	0
G	0	2	0	2	2	2
G	0	2	1	2	4	4
G	0	2	1	3	4	6
C	0	0	1			
T	0					

Top row and left column are all zeros (because they cannot be negative)

Lokal alignment

		G	A	G	G	G
	0	0	0	0	0	0
G	0	2	0	2	2	2
G	0	2	1	2	4	4
G	0	2	1	3	4	6
C	0	0	1	1		
T	0					

Top row and left column are all zeros (because they cannot be negative)

Lokal alignment

		G	A	G	G	G
	0	0	0	0	0	0
G	0	2	0	2	2	2
G	0	2	1	2	4	4
G	0	2	1	3	4	6
C	0	0	1	1	2	
T	0					

Top row and left column are all zeros (because they cannot be negative)

Lokal alignment

		G	A	G	G	G
	0	0	0	0	0	0
G	0	2	0	2	2	2
G	0	2	1	2	4	4
G	0	2	1	3	4	6
C	0	0	1	1	2	4
T	0					

Top row and left column are all zeros (because they cannot be negative)

Lokal alignment

		G	A	G	G	G
	0	0	0	0	0	0
G	0	2	0	2	2	2
G	0	2	1	2	4	4
G	0	2	1	3	4	6
C	0	0	1	1	2	4
T	0	0				

Top row and left column are all zeros (because they cannot be negative)

Lokal alignment

		G	A	G	G	G
	0	0	0	0	0	0
G	0	2	0	2	2	2
G	0	2	1	2	4	4
G	0	2	1	3	4	6
C	0	0	1	1	2	4
T	0	0	0			

Top row and left column are all zeros (because they cannot be negative)

Lokal alignment

		G	A	G	G	G
	0	0	0	0	0	0
G	0	2	0	2	2	2
G	0	2	1	2	4	4
G	0	2	1	3	4	6
C	0	0	1	1	2	4
T	0	0	0	0		

Top row and left column are all zeros (because they cannot be negative)

Lokal alignment

		G	A	G	G	G
	0	0	0	0	0	0
G	0	2	0	2	2	2
G	0	2	1	2	4	4
G	0	2	1	3	4	6
C	0	0	1	1	2	4
T	0	0	0	0	0	

Top row and left column are all zeros (because they cannot be negative)

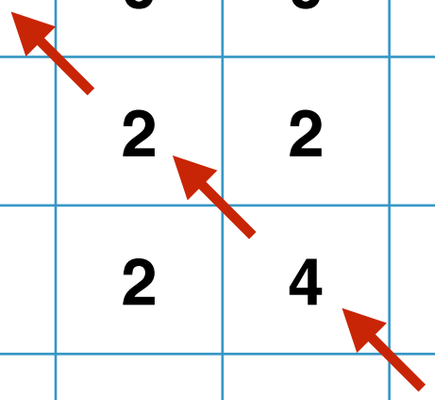
Lokal alignment

		G	A	G	G	G
	0	0	0	0	0	0
G	0	2	0	2	2	2
G	0	2	1	2	4	4
G	0	2	1	3	4	6
C	0	0	1	1	2	4
T	0	0	0	0	0	2

Top row and left column are all zeros (because they cannot be negative)

Lokal alignment

		G	A	G	G	G
	0	0	0	0	0	0
G	0	2	0	2	2	2
G	0	2	1	2	4	4
G	0	2	1	3	4	6
C	0	0	1	1	2	4
T	0	0	0	0	0	2



Alignment: **GGG**
GGG

Lokalt alignment

		G	A	G	G	G
	0	0	0	0	0	0
G	0	2	0	2	2	2
G	0	2	1	2	4	4
G	0	2	1	3	4	6
C	0	0	1	1	2	4
T	0	0	0	0	0	2

Alignment:

GGG
GGG

Globalt alignment

		G	A	G	G	G
	0	-2	-4	-6	-8	-10
G	-2	2	0	-2	-4	-6
G	-4	0	1	2	0	-2
G	-6	-2	-1	3	4	2
C	-8	-4	-3	1	2	3
T	-10	-6	-5	-1	0	1

Alignments:

GGGCT
GAGGG

G-GGCT
GAGG-

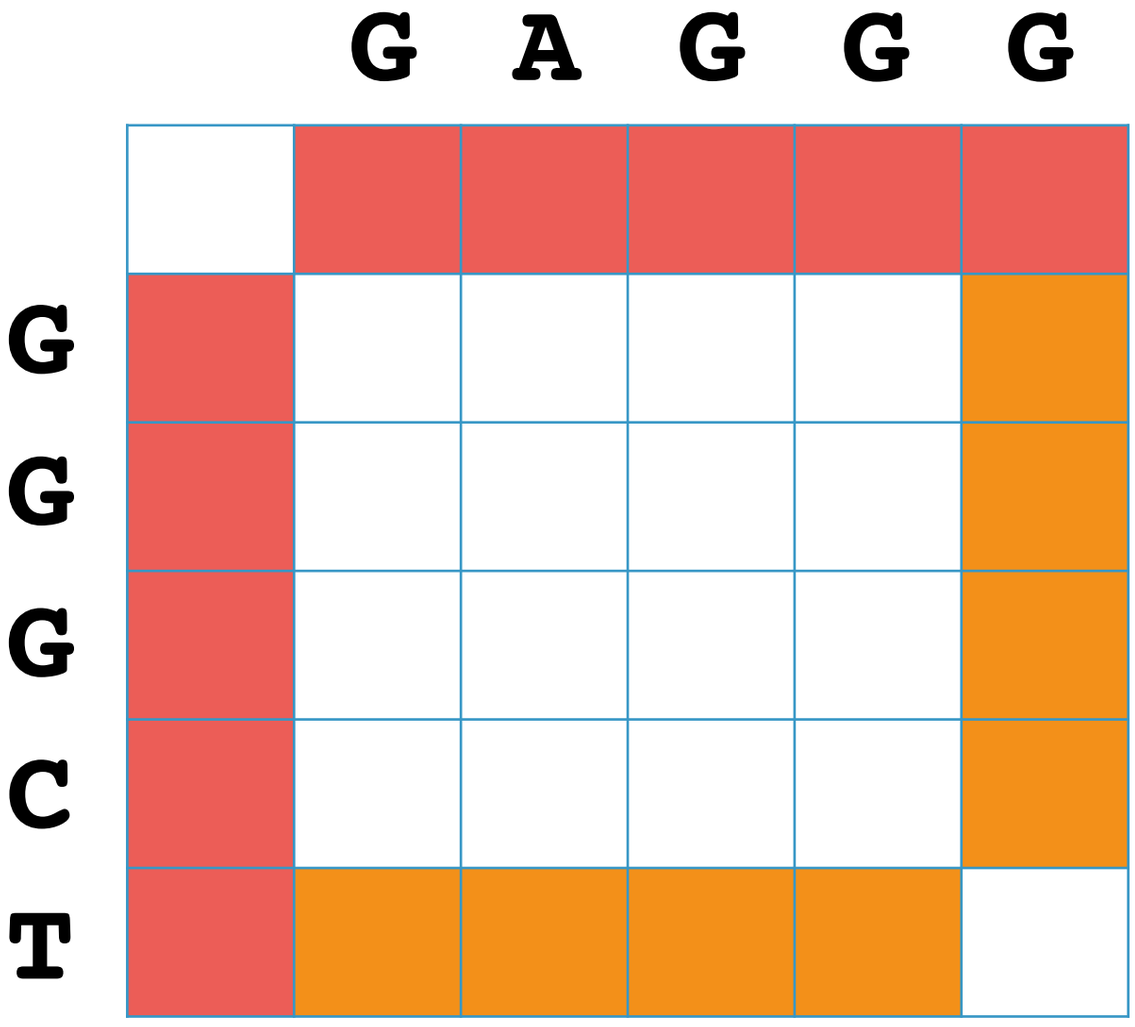
G-GGCT
GAGG-G

Improved gap scoring in global and local alignment

End gaps in global alignment

- Dedicated gap scores for ends of alignemnts.
- Should gaps at ends be “punished” more or less?
- Where in our dynamic programming table would we need to apply a special gap score.

When we compute alignment scores in the cells along the edges we should use a smaller gap penalty (e.g. instead of -2 we could use **0** to not penalize end gaps at all).



The left column and top row represent alignments with a series of gaps in the beginning.

The right column and bottom row represent alignments with a series of gaps at the end.

Linear gap score

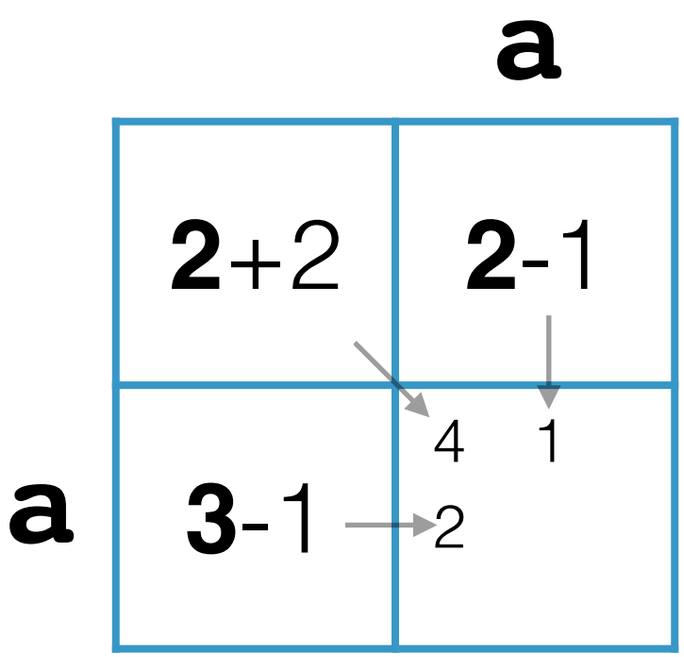
$$\gamma(g) = -dg$$

- Gap length: g
- Complexity $O(n^2)$
- If d = -10, what is the score of gap of length 11?

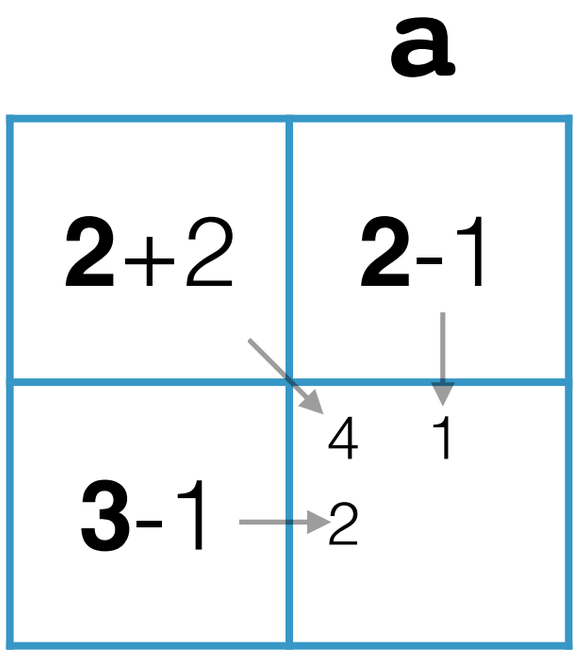
Affine gap score

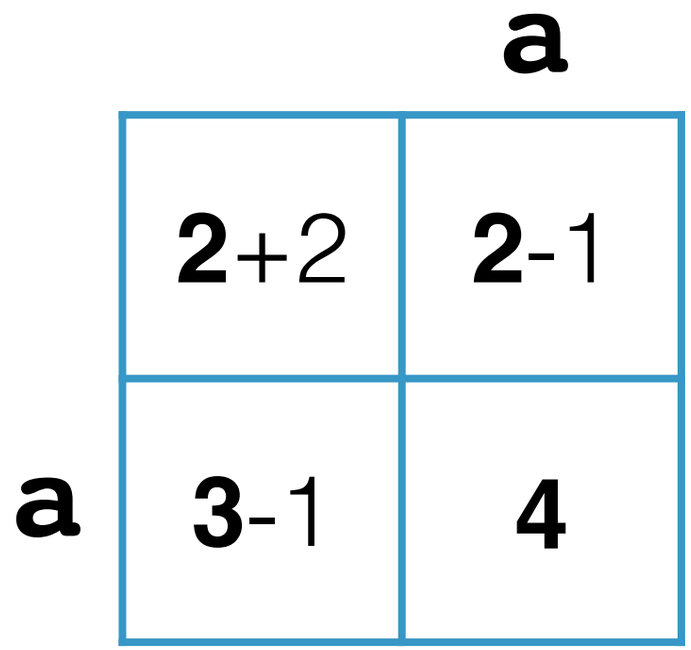
$$\gamma(g) = -d - (g-1)e$$

- Gap length: g, gap open: d, gap extend: e
- Complexity still $O(n^2)$
- Why does it make biological sense?
- If d = 10 and e = 1, what is the score of a gap of length 11?

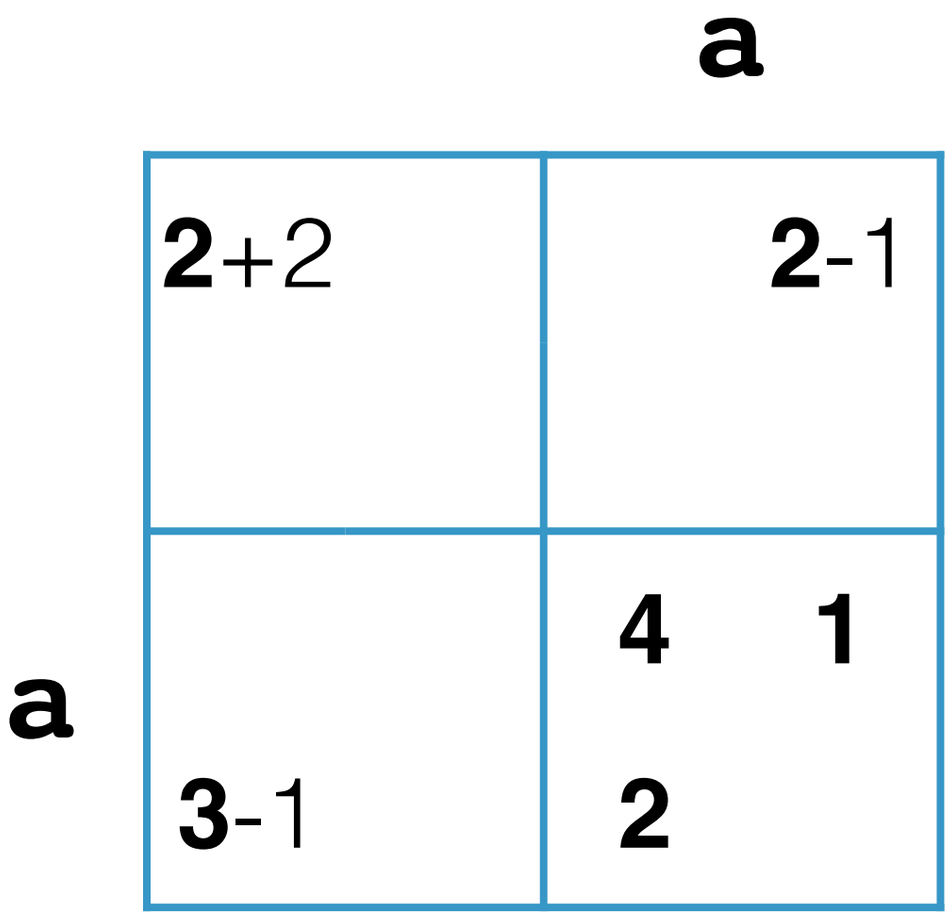


what do the three possible scores represent?



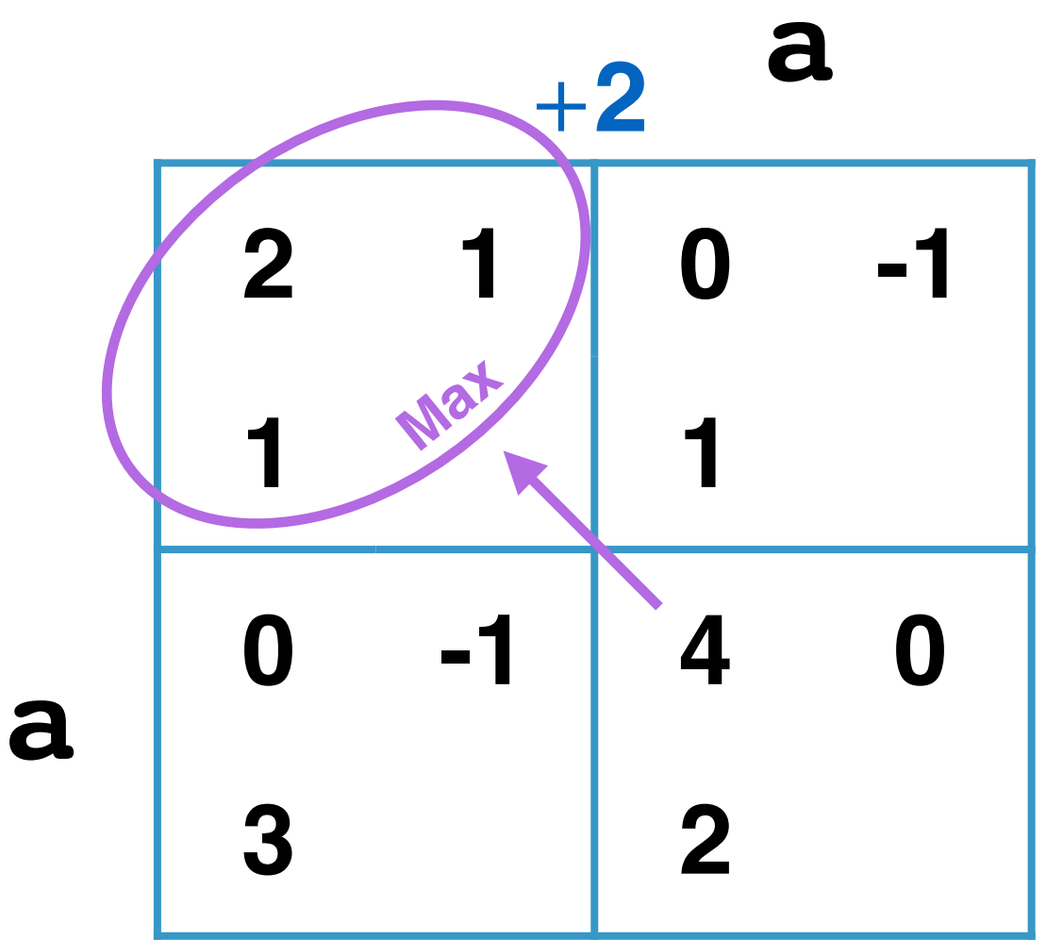


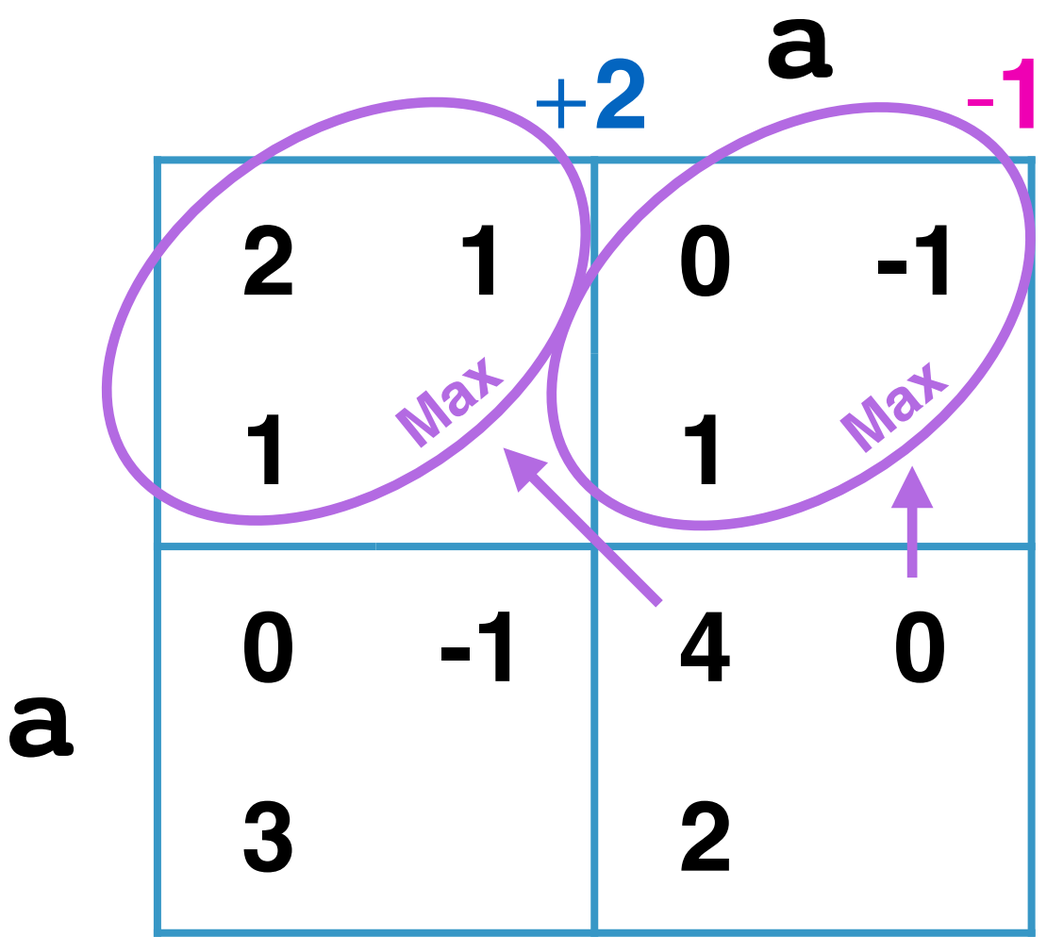
Keeping only the highest score

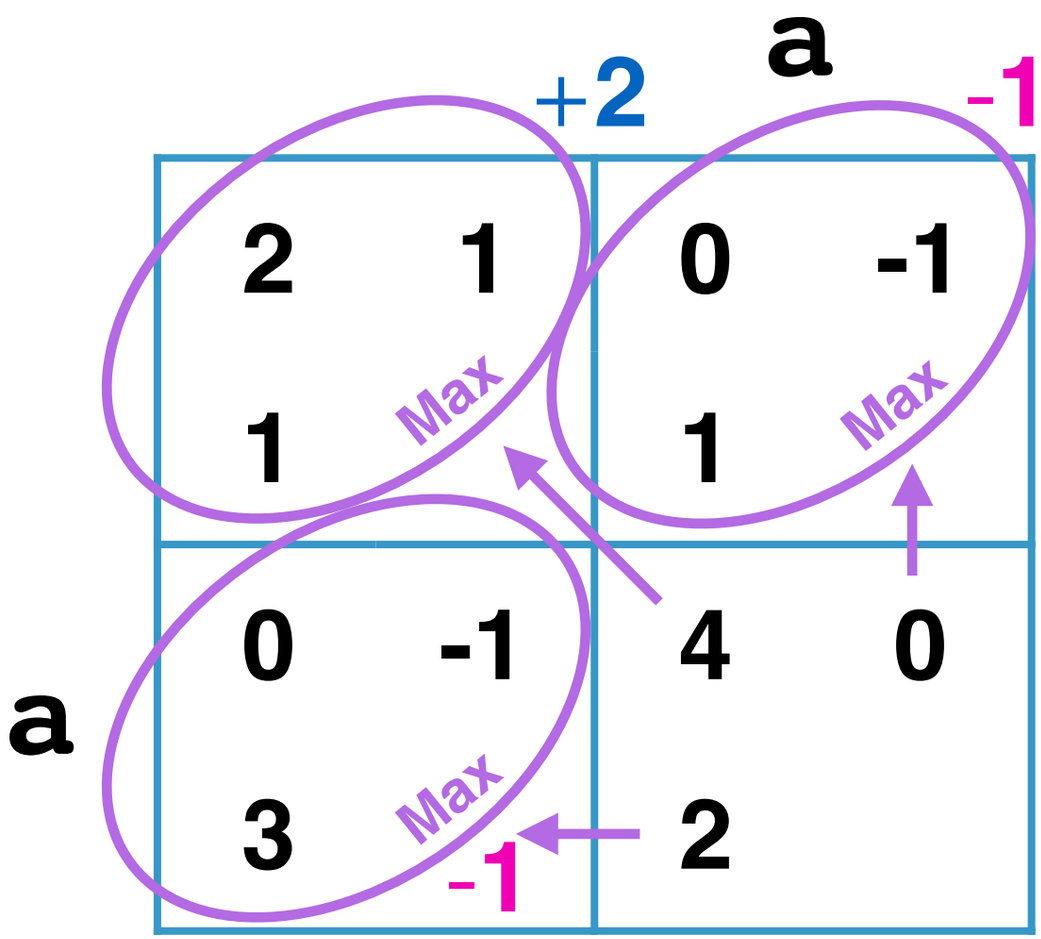


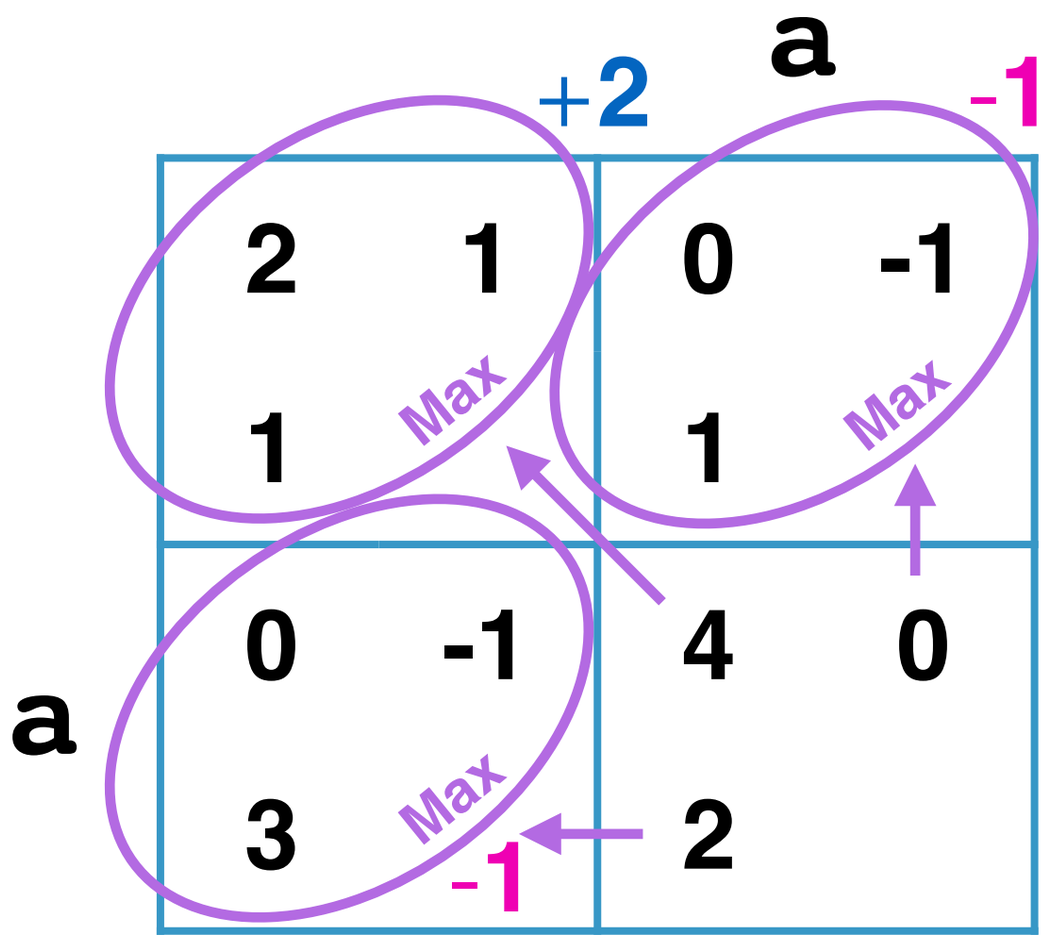
What if we kept *all three* scores?

	a		a	
a	2 1	1	0 1	-1
	0 3	-1	4 2	0

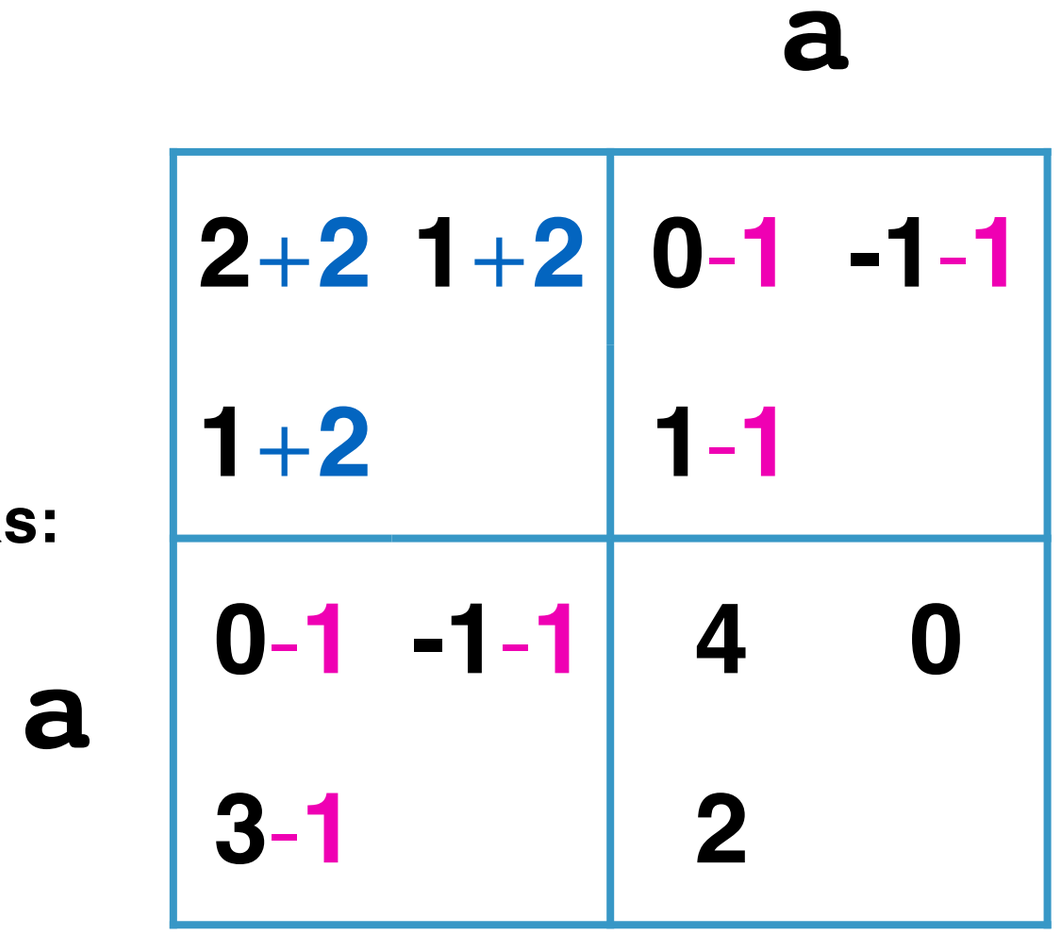

a
+2
2
1
0
-1
Max
1
1
a
0
-1
4
0
3
2

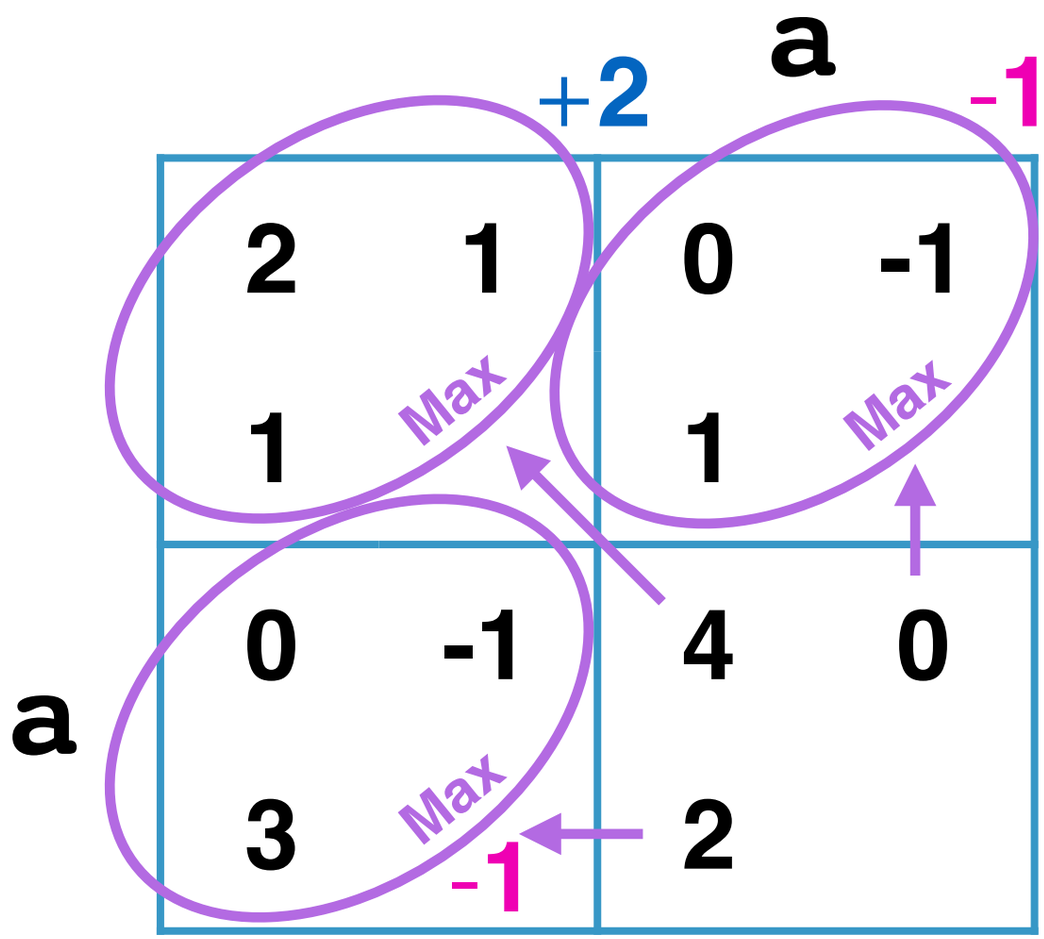

a
+2
-1
2
1
1
Max
0
-1
1
Max
a
0
-1
3
4
0
2


a
+2
-1
2
1
0
-1
Max
1
Max
1
a
0
-1
4
0
Max
3
2
-1

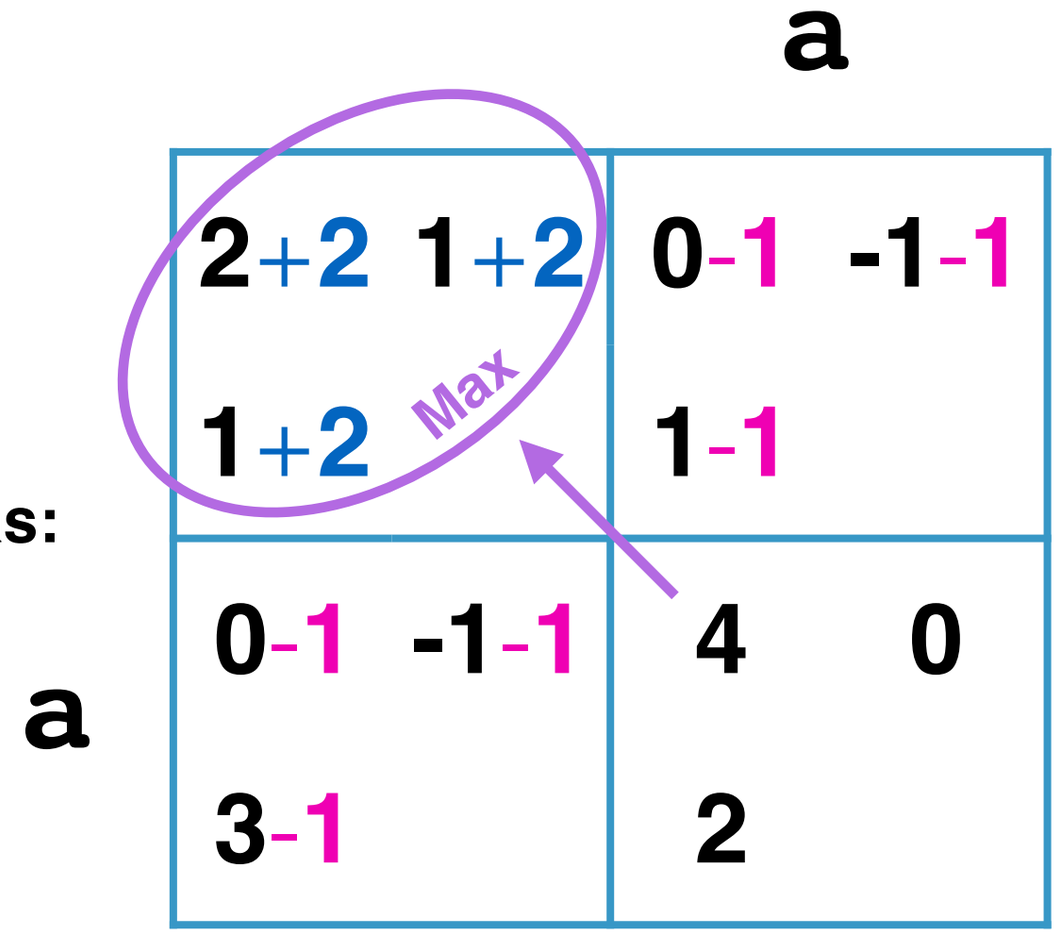


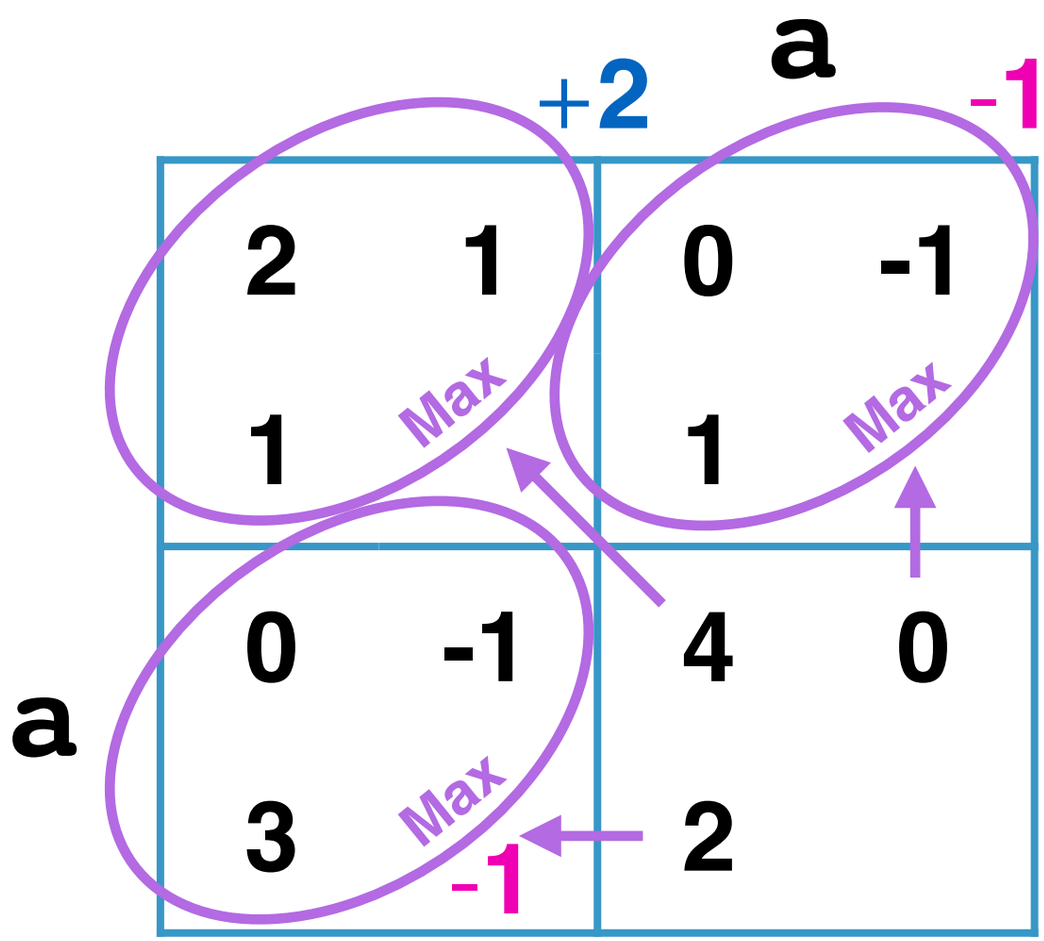
same as:



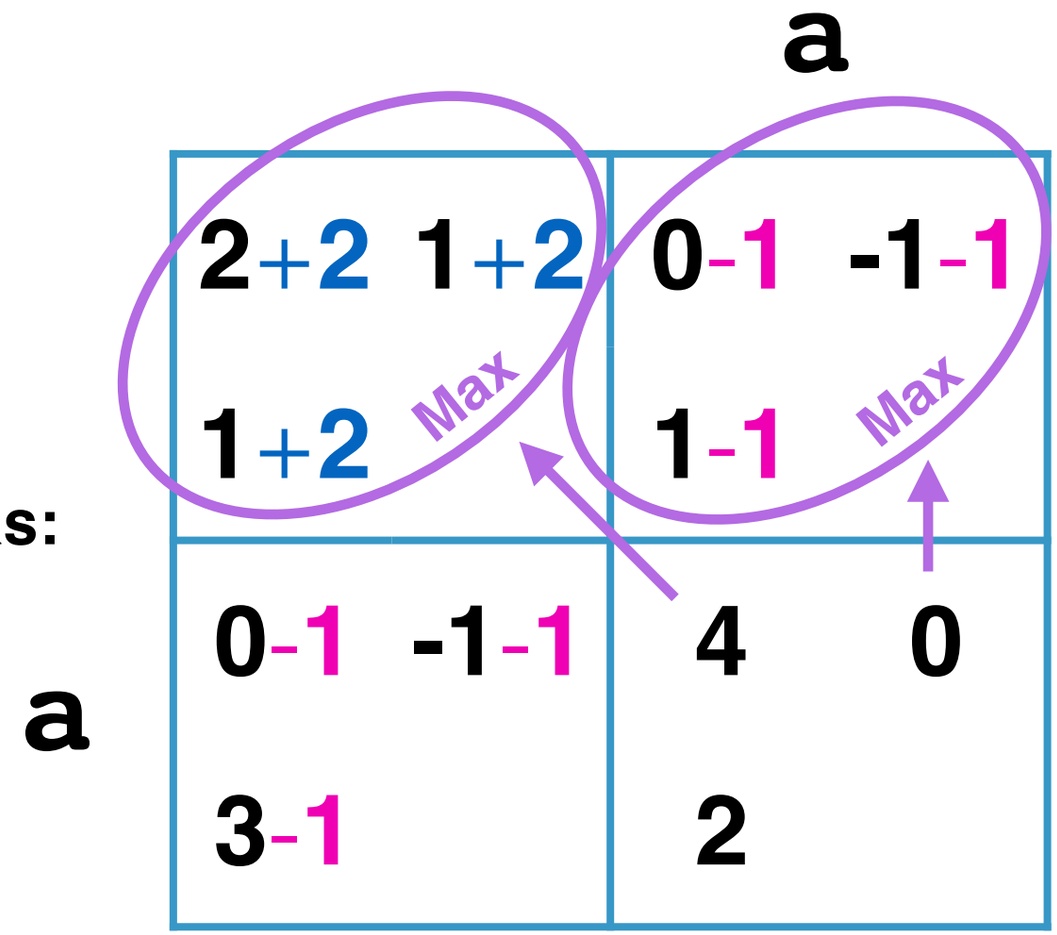

a
+2
-1
2
1
1
Max
0
-1
1
Max
a
0
-1
3
Max
-1
4
0
2

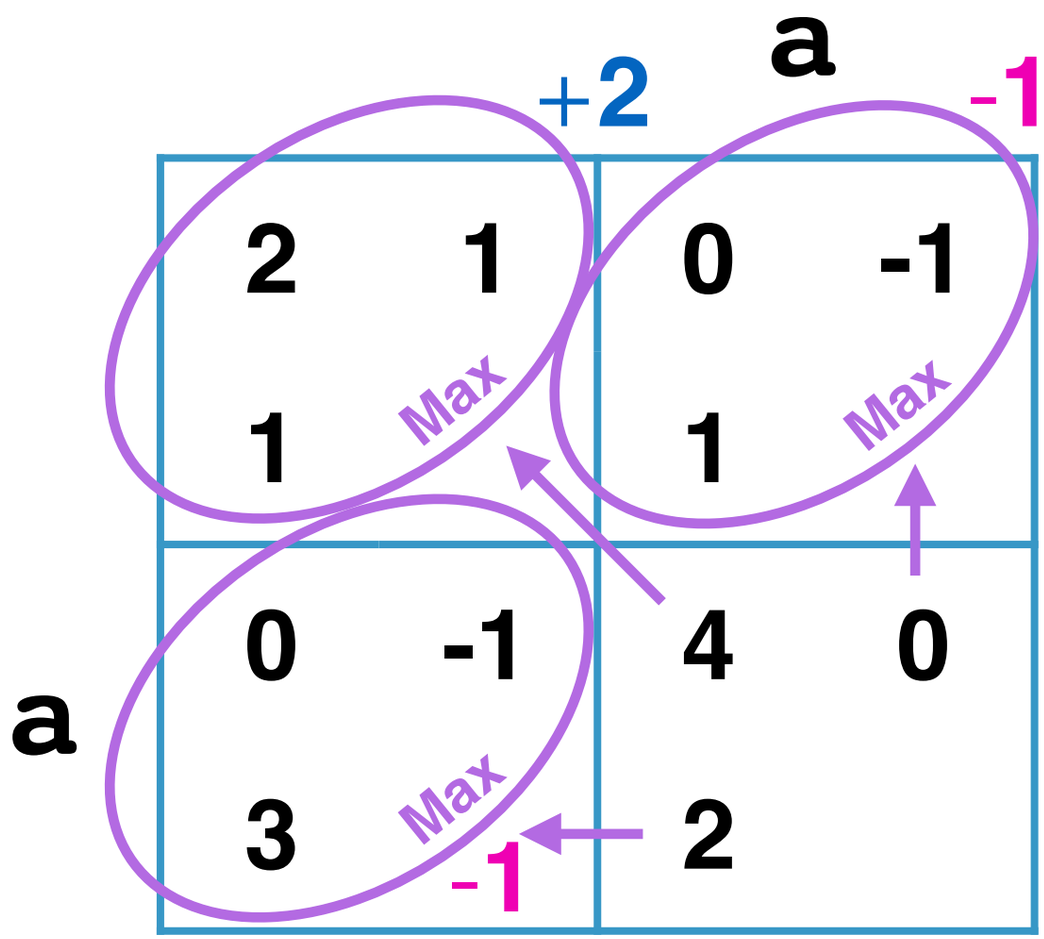
same as:


a
2+2
1+2
1+2
Max
0-1
-1-1
1-1
a
0-1
-1-1
3-1
4
0
2

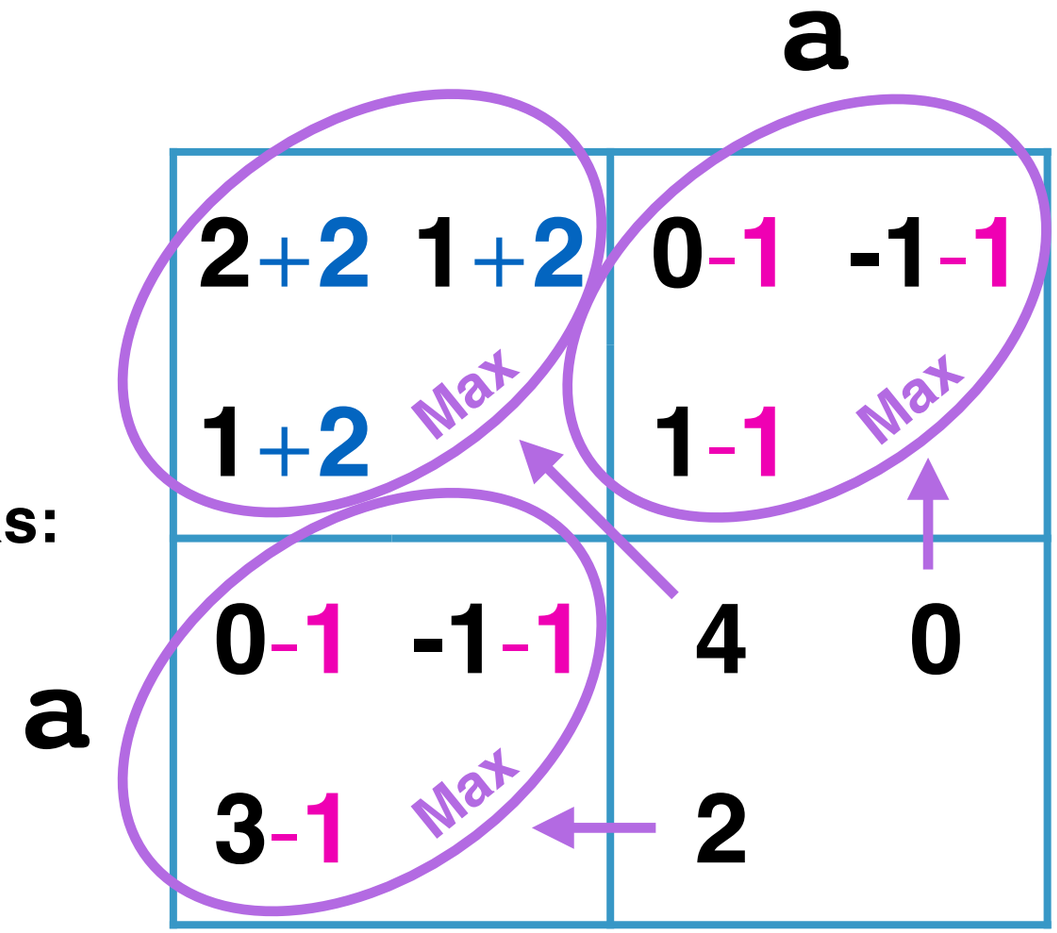

a
+2
-1
2
1
0
-1
Max
1
Max
1
0
-1
4
0
a
Max
3
2
-1

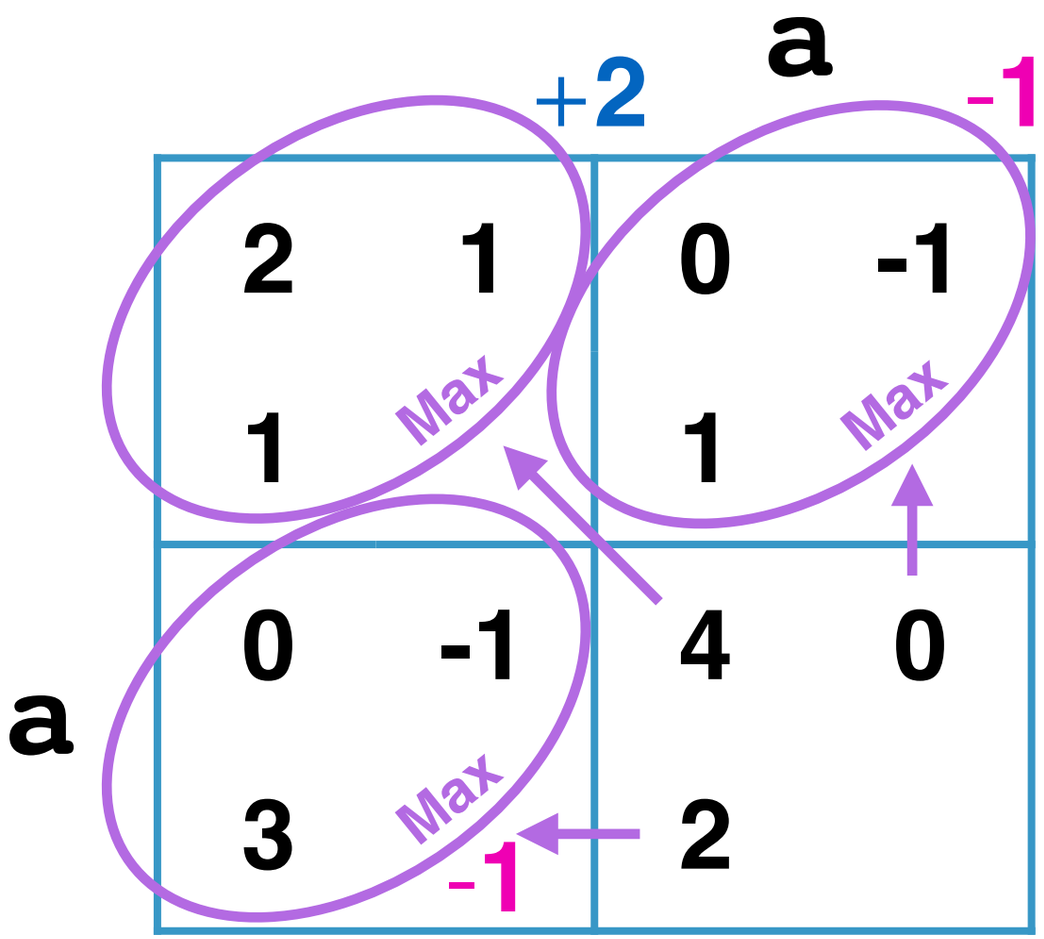
same as:


a
2+2
1+2
0-1
-1-1
Max
1+2
Max
1-1
0-1
-1-1
4
0
a
3-1
2

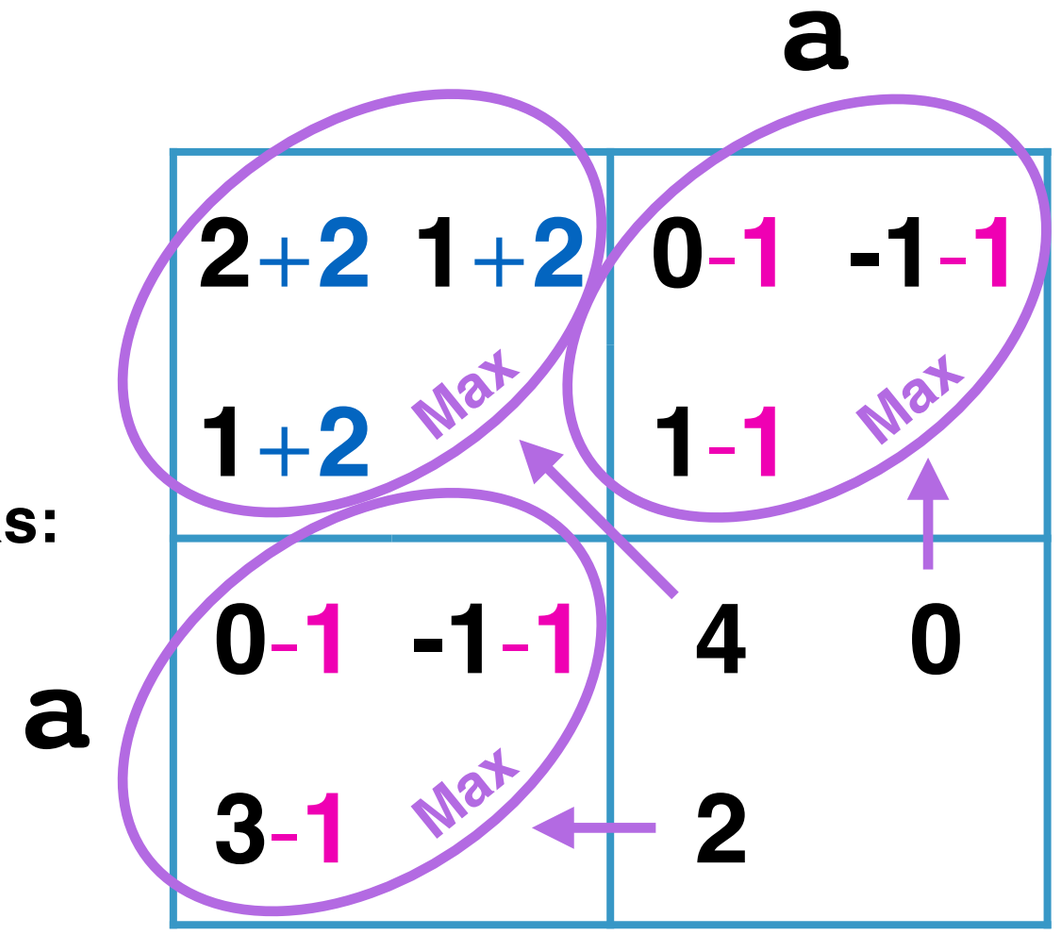

a
+2
-1
2
1
0
-1
Max
1
Max
1
0
-1
4
0
a
Max
3
-1
2

same as:

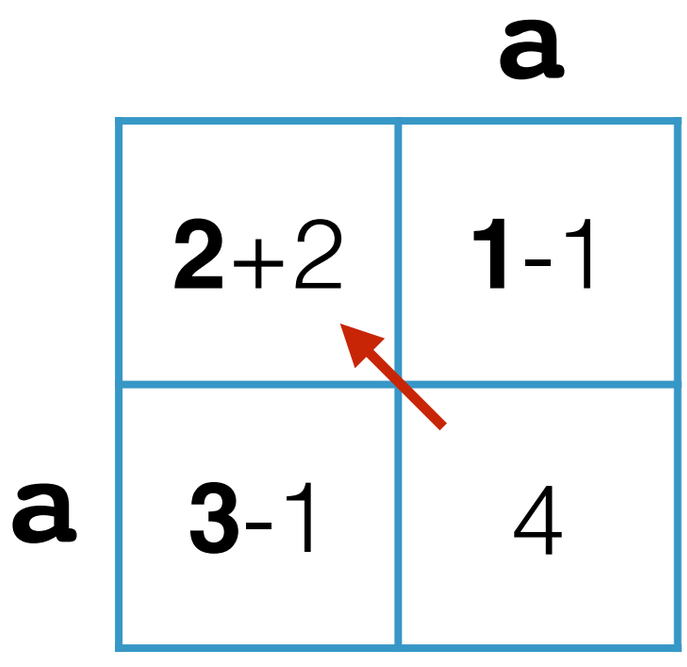

a
2+2
1+2
0-1
-1-1
Max
1+2
Max
1-1
0-1
-1-1
4
0
a
Max
3-1
2



same as:



… which is same as:



So we are still computing the same thing.

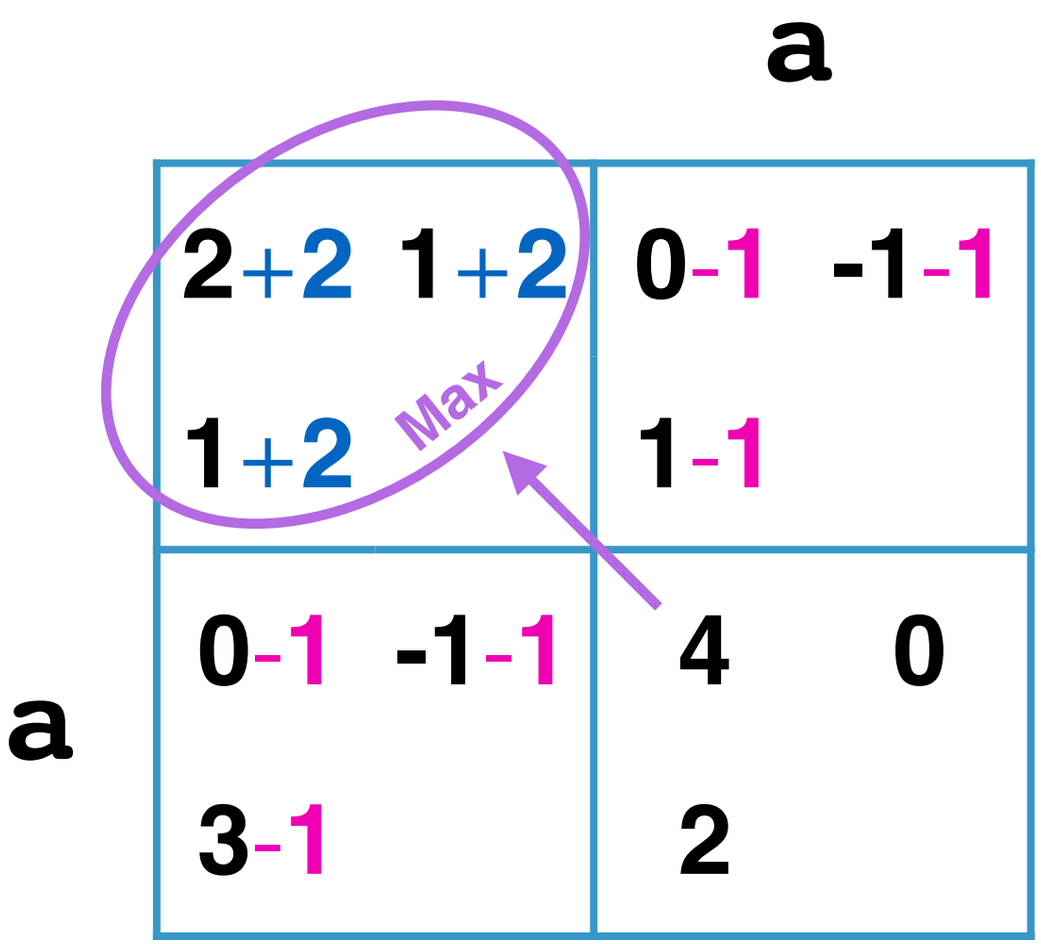
Linear:

Match score: **2**
Gap score: **-1**

		a
	2+2 1+2 1+2	0-1 -1-1 1-1
a	0-1 -1-1 3-1	4 0 2

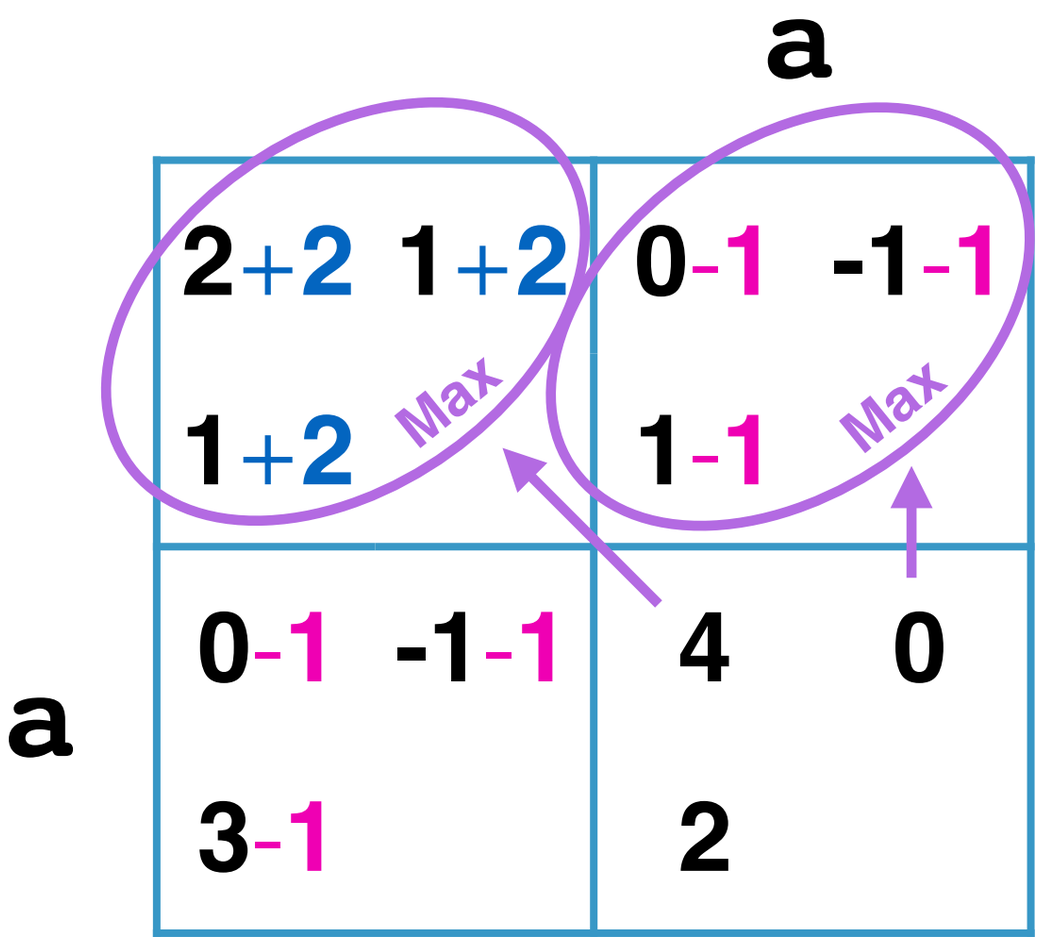
Linear:

Match score: **2**
Gap score: **-1**



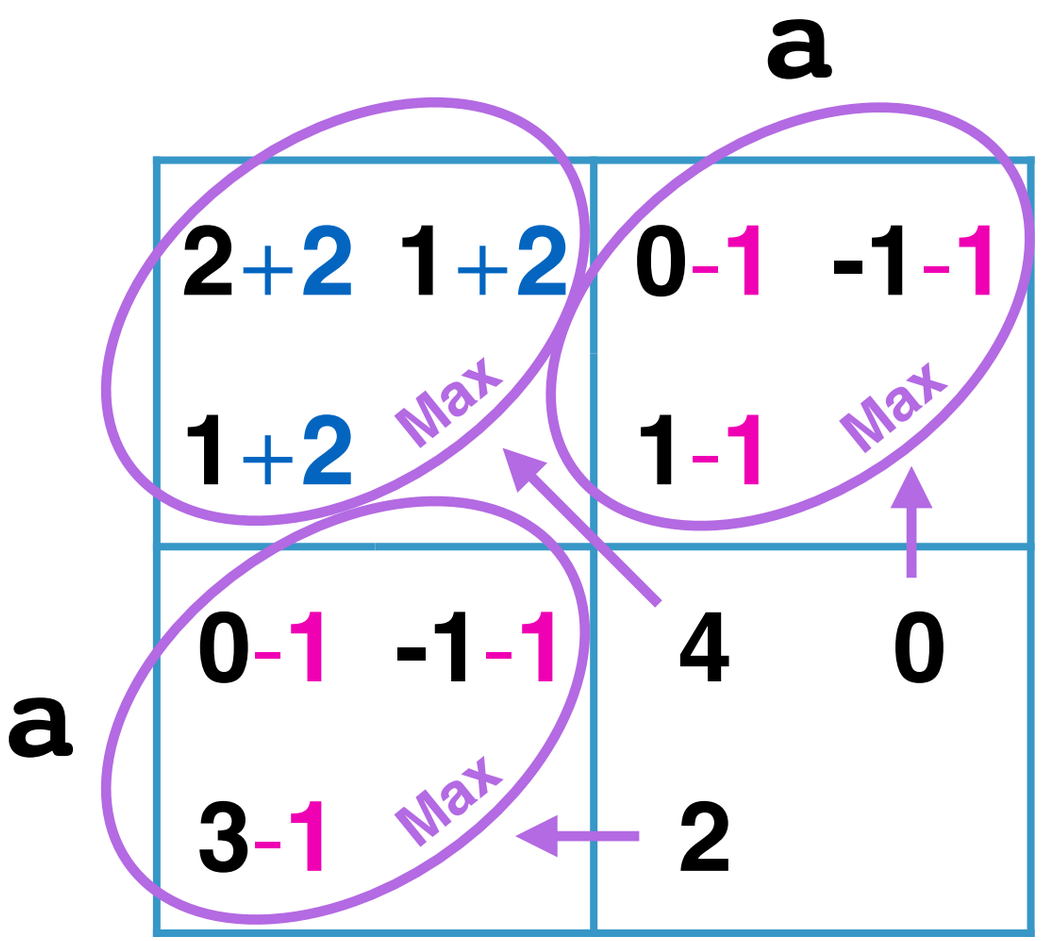
Linear:

Match score: **2**
Gap score: **-1**



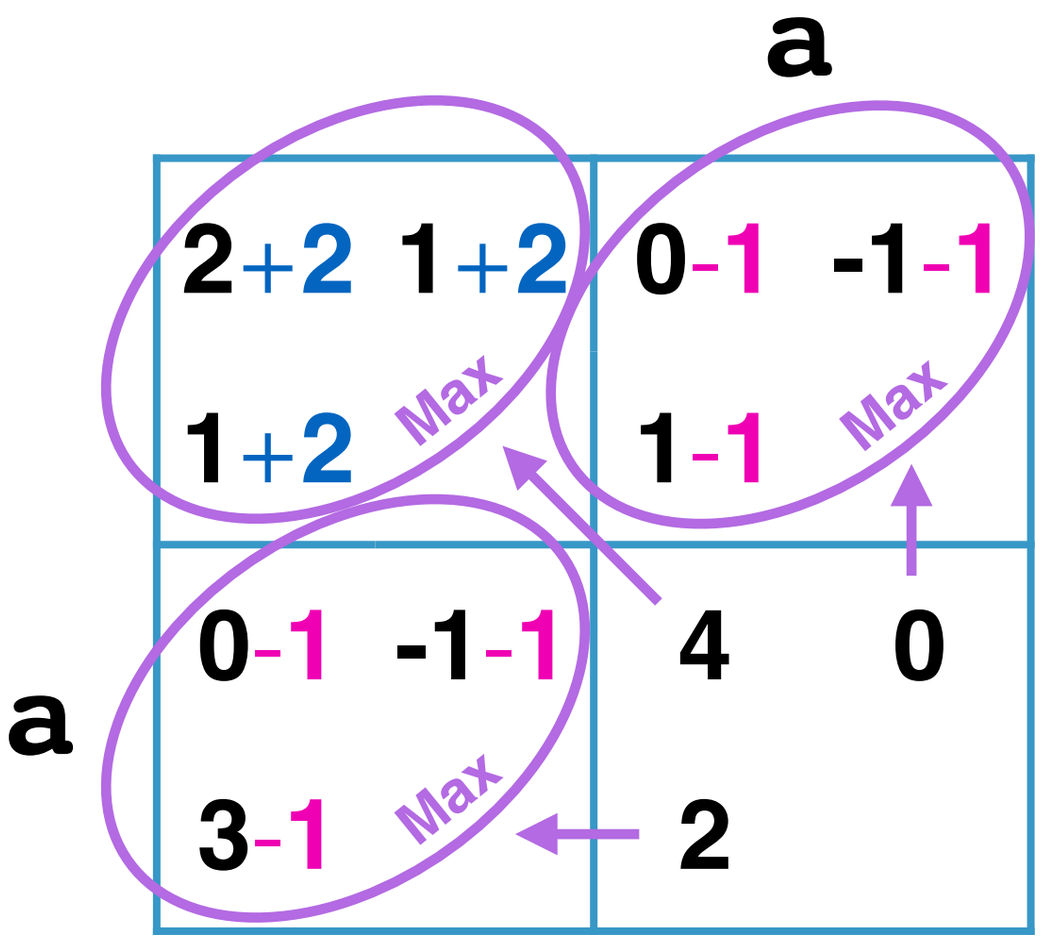
Linear:

Match score: **2**
Gap score: **-1**



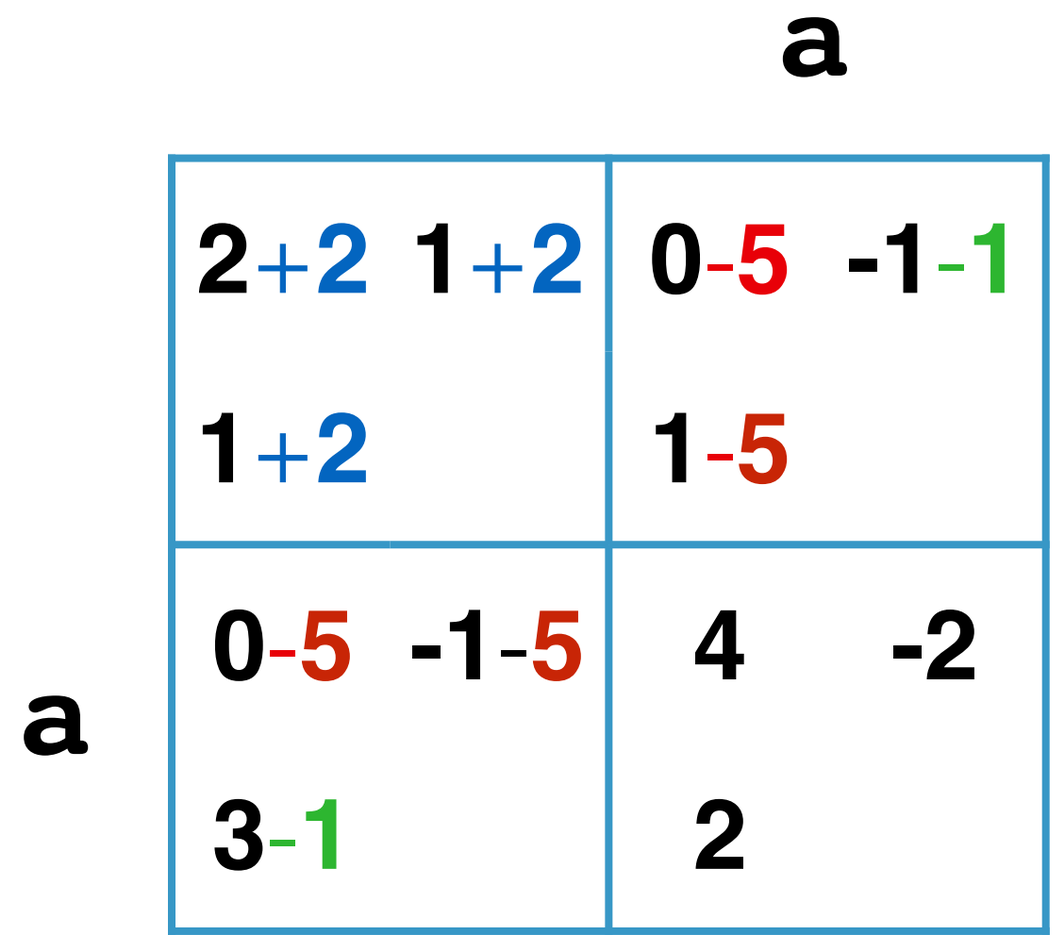
Linear:

Match score: **2**
Gap score: **-1**



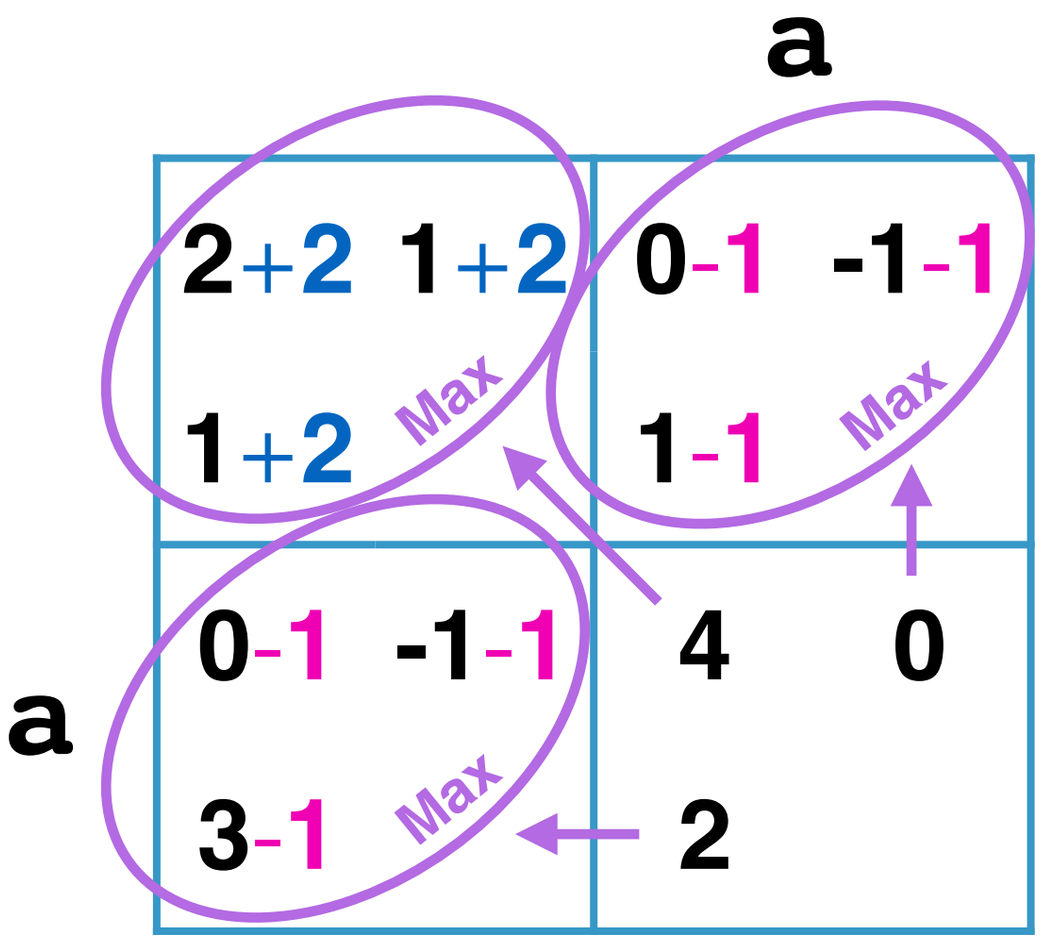
Affine:

Match score: **2**
Gap open: **-5**
Gap extend: **-1**



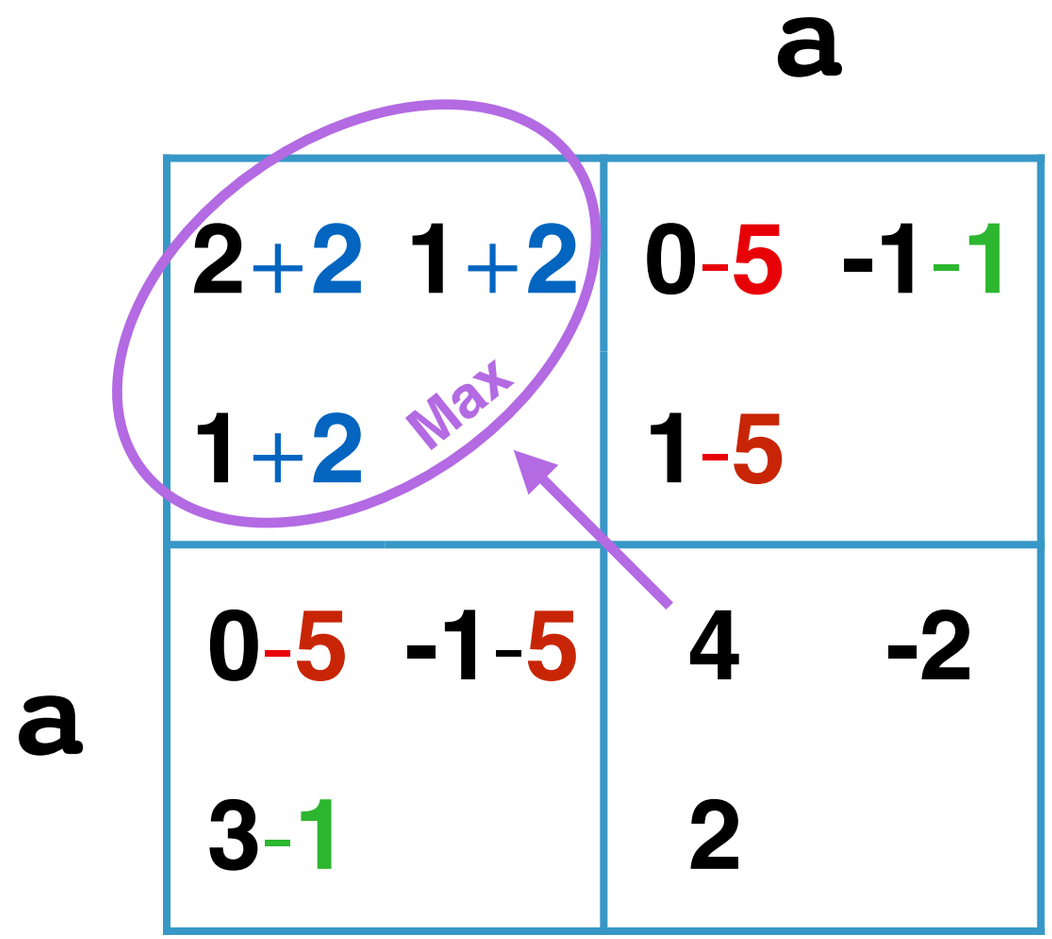
Linear:

Match score: **2**
Gap score: **-1**



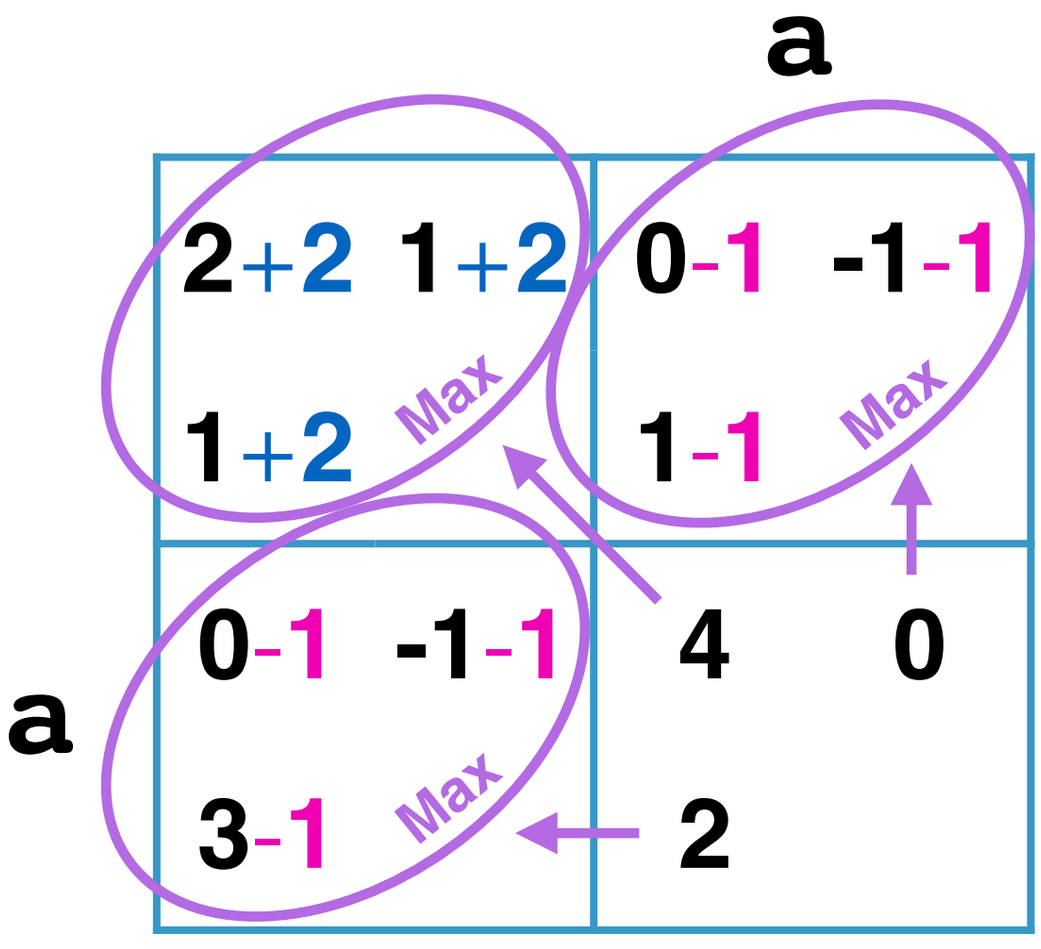
Affine:

Match score: **2**
Gap open: **-5**
Gap extend: **-1**



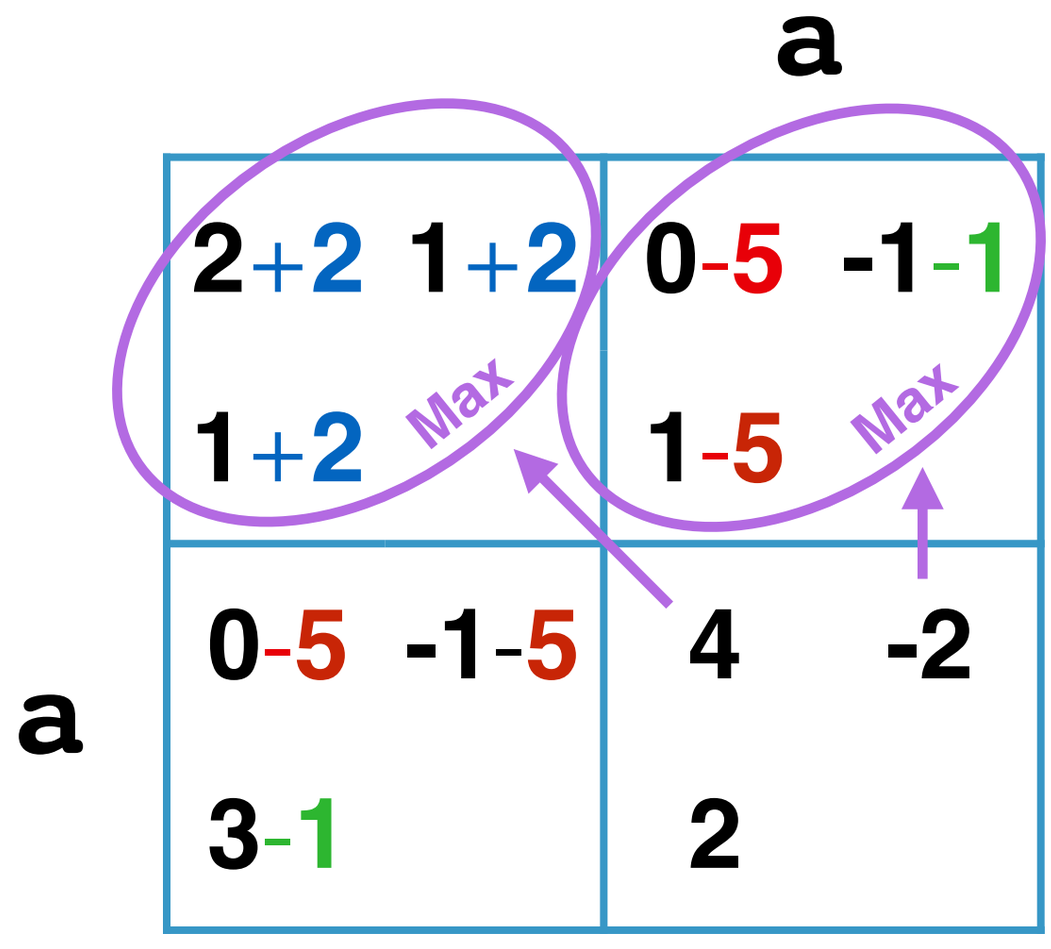
Linear:

Match score: **2**
Gap score: **-1**



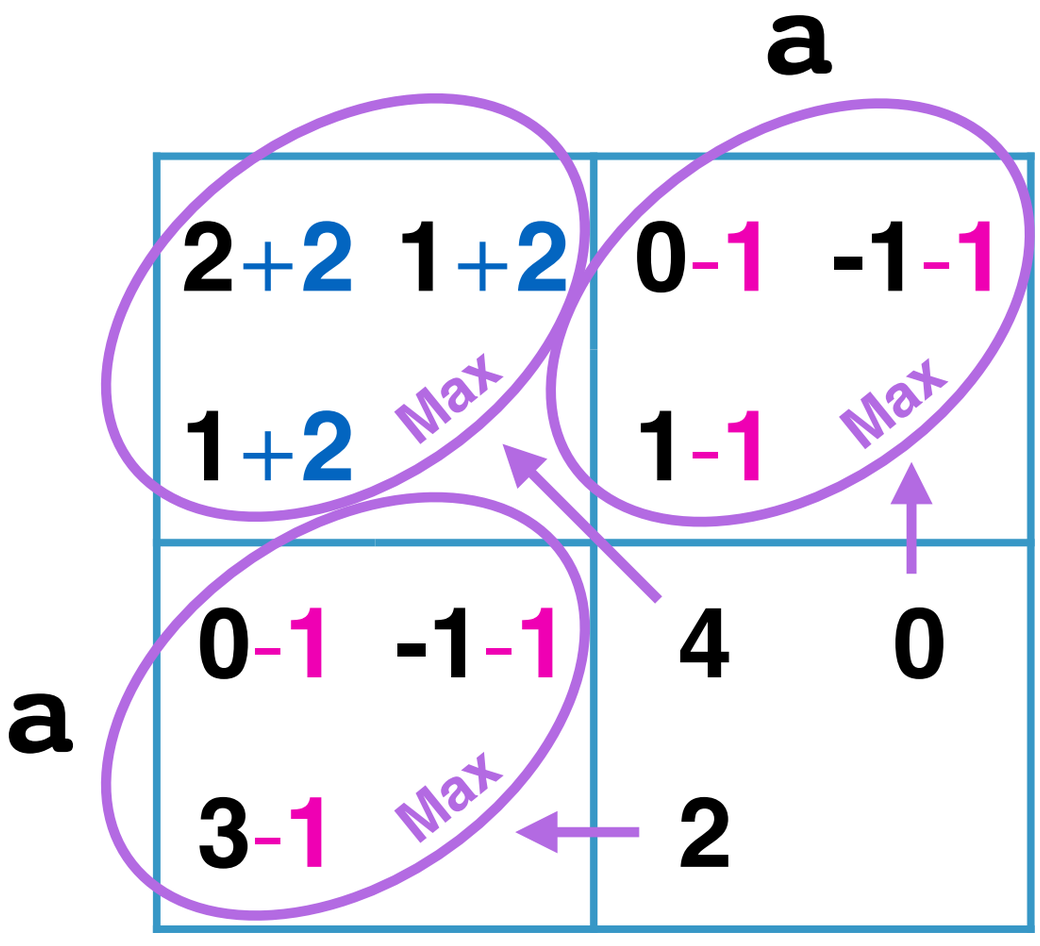
Affine:

Match score: **2**
Gap open: **-5**
Gap extend: **-1**



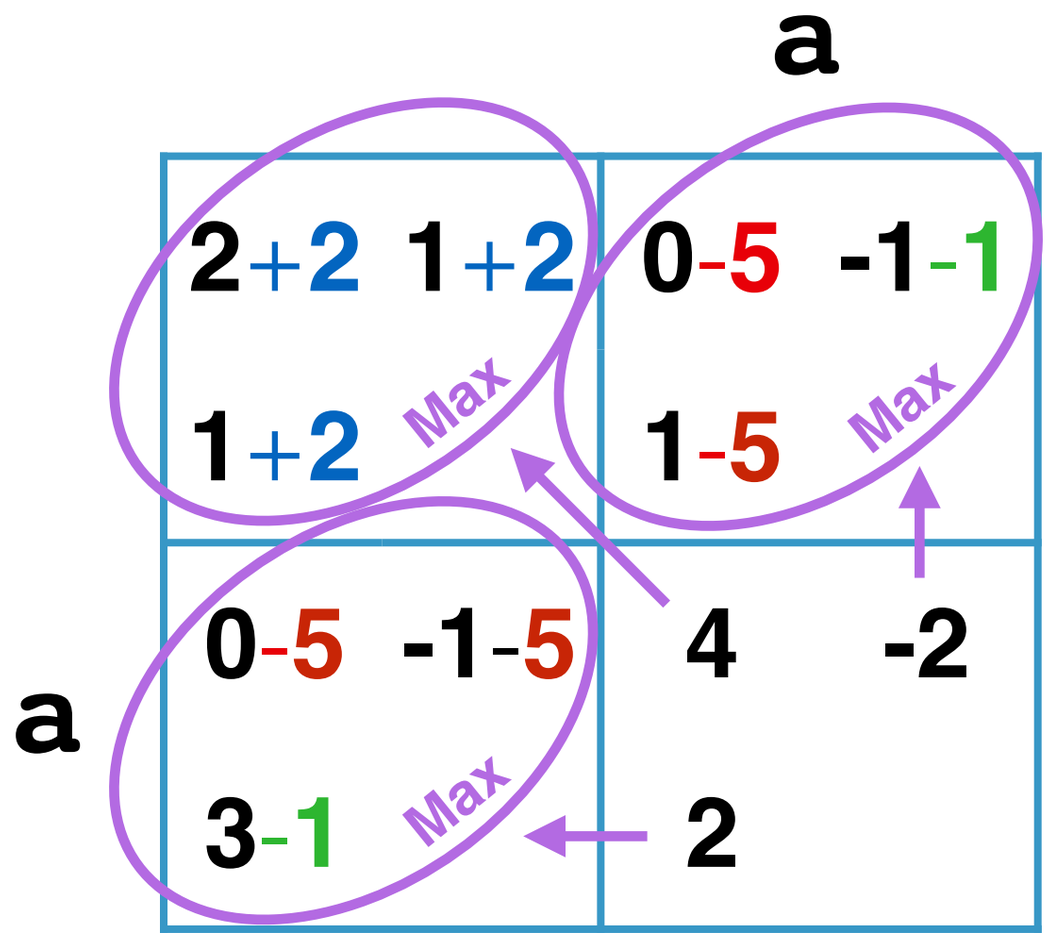
Linear:

Match score: **2**
Gap score: **-1**



Affine:

Match score: **2**
Gap open: **-5**
Gap extend: **-1**



EDNAFULL

	A	C	G	T
A	5	-4	-4	-4
C	-4	5	-4	-4
G	-4	-4	5	-4
T	-4	-4	-4	5

Gap open	-10
Gap extend	-0.5

EDNAFULL

	A	T	G	C	S	W	R	Y	K	M	B	V	H	D	N
A	5	-4	-4	-4	-4	1	1	-4	-4	1	-4	-1	-1	-1	-2
T	-4	5	-4	-4	-4	1	-4	1	1	-4	-1	-4	-1	-1	-2
G	-4	-4	5	-4	1	-4	1	-4	1	-4	-1	-1	-4	-1	-2
C	-4	-4	-4	5	1	-4	-4	1	-4	1	-1	-1	-1	-4	-2
S	-4	-4	1	1	-1	-4	-2	-2	-2	-2	-1	-1	-3	-3	-1
W	1	1	-4	-4	-4	-1	-2	-2	-2	-2	-3	-3	-1	-1	-1
R	1	-4	1	-4	-2	-2	-1	-4	-2	-2	-3	-1	-3	-1	-1
Y	-4	1	-4	1	-2	-2	-4	-1	-2	-2	-1	-3	-1	-3	-1
K	-4	1	1	-4	-2	-2	-2	-2	-1	-4	-1	-3	-3	-1	-1
M	1	-4	-4	1	-2	-2	-2	-2	-4	-1	-3	-1	-1	-3	-1
B	-4	-1	-1	-1	-1	-3	-3	-1	-1	-3	-1	-2	-2	-2	-1
V	-1	-4	-1	-1	-1	-3	-1	-3	-3	-1	-2	-1	-2	-2	-1
H	-1	-1	-4	-1	-3	-1	-3	-1	-3	-1	-2	-2	-1	-2	-1
D	-1	-1	-1	-4	-3	-1	-1	-3	-1	-3	-2	-2	-2	-1	-1
N	-2	-2	-2	-2	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1